

Convention for citing the REACH and the CLP Regulations

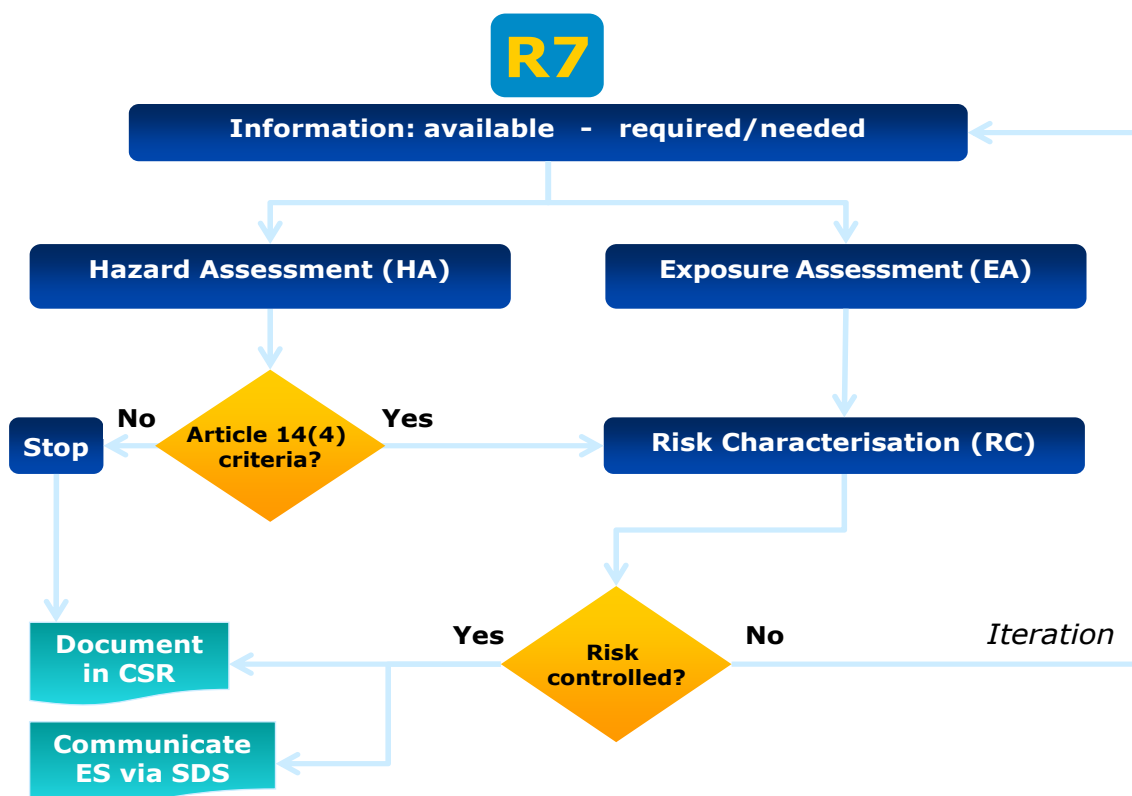
Where the REACH and the CLP Regulations are cited literally, this is indicated by text in italics between quotes.

Table of Terms and Abbreviations

See Chapter R.20

Pathfinder

The figure below indicates the location of part R.7(a) within the Guidance Document



R.7 Endpoint specific guidance

Introduction

The previous sections of the *Guidance on information requirements and chemical safety assessment* (IR&CSA) provide advice on the interpretation and application of generic aspects of the Regulation describing the overall process that should be followed in finding, assembling and evaluating all the relevant information that is required for the registration of a chemical under Regulation (EC) No 1907/2006 (the REACH Regulation). The chapters also describe factors that may have an influence on the information requirements and give advice on how the information collected from different sources could be integrated and used in an approach to allow a conclusion on whether or not the available information is sufficient for regulatory purposes, i.e. hazard assessment and risk assessment.

Under Regulation (EC) No 1272/2008 (CLP Regulation or CLP), this approach is called a *Weight-of-Evidence* (WoE) determination. According to CLP, an evaluation by applying WoE determination (i.e. all available information relevant for the evaluation of the specific hazard is considered together) using expert judgement, must always be carried out where the criteria cannot be applied directly (Article 9(3), CLP). This WoE determination should not be confused with the use of Weight of Evidence according to Annex XI, 1.2 of REACH, an adaptation rule for standard information requirements where sufficient weight of evidence may allow the conclusion/assumption that a substance has or has not a particular dangerous property.

The guidance given thus far is applicable across the field and comprises the general rules that should be followed.

Structure of Chapter R.7a

In this chapter, specific guidance on meeting the information requirements set out in Annexes VI to XI to the REACH Regulation is provided. The information requirements relate both to those physicochemical properties that are relevant for exposure and fate considerations as well as to physical hazards, human health hazards and environmental hazards. The guidance for each specified property or hazard has been developed as a specific "sub-chapter" (referred to as a "Section") in this guidance, addressing the aspects of collection, generation and evaluation of information to help registrants provide adequate and relevant information for registration under REACH.

All data sources, including non-testing data, have to be taken into account when doing the chemical safety assessment. Most of the reports follow a logical common format that complements the generic guidance and the general decision making frameworks detailed in the first paragraph above.

R.7.1 Physicochemical properties

This first "sub-chapter", underwent a guidance revision process between 2011 and 2012 and therefore follows a revised structure. The Section R.7.1 covers both classification and non-classification related properties, where the sections covering the physicochemical properties each have six or seven "sub-sections" (also referred to as "sections"), depending on the need for information on references and the sections covering the physical hazards have seven "sub-sections" (also referred to as "sections").

In the physicochemical properties sections:

- the first section details the type of property;
- the second section provides the definition of the property;
- the third section lists the preferred test method(s);

- the fourth section deals with adaptation of the standard testing regime, namely adaptation options that can be explored under each specific physicochemical property;
- the fifth section deals with impurities and uncertainties, and
- the last section outlines what kind of property-specific information should be given in the registration dossier (note that sometimes an additional section is added where relevant references are provided).

By contrast the physical hazard sections:

- start with the definition section;
- followed by a second section on classification criteria and relevant information;
- the third section explores various adaptation options, namely how the standard testing regime can be adapted;
- the fourth section outlines the impurities and uncertainties;
- the fifth section aims to help in concluding on the Directive 67/548/EEC (Dangerous Substances Directive - DSD) classification, repealed by Regulation (EC) No 1272/2008 (CLP Regulation or CLP);
- the sixth section outlines the physical hazards-specific information to be included in the registration dossier and in IUCLID, and
- the seventh section gives relevant further information and used references.

R.7.2 Human health properties or hazards

Chapters tackling human health properties or hazards in R.7a remain generally unchanged, using a similar structure. However as each section is updated the information may be re-organised to be presented in a clearer and more constructive order. In these chapters there are seven main sections to the guidance on each property or hazard:

- the introduction section (R.7.X.1 Introduction) provides an introduction in which the property or hazard is described, further defined and an explanation given as to its importance in the context of human health, or environmental fate and effect of a given substance;
- the second section (R.7.X.2 Information requirements ...) details the specific information requirements for the endpoint of interest; these will depend on the tonnage band of the substance, its usage pattern and other considerations including data on other endpoints and on related substances. Endpoint² specific guidance can be thought of as logical steps that should be taken to assemble the information that is detailed under the second section; thus
- the third section (R.7.X.3 Information sources on...) provides an inventory of all the types of data that could potentially provide useful information on the endpoint of interest and, most importantly the sources of that information;

² REACH uses the term "endpoint" both to denote a physicochemical property (example: Annex VII to REACH, Column 1 standard information required: 7.3 Boiling point, and 7.4 Relative density) and to denote hazardous properties (example: Annex VII to REACH, Column 1 standard information required: 7.11 Explosive properties and 7.13 Oxidising properties), which are subject to classification according to the applicable EU legislation. In the following, the wording of Part 7(a) of this guidance document will differentiate between these different types of properties where this appears appropriate, in order to facilitate the identification of properties which serve the regulatory purpose of classification.

- in the fourth section (R.7.X.4 Evaluation of available information for...) guidance is given on how to evaluate the information that might be available for a given substance; this advice focuses on providing the criteria to aid in the judgement and ranking of the available data for their adequacy and completeness. This section also provides an indication of the remaining uncertainty inherent in the different types of data for the given endpoint;
- the fifth section (R.7.X.5 Conclusions on...) describes how conclusions may be drawn for a given substance on the suitability of the available information for regulatory purposes. Chemical safety assessment within REACH is fundamentally dependent on an adequate conclusion on classification and PBT/vPvB assessment since exposure assessment and risk characterisation are triggered by classification and fulfilment of PBT/vPvB criteria. Therefore data need to be adequate for both classification & labelling and for chemical safety assessment if the latter is required;
- the sixth section (R.7.X.6 Integrated Testing Strategy (ITS) for...) comprises an Integrated Testing Strategy (ITS), also referred to in some sections as Testing and assessment strategy, for the given endpoint(s), providing guidance on how to define and generate relevant information on substances in order to meet the requirements of REACH. It is noteworthy that all experiments using vertebrate animals shall be designed to avoid distress and unnecessary pain and suffering to experimental animals, in accordance with Article 7(4) of Directive 86/609/EEC.

The proposed testing strategies are guidance for data generation in a stepwise approach. The strategies build on the concept that if the available information is not sufficient to meet the regulatory needs, further gathering of information at a succeeding step in the testing strategies is needed. On the other hand, if the available information is adequate and the standard information requirements are met, no further gathering of information is necessary. Standard information requirements will not need to be fulfilled by standard tests, where the available information is judged to be sufficient to adapt the standard information requirements in accordance with REACH Annex XI or an applicable Column 2 provision of REACH Annexes VII to X;

- the seventh and final section (R.7.X.7 References on...) lists all used references on the given endpoints.

Additional considerations

The following additional considerations apply generally to the endpoint specific guidance given in this chapter:

Information requirements in the light of the applicable classification regime

The main regulatory purpose of the information requirements set out in Annexes VI to X to the REACH Regulation is to assess hazards and risks related to substances and to develop and recommend appropriate risk management measures, as highlighted in Recital 19 of the REACH Regulation. According to Recital 26: *'in order to undertake chemical safety assessments of substances effectively, manufacturers and importers of substances should obtain information on these substances, if necessary by performing new tests'*. The chemical safety assessment (CSA) should be performed in accordance with the provisions set out in Annex I to the REACH Regulation. According to Section 0.6 of Annex I, the first three steps of the CSA require the carrying out of a human health hazard assessment, a human health hazard assessment of physicochemical properties and an environmental hazard assessment, including determining the classification of substances. When the REACH Regulation was adopted, the DSD was the applicable classification regime (see, more in particular, the transitional provisions set out in Article 61 of Regulation (EC) No 1272/2008). Accordingly, many REACH information requirements are inspired by the categories of danger under DSD such as points 7.10, 7.11

and 7.13 in Column 1 of REACH Annex VII (*i.e.* flammability, explosive properties and oxidising properties, respectively).

On 20 January 2009 Regulation (EC) No 1272/2008 (CLP Regulation or CLP) entered into force. The CLP Regulation has amended certain parts of the REACH Regulation (see Article 58 of CLP for amendments applicable from 1 December 2010 and Article 59 of CLP for amendments applicable from 1 June 2015). Nevertheless, the terminology used in REACH currently still comprises terms which were used under the DSD (for substances) and still apply (for mixtures until 1 June 2015) under Directive 1999/45/EC (Dangerous Preparations Directive – DPD). With respect to the updated physicochemical part of this guidance and the section dealing with the exploration of adaptation possibilities of the standard testing regime, the term ‘dangerous’ can be interpreted in a broader context (particularly, in certain contexts within this document, to include ‘hazardous’ as defined under CLP) as it does not refer strictly to the DSD.

According to the requirements of Article 10(a)(iv) of the REACH Regulation, the technical dossier required for registration purposes includes the classification and labelling of the substance as specified in Section 4 of Annex VI to REACH, resulting from the application of Titles I and II of the CLP Regulation. From 1 December 2010 until 1 June 2015 substances must be classified in accordance with both DSD and CLP and they must be labelled and packaged in accordance with CLP (Article 61(3) of CLP). Similarly, until 1 June 2015 Safety Data Sheets (SDS) must include information on classifications according to both CLP and DSD for substances and component substances in mixtures until 1 June 2015 (see updates to REACH *via* Commission Regulation (EU) No 453/2010 and the ECHA [Guidance on the compilation of Safety Data Sheets](#)).

Use of data derived from EU or other international standardised test methods

For the purposes of determining whether any of the physical hazards referred to in Part 2 of Annex I of CLP apply to a substance (or a mixture), the manufacturer, importer or downstream user must perform the tests required by the above mentioned Part 2, unless there is adequate and reliable information available (see Article 8(3) of CLP). Further in this guidance for each relevant physical hazard a reference to the corresponding test according to UN Recommendations on the Transport and Dangerous Goods, Manual of Test and Criteria (UN-MTC), starting with a UN test method name will be provided.

According to Article 8(5) of CLP, where new tests for **physical hazards** are carried out for classification and labelling purposes, they must be performed in compliance with a relevant recognised quality system (e.g. GLP) or by laboratories complying with a relevant recognised standard (e.g. with EN ISO/IEC 17025), at the latest from January 2014.

For the purpose of determining whether a substance or mixture fulfils the criteria for classification in any of the **human health and/or environmental hazard classes** (and differentiations within a hazard class, if applicable), there is no similar testing requirement. If there is already adequate and reliable information available (see Article 8(2) of CLP), this must be used. Provided that the manufacturer, importer or downstream user has exhausted all other means of generating information, new tests may however be performed (Article 8(1), CLP).

Where new tests for **human health or environmental hazards** are carried out for classification purposes, they must be performed in compliance with a relevant recognised quality system (e.g. GLP) or by laboratories complying with a relevant recognised standard (e.g. with EN ISO/IEC 17025), at the latest from January 2014. (Article 8(5), CLP). Further requirements for tests performed for the purpose of CLP are given in Article 8 of CLP.

Further, according to Article 13(3) of REACH, tests for generating information on intrinsic properties of substances must be conducted in accordance with the test methods laid down in Commission Regulation (EC) 440/2008 (Test Methods Regulation)³ or in accordance with other international test methods recognised by the Commission or the Agency as being appropriate, such as European Standards (EN) (<http://www.cen.eu/Pages/default.aspx>) or the OECD guidelines (<http://www.oecd.org/chemicalsafety/testing/oecdguidelinesforthetestingofchemicals.htm>). Regulation (EC) 440/2008 lays down the test methods to be applied for the purposes of REACH. Thus, in the following sections on specific endpoints, references given for each test method will include the OECD Test Guideline (TG) number and, where available, the test method number, as defined in the Test Methods Regulation.

According to Recital 37 of the REACH Regulation, if tests are performed, they should comply with the relevant requirements for protection of laboratory animals, as set out in Council Directive 86/609/EEC⁴. Article 13(4) of REACH states that ecotoxicological and toxicological tests and analyses must be carried out in compliance with the principles of good laboratory practice (GLP) provided for in Directive 2004/10/EC⁵ or other international standards recognised as being equivalent by the Commission or the Agency and with the provisions of Council Directive 86/609/EEC, if applicable.

Interdependence of endpoints in hazard assessment

Although guidance is provided for each specific endpoint separately, it should be remembered that different endpoints are related to each other. Information collected within one endpoint may influence hazard/risk assessment of other endpoints, for example, information on rapid primary degradation of a parent substance may result in including the degradation products in the overall assessment of the toxicity of a substance. Regarding the physicochemical properties of a substance, for example boiling point and flash point are properties used for the classification of flammable liquids, and therefore these properties are important for physical hazard assessment. Similarly, information on toxicity/specific mode of action in one endpoint may indicate possible adverse effects for organisms considered for assessment of other endpoints, for example, endocrine disrupting mode of action in mammals may indicate the same mode of action in fish. Another example may be when data on toxic effects measured in one group of organisms may be directly used in more than one endpoint, for example, data from a repeated dose toxicity study may also be used in assessment of risk for secondary poisoning of mammals exposed *via* the food chains.

Adequacy of methods for generating additional information

Before (proposing) additional animal testing, use of all other options should be considered. It is important to emphasise that testing on vertebrate animals must only be conducted or proposed as a last resort, when all other data sources have been exhausted (see Recital 47 of the REACH Regulation, Article 25 of REACH and Step 4 of REACH Annex VI). Therefore, it is

³ Council Regulation (EC) No 440/2008 laying down test methods pursuant to Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) [OJ L 142, 31.5.2008, p. 1].

⁴ Council Directive of 24 November 1986 on the approximation of laws, regulations and administrative provisions of the Member States regarding the protection of animals used for experimental and other scientific purposes (86/609/EEC).

⁵ Directive 2004/10/EC of the European Parliament and of the Council of 11 February 2004 on the harmonisation of laws, regulations and administrative provisions relating to the application of the principles of good laboratory practice and the verification of their applications for tests on chemical substances.

important to first consider all issues that may impact upon this decision whether and how to perform the testing, such as:

- applicable information requirements pursuant to REACH;
- adaptation possibilities of REACH Annex XI and Column 2 of REACH Annexes VII to X, e.g.:
 - classifications that may allow for adaptations,
 - available data on a category, a group or on individual substances for which the physicochemical and toxicological properties are likely to be similar,
 - assumption/conclusion on presence or absence of a particular dangerous property of a substance in a weight of evidence approach based on animal or human data, several independent sources,
 - absence or no significant exposure based on exposure scenarios;
- substance properties;
- available *in vitro* and *in vivo* data;
- available toxicokinetic and toxicodynamic information;
- any trigger/alert that may require testing going beyond the applicable minimum information requirements.

All these issues should be considered, not only to design fit-for-purpose *in vivo* tests, but also for justifying why an *in vivo* study is not needed under certain circumstances. Animal tests must comply with the provisions laid down in Council Directive 86/609/EEC⁶.

Degradation products and metabolites

In the context of evaluating substances for their effects, it is important to note that, once released into the environment or taken up by animals, a substance may be transformed through degradation or metabolism. These processes and their outcome may need to be taken into account in the overall assessment.

Degradation products may be formed as a result of transformation processes in the environment, either biotic or abiotic. For distinguishing the substance undergoing degradation from the degradation products, the former is often referred to as the parent substance.

Degradation products may be formed as a result of abiotic environmental processes such as hydrolysis, direct or indirect photolysis or oxidation. They may also be formed as a result of aerobic or anaerobic biodegradation, i.e. due to microbial activity. Degradation products require further investigation if the Chemical Safety Assessment indicates the need, i.e. if stable degradation products are formed in the environment within a relevant time frame, as deduced from the test system, or if they fulfil the PBT/vPvB criteria. Likewise it may be considered to assess whether degradation products fulfil the environmental hazard classification criteria (see Section R.7.9 in Chapter R.7b of the [Guidance on IR&CSA](#)).

Metabolites refer to transformation products, which are formed due to biodegradation (and then the term metabolite is synonymous with the term biodegradation product) or formed as a result of biotransformation (metabolism) within exposed organisms after uptake of the parent substance. Metabolic pathways and hence the identity of metabolites may or may not be fully

⁶ Council Directive 86/609/EEC regarding the protection of animals used for experimental and other scientific purposes [OJ L 358, 18.12.1986, p. 1].

known. The latter is frequently the case. Moreover for the same substances metabolic pathways may or may not differ between various organisms belonging to different phyla and/or trophic levels. However, the toxicity of metabolites formed within the duration of laboratory tests will be reflected by their parent substance, with the exception of delayed effects which are only evident after the observation time of the tests. Knowledge of metabolic pathways and metabolites may increase planning and focussing of toxicity testing and understanding of toxicological findings (see Section R.7.12 in Chapter R.7c of the [Guidance on IR&CSA](#)). Therefore, in some cases it may be possible to use grouping approaches for structurally closely-related substances, which undergo similar metabolic transformation (see Section R.6.2 in Chapter R.6 *Guidance on QSARs and grouping of substances* of the [Guidance on IR&CSA](#)).

When biotransformation processes include oxidation, metabolites are often less hydrophobic than the parent substance. This is a very general rule of thumb and may not always apply; however, when it does, often this has implications for the hazard profile of the metabolites. For example more polar metabolites created after oxidation processes have normally a lower adsorption potential, and thus the relevance of the metabolites for the soil and sediment compartments is normally lower than that of the parent substance. Such less hydrophobic metabolites also tend to be excreted more rapidly from organisms than the parent substance. Hence both their bioaccumulative potential and narcotic toxicity tend to be lower.

Similarities in metabolic pathways of structurally-related substances may serve as an indication for waiving for further investigation, depending on the case and nature of the metabolites.

It should be noted that metals, and in particular metal substances, do not degrade in the environment in the same way as organic substances. They transform usually through dissolution to the dissolved form.

Selection of the appropriate route of administration for toxicity testing

Having established the need for additional toxicity testing to meet the requirements of REACH for a given substance, for certain endpoints, notably acute or repeated dose toxicity but also reproductive toxicity, chronic toxicity and carcinogenicity, a decision must be made on which route(s) of exposure is(are) most appropriate. The overall objective of such testing is to determine the potential hazard of the test substance to human beings. Humans may normally be exposed to substances by one or more of three routes: inhalation, dermal and oral. In general, the final decision on which route of exposure is to be considered in a particular test should be taken in the light of the requirements for the particular endpoint concerned, the recommendation given in the respective test methods, all available information including physicochemical properties of the substance, human exposure, structure-activity relationships (SAR) or the data from available toxicity tests on the substance itself.

If no adequate experimental effect data using the relevant route of administration is available, route-to-route extrapolation might be an alternative method for evaluating the hazard. However this approach should only be used for systemic effects, and not for local effects such as irritation of the lungs following inhalation of a substance. Route-to-route extrapolation is recommended only under conditions where route-specific effects are not expected. Therefore, route-to-route extrapolation should be considered on a case-by-case basis taking into account the additional uncertainties. It is to be noted that route-to-route extrapolation is associated with a high degree of uncertainty and should be conducted with caution relying on expert judgement. In a subsequent risk assessment the uncertainties introduced through route-to-route extrapolation should be taken into account, for example by adjusting the assessment factor in the determination of the DNEL (see Section R.8.4.3, Chapter R.8 *Characterisation of dose [concentration]-response for human health* of the [Guidance on IR&CSA](#)). Further guidance on this strategic approach to toxicity testing is given in Chapter R.8 of the [Guidance on IR&CSA](#).

Assessment of the environmental impact of a substance

With regard to the evaluation of the environmental impact of a substance, the interaction of that substance with the environment is an important consideration. The fate and behaviour of a substance are largely governed by its inherent physicochemical properties. The knowledge of the physicochemical properties of the substance, together with results from multimedia fate and transport models (e.g. Mackay level 3 models), enables the identification of the environmental compartment(s) of primary concern. Such information will also determine the prioritisation of higher tiered tests. More extensive guidance and considerations on this aspect are given in Chapter R.16 *Environmental Exposure Estimation*.

R.7.1 Physicochemical properties

Advice to registrants with regard to nanomaterials characterisation can be found in *Appendix R7-1 Recommendations for nanomaterials applicable to: Chapter R7a Endpoint specific guidance* of the [Guidance on IR&CSA](#), section 2 Recommendation for physicochemical properties arising from RIP-oN 2 for nanomaterials.

R.7.1.1 Introduction on physicochemical properties

According to Article 12 of the REACH Regulation, for registration purposes all physicochemical information that is relevant and available to the registrant must be included in the technical dossier, i.e. information such as:

- data on intrinsic properties of the substance (e. g. melting point/freezing point, boiling point, vapour pressure, density);
- data necessary to assess the physical hazards of a substance (e. g. flammability), with the view to determine its classification and labelling according to CLP (and until 1 June 2015, according to DPD, see Article 61 of CLP);
- supplementary data for hazard assessment and health and environmental classification (e. g. viscosity, n-octanol/water partition coefficient).

Some physicochemical properties - notably explosive, flammable and oxidising properties - are intimately linked to physical hazards. The most straight-forward way of assessing these properties is through the classification of the substance for the corresponding physical hazards. There is thus a link between the physical hazards classification and the information requirements on explosive, flammable and oxidising properties. This is further elaborated below (see [Table R.7.1-1](#)) and in the various chapters addressing these endpoints. For substances manufactured or imported in quantities of 100 tonnes or more per annum, some additional physicochemical data are required; in accordance with Annex IX to REACH (see also [Table R.7.1-1](#)).

Further details are given in the sections dedicated to specific endpoints.

R.7.1.1.1 Information requirements on physicochemical properties

Commission Regulation (EU) No 252/2011⁷ has amended Annex I to REACH in order to adapt the chemical safety assessment provisions to the criteria for classification laid down in the CLP Regulation. The relevant amendments have been applied since 5 May 2011; however, for registrations submitted prior to this date, the chemical safety report shall be updated in accordance with Regulation No 252/2011 by 30 November 2012 at the latest.

The information needed under Article 12, REACH on one hand and according to section 4 of Annex VI to REACH on the other (namely hazard classification according to Title I and II CLP) is often complementary but in some cases may be different. The reason is that the classification criteria and/or test methods under DSD and CLP regimes are different. This is also expressed by the fact that CLP classifications are distributed over a different grid of hazard classes and categories compared to the DSD regime, e.g. substances and mixtures

⁷ Commission Regulation (EU) No 252/2011 of 15 March 2011 amending Regulation (EC) No 1907/2006 of the European Parliament and of the Council on the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) as regards Annex I.

classified as explosive under DSD may be classified as explosives or self-reactives or organic peroxides under CLP, or they may even be classified as flammable solids, oxidizing solids or not at all. A translation table from DSD to CLP classification is provided in Annex VII, CLP and an indication of potential classification outcomes under CLP compared to DSD classifications is provided by Table 1.7.2.1(a) in the [Guidance on the Application of the CLP criteria](#).

The CLP classification regime is not explicitly considered in Annex VII to REACH and therefore has to be understood as part of the information requirements under REACH. In particular, certain headlines set out in column 1 of Annex VII to REACH, namely 'explosive properties', 'flammability' and 'oxidizing properties', must be interpreted as covering the CLP hazard classes that are referred to in Article 58(11) of CLP.

The physical hazard classes according to CLP are structured differently from the corresponding classifications according to DSD. Despite this, most of the CLP physical hazard classes can unambiguously be assigned to specific heading of the information requirements according to Annexes VI to IX to REACH. However, for some CLP physical hazard classes - notably the hazard class 'self-reactive substances and mixtures' and the hazard class 'organic peroxides' - the assignment to a specific heading is not straight-forward, since they may have both explosive and/or flammable properties. Therefore, some of the hazard classes are listed twice in [Table R.7.1-1](#) below. It should be noted that this assignment is provided only as example and is done for the purposes of structuring this guidance document according to CLP but nevertheless also allowing the assignment to the respective information requirements according to Annexes VII to IX to REACH.

According to Article 1(6) CLP, CLP Regulation does not apply to the transport of dangerous goods by air, sea, road, rail or inland waterways (save where the specific rules for labelling of packaging applies under Article 33 of CLP). The transport of dangerous goods is, covered by the UN Model Regulations for Transport of Dangerous Goods (UN-RTDG) and related legal instruments (ADR, RID, ADN, IMDG Code and ICAO TI); the criteria listed in these instruments and in CLP Regulation for classification purposes are intended to be the same. Thus, a substance (or mixture) classified in a hazard class which is common to both CLP and the transport legislation will normally be classified the same according to both systems. Therefore the transport classification of a substance could be a source of information for the classification and labelling of substance (or a mixture) under CLP for physical hazards. However it should be kept in mind that the transport classifications do not cover all hazard categories which are relevant for CLP and it may be based on other considerations than just the test data and criteria (e.g classifications which are based on experience rather than testing or which apply only in connections with certain special provisions). As a result, the transport classifications may be different for the classification according to CLP. Similarly, the absence of a transport classification does not necessarily mean the substance (or mixture) should not be classified under CLP. Consequently in the case of a substance which has been tested for the purposes of the UN-RTDG and for which the same procedure was followed as required by the CLP Regulation, the same information could be used to comply with the REACH Regulation on a case-by-case basis. The limitations to the approach described above are described in detail in the [Guidance on the Application of the CLP criteria](#), Section 1.7.2.1.

For the preparation of the registration dossier, registrants are required to submit all the information listed in Article 10 of REACH. Article 14(1) in conjunction with Annex I and Article 10(a)(vii) of the REACH Regulation, require the provision of a Robust Study Summary (RSS) for information derived from the application of Annexes VI to XI to REACH. In order to facilitate the evaluation conducted by the European Chemicals Agency and the Member States, as well as to save registrant's resources in case of a tonnage update, it **is recommended** that registrants also use the RSS for covering physicochemical endpoints under section 4 of the IUCLID file. This guidance includes under each physicochemical property chapter a list of detailed information to be given for each respective endpoint. Note that no further guidance is provided on the general aspects related to information common for all endpoints in IUCLID. For these aspects, further guidance can be found in *Practical guide 3: How to report robust study*

summaries available on the ECHA Website (at: <http://echa.europa.eu/practical-guides>) and in the *IUCLID 5 End User Manual* available on the IUCLID Website at: <http://iuclid.eu/index.php?fuseaction=home.documentation#usermanual>.

Those endpoints, such as explosive properties and oxidising properties, which are intimately linked to classification, should be assessed according to CLP. For these endpoints, the test methods of CLP should preferably be used, in order to avoid double testing. For endpoints not linked to classification the preferred test methods are those found in the Test Method Regulation. For some endpoints (for example flammability), more than one test procedure is referred to in the standard test method reported in the Test Method Regulation. The one chosen should be suitable for the substance in question and be operating within its validity range.

Note that in the table below ([Table R.7.1-1](#)) in order to distinguish the physicochemical properties that are directly linked to physical hazard classifications from those that are not, the former have been shaded in gray and that in addition the preferred test methods for the different endpoints have been put in bold text.

Table R.7.1-1 Information requirements as specified in Annexes VII to IX to REACH and corresponding tests methods according to the Test Method Regulation and CLP

Information requirement according to Art. 10 (a) (vi) of the REACH Regulation (EC) No. 1907/2006 (the no. in brackets is the respective no. in the table in Annexes VII to IX to REACH)	Corresponding test method according to The Test Method Regulation No. 440/2008	Chapter in revised R.7(a) guidance	CLP Regulation (EC) No. 1272/2008 (the no. in brackets is the respective chapter no. in Annex I to CLP)	Corresponding test method according to CLP Regulation
Melting/ Freezing point (7.2)	A.1 Melting/Freezing temperature	R.7.1.2	n.a.	n.a.
Boiling point (7.3)	A.2 Boiling temperature	R.7.1.3	n.a.	n.a.
Relative density (7.4)	A.3 Relative density	R.7.1.4	n.a.	n.a.
Vapour pressure (7.5)	A.4 Vapour pressure	R.7.1.5	n.a.	n.a.
Surface tension (7.6)	A.5 Surface tension	R.7.1.6	n.a.	n.a.
Water solubility (7.7)	A.6 Water solubility	R.7.1.7	n.a.	for metals - Transformation/Dissolution Protocol (Annex 10 to UN GHS)
Partition coefficient n-octanol/water (7.8)	A.8 Partition coefficient	R.7.1.8	n.a.	n.a.

Flash point (7.9)	A.9 Flash-point	R.7.1.9	n.a.	CLP Annex I chapter 2.6.4.4
Flammability (7.10)	A.11 Flammability (gases)	R.7.1.10.1	Flammable gases ⁸ (2.2)*	ISO 10156 EN 1839
	for liquids: see Flash point	R.7.1.10.2	Flammable liquids (2.6)*	see CLP, Annex I, Chapter 2.6.4.4, Table 2.6.3
	A.10 Flammability (solids)	R.7.1.10.3	Flammable solids (2.7)*	UN Test N.1
	n.a.	R.7.1.10.4	Self-reactive substances and mixtures (2.8)*	UN Test series A to H
	A.13 Pyrophoric properties of solids and liquids	R.7.1.10.5	Pyrophoric liquids (2.9)*	UN Test N.3
		R.7.1.10.6	Pyrophoric solids (2.10)*	UN Test N.2
	n.a.	R.7.1.10.7	Self-heating substances and mixtures (2.11)*	UN Test N.4
	A.12 Flammability (Contact with water)	R.7.1.10.8	Substances and mixtures which in contact with water emit flammable gases (2.12)*	UN Test N.5
	n.a.	R.7.1.10.9	Organic peroxides (2.15)*	UN Test series A to H
Explosive properties (7.11)	A.14 Explosive properties	R.7.1.11.1	Explosives (2.1)*	UN Test series 1 to 3 (further test series 4 to 6 are necessary for classification)
	n.a.	R.7.1.11.2 see R.7.1.10.4	Self-reactive substances and mixtures (2.8)*	A.14 (existing data only)
	n.a.	R.7.1.11.3 See R.7.1.10.9	Organic peroxides (2.15)*	A.14 (existing data only)

⁸ The Commission Regulation (EU) No 487/2013 of 8 May 2013 amending, for the purposes of its adaptation to technical and scientific progress, Regulation (EC) No 1272/2008 of the European Parliament and of the Council on classification, labelling and packaging of substances and mixtures (hereinafter referred to as '4th Adaptation to Technical Progress (ATP) to the CLP Regulation') amends the criteria in the CLP Annex I, Section 2.2 Flammable gases by including subclassifications for chemically unstable gases. The 4th ATP to the CLP Regulation will apply in respect of substances from 1 December 2014 and in respect of mixtures from 1 June 2015.

Self ignition temperature (7.12)	A.15 Auto-ignition temperature (liquids and gases)	R.7.1.12.1	For gases and liquids*	n.a.
	A.16 Relative self-ignition temperature for solids	R.7.1.12.2 , R.7.1.10.7	For solids* Note: the UN Test N.4 is preferable to generate the information for this endpoint. Refer to R.7.1.10.7 .	n.a.
Oxidising properties (7.13)	n.a.	R.7.1.13.1	Oxidising gases (2.4) *	ISO 10156
	A.21 Oxidising properties (liquids)	R.7.1.13.2	Oxidising liquids (2.13) *	UN Test O.2
	A.17 Oxidising properties (solids)	R.7.1.13.3	Oxidising solids (2.14) *	UN Test O.1
Granulometry (7.14)	n.a.	R.7.1.14	n.a.	n.a.
Adsorption/Desorption (7.15)	n.a.	R.7.1.15	n.a.	n.a.
Stability in organic solvent and degradation products (7.16)	n.a.	R.7.1.16	n.a.	n.a.
Dissociation constant (7.17)	n.a.	R.7.1.17	n.a.	n.a.
Viscosity (7.18)	n.a.	R.7.1.18	n.a.	n.a.

* Note that regardless of whether the hazard class or category is listed in Article 14 (4) (a) of REACH, the chemical safety assessment (when required) must be performed in accordance with Article 14 (3) of REACH. Furthermore, according to Article 10 (a) (iv) of REACH the technical dossier of a registration for a substance under the REACH Regulation must include information on classification and labelling of the substance as specified in section 4 of Annex VI to the REACH Regulation.

In addition the CLP Regulation has the following hazard classes ([Table R.7.1-2](#)) for which the REACH Regulation does not require the generation of information (Article 10(a)(vi) and (vii) REACH):

Table R.7.1-2 CLP Regulation hazard classes for which the REACH Regulation does not require the generation of information

CLP Regulation (EC) No. 1272/2008 (the no. in brackets is the respective chapter no. in Annex I to CLP)	Corresponding test method according to the Test Method Regulation	Chapter in revised R.7(a) guidance	Information requirement according to Art. 10(a)(vi) of the REACH Regulation	Corresponding test method according to CLP Regulation
Flammable aerosols ⁹ (2.3)	n.a.	0	n.a.	Test methods according to 75/324/EC amended by 2008/47/EC (harmonised with UN-MTC Section 31)
Gases under pressure (2.5)	n.a.	R.7.1.21.2	n.a.	n.a.
Corrosive to metals (2.16)	n.a.	R.7.1.21.3	n.a.	UN Test C.1 (UN-MTC Section 37.4)

In order to comply with the REACH information requirements, registrants have to take due account of specific rules for adaptation according to column 2 of the tables in Annexes VII to XI to REACH, including the provisions given within the individual test methods of the Test Method Regulation, which have to be interpreted and applied in relation to the appropriate CLP hazard class. Further adaptations according to Annex XI to REACH must then be read together with the adaptation possibilities provided for by Article 8(2) of CLP and the CLP criteria themselves, namely those in Part 2 of Annex I to CLP.

Physicochemical data are mostly numeric and should be provided in SI units. Normally a numeric value or range is required. Where relevant, additional information should be provided on test conditions, such as temperature and/or pressure and/or concentration level or range etc., and estimated uncertainty in the numerical value. Furthermore details of any observations made during testing should be reported, e.g. decomposition during melting or boiling, emulsion formation during partitioning.

R.7.1.1.2 Available information on physicochemical properties

There are many published sources of data for basic substance characterisation and of supplementary information for hazard assessment. The relevant references are listed under the respective endpoint.

⁹ The 4th ATP to the CLP Regulation amends the criteria in the CLP Annex I, Section 2.3 Flammable aerosols by changing the scope and title to Section 2.3 Aerosols.

R.7.1.1.3 Evaluation of available information on physicochemical properties

Advice to registrants with regard to nanomaterials characterisation can be found in *Appendix R7-1 Recommendations for nanomaterials applicable to: Chapter R7a Endpoint specific guidance* of the [Guidance on IR&CSA](#), section 2.1.3 Evaluation of available information.

Experimental data

Further, according to Article 13(3) of the REACH Regulation, tests to generate information on intrinsic properties of substances must be conducted in accordance with the test methods laid down in a Commission Regulation or in accordance with other international test methods recognised by the Commission or the Agency as appropriate, such as European standards (<http://www.cen.eu/Pages/default.aspx>) or OECD guidelines (<http://www.oecd.org/chemicalsafety/testing/oecdguidelinesforhetestingofchemicals.htm>). Data obtained from the tests in accordance with section 1.1.1 of Annex XI to REACH can be considered to be equivalent to data generated by the corresponding test methods referred in Article 13 (3) of REACH. Data for the purpose of physical hazard classification can be obtained using the test methods specified in the Articles 5 (1) and 8 (3) CLP. The test methods for the physicochemical properties are described in Regulation (EC) No 440/2008, whereas preferred tests for the purposes of physical hazard classification are referred to in Part 2 of Annex I to CLP, *via* references to the UN-MTC and to applicable standards. In [Table R.7.1-1](#), the preferred test method for each endpoint is highlighted in bold. The test methods referred to in the CLP Regulation are also used for the transport of dangerous goods. Therefore, available information on physicochemical properties and physical hazards may also originate from tests that were carried out for the purposes of classification for transport. Such test data may be used for the information requirements according to the REACH Regulation. It should, however, be kept in mind that the classification for transport does not cover all hazard categories which are relevant for CLP and it may be based on other considerations than just the test data and criteria (e.g. classifications which are based on experience rather than testing or which apply only in connection with certain special provisions). As a result transport classifications may be different from the classification according to CLP. Such limitations are described in detail in the [Guidance on the Application of the CLP criteria](#), Section 1.7.2.1.

Where relevant recognised standards for testing are applicable, the use of the most recent updates is advised; they are accessible *via* numerous websites, for example:

1. EN standards;
2. [ISO standards](#);
3. [IEC standards](#).

The national editions of the EN or ISO standards are available *via* the national standardization organizations accessible *via* the [CEN Website](#).

Measured values which are evaluated in reviews and assigned recommended values are given precedence over calculated values. The major criteria that characterise the analysis of the available information are:

- **Experimental data.** When assessing physicochemical properties, priority is given to first hand experimental results (primary references) provided that the methods are suitable for the substance under investigation and that they operate within their validity range. Proper documentation on the methods and the inherent uncertainty of the measurements should also be provided.
- **Non-testing information.** If the information described in point (a) is not available, QSPRs, read-across or secondary data sources (e.g. handbook data) can be used in

accordance with the limitations described in the individual endpoint chapters (7.2 to 7.19 in this guidance) instead, and within the constraints of Annex XI to REACH.

Measurement uncertainty

Test data have an uncertainty of measurement. Some test methods include information about their uncertainty, which then may be referred to for test data generated using these test methods. Where the uncertainty of measurement is not specified by the test method, it is recommended to determine uncertainty by generally accepted processes of measurement uncertainty estimation (e.g. according to ISO/IEC Guide 98-3:2008).

Quality assurance for the determination of physicochemical properties

Test data on physicochemical properties should be of sufficient quality i.e. they must be reliable. Normally this can only be achieved by testing that is carried out in compliance with a relevant recognised quality system (e.g. GLP) or by laboratories complying with a relevant recognised standard (e.g. EN ISO/IEC 17025). Under Article 8 (5) of CLP, where new tests for physical hazards are required for the purposes of CLP they have to be carried out in compliance with a relevant recognised quality system or by laboratories complying with a relevant recognised standard at the latest from 1 January, 2014.

Non-experimental data

Quantitative Structure Property Relationships (QSPR) models exist for some of the physicochemical endpoints¹⁰. Where applicable, the details of any specific QSPR models are given under each endpoint.

The majority of QSPR models have been built using training sets of substances. The model will have been optimised to calculate values for the training substances that most closely match measured ones. Therefore, the use of QSPR estimation techniques requires expert judgement. The calculated values need to be checked to ensure that they are reasonable and that the model used is appropriate.

A valid model will give values that are in reasonably close agreement with the measured ones for your chosen analogue substances (i.e. the substance with a data gap should have similar substances in the training set of the model). The models may not predict very well the properties of substances which are too dissimilar to the reference set for the model. Thus, the model can be used to provide a predicted value for your substance without the need for testing. Another check is that the values are realistic. This can be done by cross-referencing the calculated value to measured values for similar substances and related endpoints. If a QSAR method is used as a stand alone method to determine a value to meet the endpoint data requirements, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Assessing the quality of QSPR models

The European Commission and the OECD member countries adopted five principles for the validation of (Q)SAR/(Q)SPR models in 2004 (OECD^a, 2004). According to these principles, a valid (Q)SPR model should have 1) a defined endpoint whose experimental conditions are clearly specified; 2) an unambiguous algorithm; 3) a defined domain of applicability that defines for what kind of substances predictions can be made; 4) appropriate measures of

¹⁰ A comprehensive review paper with a title: "QSPR prediction of physico-chemical properties for REACH" was published in the SAR and QSAR in Environment Research in 2013 (Dearden, J.C., Rotureau, P., Fayet G. (2013). QSPR prediction of physico-chemical properties for REACH, SAR and QSAR in Environmental Research, Vol. 24, No.4, 279-318).

goodness-of-fit, robustness and predictivity; and 5) a mechanistic interpretation if possible. These principles are outlined on the ECB website and more extensively covered in the [Guidance on IR&CSA](#) Chapter R.6: 'QSAR and grouping of substances', Section R.6.1.3. Moreover, a practical overview of these principles is given in the report from the expert group on (Q)SARs (OECD^b 2004).

Assessing the quality of read-across predictions

This paragraph reports the basic concepts of a read-across approach. Thorough information on this topic can be found in the guidance on the grouping of substances (see [Guidance on IR&CSA](#) Chapter R.6: 'QSAR and grouping of chemicals', Section R.6.2).

A read-across/analogue approach assesses the relevance of a given property on one or more chemical structures and then makes some assessment (qualitative or quantitative) on the relevance of this information for another substance (see Annex XI, REACH). Since a read-across may involve two substances¹¹ it is of paramount importance to detail the reasoning behind the inference on the substance whose property is unknown. An analogue must:

- contain the same major structural features and the same functional groups as the substance under investigation;
- have a physicochemical profile comparable to that of the substance under examination as far as the known physicochemical properties are concerned;
- have comparable values for the relevant molecular descriptors (i.e. excess molar refractivity and hydrogen bond donor and acceptor abilities for water solubility predictions) generally used for the quantification of the property of interest;
- have approximately the same molecular weight.

The interpretative analysis of a read-across is usually the result of an expert judgement evaluation and detailed documentation should therefore always be provided to support the conclusions. It is important to point out that, in practice, read-across for physicochemical properties is not generally recommended, since reliable data should normally be available or easily obtainable. This is particularly true for physical hazard related physicochemical properties for which reliable test data must be available according to Article 8 (2) of CLP. Therefore, if read-across is used as a stand alone method to generate a value to meet the endpoint data requirements, the criteria given in section 1.5 of Annex XI to REACH must be met.

Use of secondary and historical data sources for physicochemical properties

The reliability of data must be demonstrated by providing information on the identity and purity of the test substance, the methodology used to make the measurement, and whether or not this was performed in compliance with a relevant recognised quality system (e.g. GLP) (Annex VI, REACH).

Numerical physicochemical data is particularly prone to data recycling (transfer from one database to another, often with loss of the original source and contextual information). Data from secondary and historical sources must be adequate and is especially important where the

¹¹ A read-across can also involve more than two substance: one-to-one (one analogue used to make an estimation for a single substance) b) many-to-one (two or more analogues used to make an estimation for a single substance) c) one-to-many (one analogue used to make estimations for two or more substances) d) many-to-many (two or more analogues used to make estimations for two or more substances).

endpoint is relevant for classification, PBT/vPvB assessment, is the basis of waiving arguments for other endpoints, or has a large influence on the outcome of the risk assessment. The criteria in section 1.1.1 of Annex XI or section 1.2 of Annex XI to REACH must be met.

R.7.1.1.4 Overall consistency of the physicochemical information

The physicochemical data for a given substance cannot contain incompatible values for two or more properties (i.e. high boiling point and high vapour pressure at normal temperature). This consistency check should be always done and it can turn out to be particularly useful when measured values are significantly at odds with predictions from QSPR models. Indeed, in this case a wider assessment of the known physicochemical properties should be performed in order to determine the possible cause of the inconsistencies.

Concluding on classification and labelling and chemical safety assessment

Data on physicochemical properties not only determine the presence or absence of a physical hazard but also have also an impact on the sections of the chemical safety assessment concerning the environment and human health. The assessment determines the risk posed to humans and the environment from all stages of the substance's lifecycle. This includes its manufacture, transfer, use and disposal. Firstly, the physicochemical data set provides the input parameters for the purpose of the human and environmental exposure estimation. For example, the vapour pressure and particle size information are required to estimate the likely exposure of humans, both in the workplace and in consumer use as well as to estimate the likelihood of forming flammable/explosive vapour/dust-air mixtures. The volatility (vapour pressure) or the size and nature of particles are indicators of the potential for inhalation exposure. Particle size is also important for determining the likely dermal exposure and the presence of a dust explosion hazard. Viscosity is a key parameter in determining aspiration hazards. The physical state of a substance at the process temperature is important for determining possible hazards. Further, physico-chemical data are essential for the correct planning of (eco)toxicological studies and for the optimisation of the test conditions.

R.7.1.1.5 References for introduction of Physicochemical properties

Recommendations on the transport of dangerous goods, Manual of Test and Criteria, United Nations. http://www.unece.org/trans/danger/publi/manual/manual_e.html

Guidance on the Application of the CLP Criteria, Version 4.0 - 2013, ECHA.
<http://echa.europa.eu/web/guest/guidance-documents/guidance-on-clp>

OECD^a (2004) Principles for the Validation of (Q)SARs
<http://www.oecd.org/chemicalsafety/risk-assessment/validationofqsarmodels.htm>

OECD^b (2004) series on testing and assessment Number 49 The report from the expert group on (quantitative) structure activity relationships [(Q)SARs] on the principles for the validation of (Q)SARs. 2nd Meeting of the ad hoc Expert Group on QSARs
[http://www.oecd.org/officialdocuments/displaydocumentpdf/?cote=env/jm/mono\(2004\)24&doclanguage=en](http://www.oecd.org/officialdocuments/displaydocumentpdf/?cote=env/jm/mono(2004)24&doclanguage=en)

R.7.1.2 Melting point/freezing point

R.7.1.2.1 Type of property

The melting point contributes to the identification of a substance and to the designation of its physical state (liquid or solid¹²) of a substance. A number of physical hazard classes are distinguished based on the physical state. Therefore the melting point of a substance and the consequent designation as liquid or solid has also consequences for the assignment of the correct hazard class. Furthermore, the melting/freezing point together with vapour pressure serves as an indicator for the physical state (liquid or solid) of a substance under specific conditions (e.g. environmental conditions, manufacturing process conditions). As a result, with regard to environmental relevance the melting point can give an indication of the distribution of the substance within and between the environmental media (water, soil and air).

R.7.1.2.2 Definition

The melting temperature is defined as the temperature at which the phase transition from the solid to the liquid state occurs at atmospheric pressure and this temperature ideally corresponds to the freezing temperature. As the phase transition of many substances takes place over a temperature range, it is often described as the melting range. For some substances, the determination of the freezing or solidification point is more appropriate. Where, due to the particular properties of the substance, none of the above parameters can be conveniently measured, a pour point may be appropriate.

R.7.1.2.3 Test method(s)

Method A.1 of Regulation (EC) 440/2008 or OECD Test Guideline 102 should be generally used for testing. Any procedure given in A.1 may be used within the scope and applicability specifications. However, it is advisable to use the Differential Scanning Calorimetry (DSC) or Differential Thermo-Analysis (DTA) method since they give additional information about the thermal stability of the substance like decomposition onset and energy. If decomposition occurs during the melting point study, determination of the boiling point need not be carried out. In this case, if DSC has been used, conducting the experiment under inert gas should be considered.

R.7.1.2.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for melting/freezing point:

'The study does not need to be conducted below a lower limit of - 20°C.'

Therefore, Annex VII to REACH does not require determination of the melting point below a lower limit of -20°C. The lower limit should be confirmed through testing, except where a (Q)SAR indicates a melting point of -50°C or lower.

¹² Definitions of physical states can be found in Section 1.0. of Annex I to the CLP Regulation.

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria of Annex XI, section 1.1.1, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in Annex XI, section 1.3.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria of Annex XI, section 1.1.1 or where several sources of similar reliability with deviating results exist, a weight-of-evidence approach may be used. The criteria of Annex XI, section 1.2 must then be met¹³.

(Q)SAR

For the determination of the melting point, (Q)SAR approaches are discouraged, because the accuracy is not sufficient ($\pm 25^{\circ}\text{C}$ or more) for the purposes of classification / risk assessment.

Grouping of substances and read-across approach

For the determination of the melting point read-across is usually not possible. However interpolation may still be possible within homologous series.

Testing is technically not possible

Some substances will decompose or sublime before the melting point is reached.

Further adaptation possibilities

Not foreseen.

R.7.1.2.5 Impurities; uncertainties

Impurities can have a significant influence on the melting point, as they will generally lower the melting point noticeably. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

R.7.1.2.6 Endpoint specific information in the registration dossier/ IUCLID

Materials and methods

- type of method or reference to the standard or the test method applied.

Results and discussion

- melting point value ($^{\circ}\text{C}$) as measured;

¹³ National Institute of Standards and Technology (NIST) have a useful statistical approach which has been used for the evaluation of literature melting point data (ref.: <http://webbook.nist.gov/chemistry/site-cal.html#AVG>).

- rate of temperature increase if available;
- decomposition or sublimation temperature (if applicable);
- measurement uncertainty if available;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on melting point/freezing point can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.2	VII 7.2	Melting point/freezing point	E.4.3	3.2

R.7.1.3 Boiling point

R.7.1.3.1 Type of property

The boiling point is a property:

- which contributes to the characterisation of a substance and to the designation of its physical state (gas or liquid);
- which is the basis for the assignment of the correct hazard class because a number of physical hazard classes are distinguished based on the physical state;
- which is needed for the classification of flammable liquids into categories;
- which gives an indication of the distribution of the substance within and between the environmental compartments (air, soil and water);
- which have correlations with vapour pressure and therefore gives indications whether a substance may be available for inhalation as a vapour or may form flammable/explosive vapour-air mixtures, too;
- which is important for physical hazard assessment.

R.7.1.3.2 Definition

The normal boiling point is the temperature at which the vapour pressure of a liquid equals 101.3 kPa.

Note: If the vapour pressure equals 101.3 kPa or more at a given temperature this means the substance is completely gaseous at that temperature. If this is the case at temperatures $\leq 20^{\circ}\text{C}$ the substance is a gas also according to the CLP Regulation.

R.7.1.3.3 Test method(s)

Method A.2 of Regulation (EC) 440/2008 or OECD Test Guideline 103 should be used for testing. Any determination method may be used within the scope and applicability specifications. DSC allows the determination of the melting and boiling point in a single test. Likewise, for some substances a single test can be used to determine both 'boiling point' and 'vapour pressure', as when the dynamic method is applied.

For high-boiling liquids or liquids which may decompose, auto-oxidize etc. before the boiling point at 101.3 kPa or more is reached, it is recommended to determine the boiling point either under inert gas or at reduced pressures, in order to derive the boiling point at reduced pressures from the vapour pressure curve.

If explosive substances, pyrophoric substances or self-reactive substances are to be characterized, determination of the boiling point is in general not practicable. For pyrophoric substances testing under inert gas or reduced pressures should be considered.

Where standards are applicable, the use of the most recent updates is advised; they are accessible *via* numerous websites, see above in Section [R.7.1.1.3](#).

R.7.1.3.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for boiling point:

'The study does not need to be conducted:

- *for gases; or*
- *for solids which either melt above 300 °C or decompose before boiling. In such cases the boiling point under reduced pressure may be estimated or measured; or*
- *for substances which decompose before boiling (e.g. auto-oxidation, rearrangement, degradation, decomposition, etc.).'*

Therefore the Annex VII to REACH does not require determination of the boiling point if:

- the substance is a gas;

However, for some gases the boiling point may be relevant. In the CLP Regulation, the boiling point is the main criterion to distinguish gases from liquids (see Annex I, section 1.0: Gas means a substance which (i) at 50°C has a vapour pressure greater than 300 kPa (absolute); or (ii) is completely gaseous at 20°C at a standard pressure of 101.3 kPa). Therefore it is important to report the boiling point in borderline cases where the transition from liquid to gas occurs close to 20°C.

- the melting point of the substance is above 300°C or when any chemical change occurs during the melting point study;
- the substance decomposes before boiling at ambient pressure.

In such cases the boiling point under reduced pressure (down to 0.2 kPa) should be determined if possible without decomposition.

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria of Annex XI, section 1.1.1, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in Annex XI, section 1.3.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria of Annex XI, section 1.1.1 or where several sources of similar reliability with deviating results exist, a weight-of-evidence approach may be used. The criteria of Annex XI, section 1.2 must then be met¹⁴.

(Q)SAR

For the determination of the boiling point, (Q)SAR approaches are discouraged for the purpose of classification / risk assessment, except when the mean absolute error of the method is lower than 2 K.

Grouping of substances and read-across approach

For the determination of the boiling point read-across is usually not possible. However interpolation may still be possible within homologous series.

Testing is technically not possible

Testing is not possible if:

- the substance is an explosive;
- the substance is self-reactive;
- any chemical change occurs during the melting point study;
- the liquid decomposes before the boiling point is reached even at reduced pressures below 0.2 kPa.

In such cases the decomposition temperature in relation to the (reduced) pressure should be reported, in order to allow determination of whether it is the substance itself or its decomposition products that should be considered under environmental conditions for the purpose of risk assessment. The details of the determination method should also be reported.

Further adaptation possibilities

Data generated with the same tests and classification principles as specified in the CLP Regulation on boiling point generated in conjunction with transport classification can be deemed to satisfy the REACH requirements on a case-by-case basis. As stated in Annex IX of the REACH Regulation, when for certain endpoints, it is proposed to not provide information for

¹⁴ The NIST have a useful statistical approach which has been used for the evaluation of literature boiling point data (ref.: <http://webbook.nist.gov/chemistry/site-cal.html#AVG>).

other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used.

R.7.1.3.5 Impurities; uncertainties

Impurities can have a significant influence on the boiling point. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

R.7.1.3.6 Endpoint specific information in the registration dossier / in IUCLID

Materials and methods

- type of method or reference to the standard or the test method applied.

Results and discussion

- boiling point value (°C) as measured;
- pressure value and unit;
- rate of temperature increase if available;
- decomposition (if applicable);
- measurement uncertainty if available;
- boiling point value in °C (corrected to standard pressure, except where the boiling point has been determined at specified reduced pressures) (as above, but in a separate block of fields);
- if testing is waived, the reasons for waiving must be documented in the dossier.

 Note: In cases where the boiling point is determined at reduced pressure a determination at ambient pressure is obviously not possible. A boiling point at standard pressure could then only be derived by extrapolation of the vapour pressure curve in cases where a vapour pressure curve is known. Even in such cases this corrected/extrapolated boiling point could only be nominal one and would be potentially misleading because it is not possible to determine it at ambient pressure.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on boiling point can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.3	VII 7.3	Boiling point	E.4.4	3.3

R.7.1.4 Relative density

R.7.1.4.1 Type of property

For gaseous materials, relative density is of value in determining the tendency to settle or to disperse when discharged at high concentrations into the atmosphere. The relative density of gaseous substances can be calculated from molecular weight using the Ideal Gas Law.

For insoluble liquids and solids, (absolute) density will be a determining factor in the settling of the substance.

R.7.1.4.2 Definition

Density (ρ) of a substance is the quotient of the mass m and its volume V :

$$\rho = m/V \quad \text{SI units (kg/m}^3\text{)}$$

The relative density is related to a standard, the density of which is set to 1. It has no dimension. For gases air is used as standard so that gases with a relative density of less than 1 are lighter than air (and those with a value above 1 heavier).

The relative density, D_4^{20} , of solids or liquids is the ratio between the mass of a volume of substance to be examined, determined at 20°C, and the mass of the same volume of water, determined at 4°C (at which temperature, water has its maximum density, i.e. 999.975 kg/m³).

R.7.1.4.3 Test method(s)

Test methods for determining (absolute) density are applicable to solids and liquids. [Table R.7.1-3](#) lists the respective test methods.

Table R.7.1-3 Test methods for determining density

Method	Applicability	Maximum Dynamic Viscosity (Liquids only)/Pa.S
Hydrometer	Liquids	5
Hydrostatic balance	Solids and Liquids	5
Immersion ball	Liquids	20
Pycnometer	Solids and Liquids	500
Air comparison pycnometer	Solids	-
Oscillating densitometer	Liquids	5

EU Test guideline A.3 for relative density Regulation (EC) No 440/2008 includes a list of standards with technical information about the different methods and actual measuring of different types of substances.

R.7.1.4.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for relative density:

'The study does not need to be conducted if:

- the substance is only stable in solution in a particular solvent and the solution density is similar to that of the solvent. In such cases, an indication of whether the solution density is higher or lower than the solvent density is sufficient; or*
- the substance is gaseous at room temperature. In this case, an estimation based on calculation can be made from its molecular weight and the Ideal Gas Laws.'*

For liquids, it is useful to have some indication of the dynamic viscosity as this can affect the choice of method. The physical state of test substances should always be homogeneous, this is particularly relevant for highly viscous substances where internal bubbles can be formed; in these cases, the test substance should be allowed to rest until all internal bubbles have disappeared.

The summary should include the numerical value for density and temperature at which it was measured, test material identity, purity of the sample used, physical state, method and guideline used and reference substance (if any).

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met¹⁵.

(Q)SAR

(Q)SAR is generally not applicable for determination of relative density.

For this endpoint there are often experimental measurements and therefore QSPR models for this property have not received special attention in the environmental literature. Several software programs can be used to calculate the density of a given substance but the documentation and validation of the methods is limited.

¹⁵ The NIST have a useful statistical approach which has been used for the evaluation of literature data (ref.: <http://webbook.nist.gov/chemistry/site-cal.html#AVG>).

Grouping of substances and read-across approach

For the determination of the relative density read-across is usually not possible. However interpolation may still be possible within homologous series.

Testing is technically not possible

Testing should always be considered, if none of the waiving possibilities applies. Waiving relative density testing on the basis of not being technically possible is not applicable.

Further adaptation possibilities

Not foreseen.

R.7.1.4.5 Impurities; uncertainties

Impurities can have a significant influence on the density. This influence depends on the amount and density of the impurity; thus, the higher the amount of impurity and the higher the difference between the densities of the main component and the impurity, the higher the influence. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

Density is temperature dependant. Whenever possible, determinations should be performed at 20°C.

R.7.1.4.6 Endpoint specific information in the registration dossier / in IUCLID

Materials and methods

- type of method or reference to the standard or the test method applied.

Results and discussion

- temperature (°C);
- relative (for gases)/ absolute (for liquids and solids) density value (dimensionless);
- measurement uncertainty if available;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on relative density can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.4	VII 7.4	Relative density	E.4.5	3.4

R.7.1.5 Vapour pressure

R.7.1.5.1 Type of property

Vapour pressure is a property:

- for substance characterisation;
- which serves as a key parameter for assessing some toxicological and environmental hazards;
- which gives indications whether a substance may be available for inhalation as a vapour or may form flammable/explosive vapour-air mixtures;
- which allows determination of the volatility of a substance from an aqueous medium or soil, in terms of the Henry's Law constant ([Appendix R.7.1-1](#)) and partition coefficient air/soil, respectively;
- which allows determination of the right container/vessel to ensure safety during storage, transport and use;
- which is important for physical hazard assessment.

R.7.1.5.2 Definition

The vapour pressure of a substance is defined as the saturation pressure above a solid or a liquid substance at constant temperature. At the thermodynamic equilibrium, the vapour pressure of a pure substance is a function of temperature only.

R.7.1.5.3 Test method(s)

Method A.4 of Regulation (EC) 440/2008 or OECD Test Guideline 104 (Vapour pressure) should be used for testing. It is useful to have preliminary information on the structure, the melting point and the boiling point of the substance to perform this test.

There is no single measurement procedure applicable to the entire range of vapour pressure values. Therefore, several methods are recommended to be used for the measurement of vapour pressure from $< 10^{-10}$ to 10^5 Pa. For the selection of the test method the scope and applicability specifications have to be taken into account. The results should be checked for consistency with other physical data like boiling point, flash point etc.

It is recommended to determine the vapour pressure at least for two temperatures, for volatile substances (boiling point up to 150°C) preferably at 20°C and at 50°C.

Where standards are applicable, the use of the most recent updates is advised, please check Section [R.7.1.1.3](#) for further information.

R.7.1.5.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for vapour pressure:

'The study does not need to be conducted if the melting point is above 300 °C.'

If the melting point is between 200 °C and 300 °C, a limit value based on measurement or a recognised calculation method is sufficient.'

Vapour pressure testing is also not required for substances with a standard boiling point of < 30 °C, as these substances will have vapour pressures above the limit of measurement (i.e. 10^5 Pa).

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

For the determination of the vapour pressure, (Q)SAR approaches may be used if determination by experiment is not possible.

The vapour pressure depends on the temperature. This dependence was modelled by Grain (Grain, 1982), based on thermodynamic principles. The estimation methods differ for vapour pressure that can be applied for compounds that are liquid or gaseous at the temperature of interest, and for solid and liquid compounds. The former can be estimated by the Antoine equation, while the latter could be predicted by the Watson correlation, which accounts also for the heat of vaporisation. Another method, described by Mackay *et al.* (1982), is applicable only for hydrocarbons and halogenated hydrocarbons. Further, the Grain model was modified to be applicable for all solids, liquids, and gases. These methods are still in practical use today.

The OECD guideline 104 reports that the Watson correlation is applicable over the pressure range from 10^5 Pa to 10^{-5} Pa. It should in any case be pointed out that estimated values for vapour pressure can be subjected to great uncertainty if the computed pressure is lower than 1 Pa, especially when the boiling point has not been experimentally determined (OECD monograph 67). The uncertainty is even greater if the estimated value is used together with water solubility in order to estimate the Henry's Law constant.

The environment monograph 67 of the OECD describes all of the above mentioned methods and the OECD guideline 104 supports the use of the Watson correlation for the calculation of vapour pressure, but does not specifically reject other calculation methods.

The handbook for estimating the physico-chemical properties of organic compounds (Reinhard and Drefahl, 1999) reports another method based on thermodynamic properties and elaborated by Mishra and Yalkowsky that discussed the application of the method of Mackay (Mackay *et al.*, 1982).

The equation by Mishra and Yalkowsky gave significantly better estimates than the method of Mackay on the same data set (Mishra and Yalkowsky, 1991).

Another methodology that proved to be effective in estimating vapour pressure relies on group contribution approaches. Several models using this strategy have been proposed (Reinhard and Drefahl, 1999; see [Table R.7.1–4](#)).

Table R.7.1–4 Group contribution approach and vapour pressure

Compounds	Authors	Methodology	Statistics
Alkyl aromatic compounds	Amidon and Anik	Group contribution approach	Standard error ... 1.1 kJ on the estimation for the free energy of vaporisation
Mono-, di-, tri- and tetra substituted	Hoshino <i>et al.</i>	Group contribution approach	Average error 3.7% Max. error 30.9%
Perfluorinated saturated hydrocarbons	Kelly <i>et al.</i>	Group contribution approach	Arithmetic mean deviation <0.5%

Numerous other models are available for the estimation of vapour pressure, and Schwarzenbach *et al.* (1993), Delle Site (1996), Sage and Sage (2000) and Dearden (2003) have reviewed many of these. The descriptors used in vapour pressure QSPRs include physico-chemical, structural and topological descriptors, and group contributions. Katritzky *et al.* (1998) used 4 CODESSA descriptors to model the vapour pressure (in atmospheres at 25°C) of 411 diverse organic chemicals, with $r^2 = 0.949$ standard error = 0.331 log unit. A number of studies (Andreev *et al.* 1994, Kühne *et al.* 1997, Yaffe & Cohen 2001) allow of the estimation of vapour pressures over a range of temperatures.

Grouping of substances and read-across approach

For the determination of vapour pressure read-across is usually not possible. However interpolation may still be possible within homologous series.

Testing is technically not possible

Vapour pressure testing is not required for substances with a standard boiling point of < 30°C, as these substances will have a vapour pressure value above the limit of measurement (i.e. 10^5 Pa).

For substances which decompose during measurement or which are unstable or explosive, determination of the vapour pressure may not be technically possible. This also applies to self-reactive substances and organic peroxides.

Pyrophoric substances may be difficult to handle experimentally. If fully inert conditions cannot be maintained during sample preparation and measurement, use of an appropriate calculation method is recommended.

A calculation method should also be applied in the case of some corrosive substances which would destroy essential metallic parts of the measurement apparatus.

Further adaptation possibilities

Not foreseen.

R.7.1.5.5 Impurities; uncertainties

Impurities can have a large influence on vapour pressure. The influence depends on the amount of the impurity and the vapour pressure of that impurity. Small amounts of volatile impurities may increase the vapour pressure by several orders of magnitude. This has to be kept in mind when performing the measurements and for the interpretation of results. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

Where there are volatile impurities in the sample which could affect the result, the substance may be purified. Test method A.4 states that it may also be appropriate to quote the vapour pressure for the technical material. However, in consideration of the large effect that impurities may have (see above), doing so is strongly discouraged.

R.7.1.5.6 Endpoint specific information in the registration dossier / in IUCLID

Materials and methods

- type of method or description of the apparatus or reference to the standard or the test method applied.

Results and discussion

- if testing is waived, the reasons for waiving must be documented in the dossier;
- measured value of the vapour pressure for at least two temperatures;
- estimate of the vapour pressure at 20 or 25°C (if not measured at these temperatures);
- if a transition (change of state, decomposition) is observed, the following should be noted:
 - nature of change;
 - temperature at which change occurs.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on vapour pressure can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.6	VII 7.5	Vapour pressure	E.4.7	3.6

R.7.1.5.7 References on vapour pressure

OECD Guidelines for the Testing of Chemicals / Section 1: Physical-Chemical properties, Test No. 104: Vapour Pressure, OECD Code: 979910401E1, July 2006.

Andreev N.N, Kuznetsov S.E, Storozhenko S.Y. (1994) Prediction of vapour pressure and boiling points of aliphatic compounds. Mendeleev Commun. 173-174.

Grain C.F., (1982) Handbook of chemical property estimation methods. New York, Mc Graw-Hill

Delle Site A. (1996) The vapour pressure of environmentally significant organic chemicals: a review of methods and data at ambient temperature. J. Phys. Chem. Ref. Data 26:157-93.

Dearden JC. (2003) Quantitative structure-property relationships for prediction of boiling point, vapour pressure, and melting point. Environ Toxicol Chem 22(8):1696-709.

Katritzky AR, Y. W, Sild S, Tamm T, Karelson M. (1998) QSPR studies on vapour pressure, aqueous solubility, and the prediction of water-air partition coefficients. J. Chem. Inf. Comput. Sci. 38:720-5.

Kühne R, Ebert RU, Schüürmann G. (1997) Estimation of vapour pressures for hydrocarbons and halogenated hydrocarbons from chemical structure by a neural network. Chemosphere 34:671-86.

Mackay D, Bobra A, Chan W, Shiu WY. (1982) Vapour pressure correlation for Low-Volatility Environmental Chemicals. Environ. Sci. Technol. 16:645-9.

Mishra DS, Yalkowsky SH. Estimation of vapour pressure of some organic compounds. Ind. Eng. Chem. Res. 1991;30:1609-12.

OECD Guidelines for Testing of Chemicals, Method 104 "Vapour Pressure Curve"

Reinhard M, Drefahl (1999).A. Handbook for Estimating Physico-Chemical Properties of Organic Compounds. New York: Wiley.

Sage M.L, Sage G.W.(2000) Handbook of Property Estimation Methods for Chemicals. Boca Raton, FL: Lewis.

Schwartzenbach, R.P., Gswend, P.M., Imboden, D.M. (1993). Environmental Organic Chemistry. John Wiley and Sons.

Yaffe D, Cohen, Y (2001) Neural network based temperature-dependent quantitative structure property relationships (QSPRs) for predicting vapour pressure of hydrocarbons. J. Chem. Inf. Comput. Sci. 41:463-477.

R.7.1.6 Surface tension

R.7.1.6.1 Type of property

Surface tension measurements of aqueous solutions are significant since decreasing the surface tension of water may impact on the properties of the solution and other physicochemical measurements.

R.7.1.6.2 Definition

- Surface tension:

'The free surface enthalpy per unit of surface area is referred to as surface tension' (Council Regulation (EC) No 440/2008).

The surface tension is given as: N/m (SI unit) or mN/m (SI sub-unit). $1 \text{ N/m} = 10^3 \text{ dyne/cm}$ or $1 \text{ mN/m} = 1 \text{ dyne/cm}$ in the obsolete cgs system.

The surface tension of an aqueous solution of a substance can be used to determine whether the substance is surface active.

- Surface active substance (surfactant):

"Surfactant' means any organic substance and/or preparation [mixture] used in detergents, which has surface-active properties and which consists of one or more hydrophilic and one or more hydrophobic groups of such a nature and size that it is capable of reducing the surface tension of water, and of forming spreading or adsorption monolayers at the water-air interface, and of forming emulsions and/or microemulsions and/or micelles, and of adsorption at water-solid interfaces' (see Article 2(6) of Council Regulation (EC) No 648/2004).

R.7.1.6.3 Test method(s)

Testing should be done in accordance with one of the methods specified under section A.5 of Regulation (EC) No 440/2008. These methods are applicable to most chemical substances.

It is useful to have preliminary information on the water solubility, the structure, the hydrolysis properties and the critical concentration for micelles formation of the substance before performing the test.

Surface tension measurements require a test material that is stable against hydrolysis during the test period and soluble in water at concentrations of $> 1 \text{ mg/l}$. Measurements should be performed on a solution at either 90 % of the solubility limit or 1 g/l (where viscosity permits), whichever is smaller.

R.7.1.6.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for surface tension:

'The study need only be conducted if:

- *based on structure, surface activity is expected or can be predicted; or*
- *surface activity is a desired property of the material.*

If the water solubility is below 1mg/l at 20°C the test does not need to be conducted.'

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

At the time of writing, no reliable (Q)SAR methods exist for sufficiently accurate predictions of surface tension.

Grouping of substances and read-across approach

For the determination of the surface tension read-across is usually not possible. However interpolation may still be possible within homologous series.

Testing is technically not possible

Testing should always be considered, if none of the waiving possibilities applies. Testing may not be possible for reactive substances which react with water or air (hydrolyse, are pyrophoric, evolve gas, etc).

Further adaptation possibilities

Not foreseen.

R.7.1.6.5 Impurities; uncertainties

For the measurement of surface tension the ring or plate tensiometer methods are preferred. The error on the measurement is in the order of 0.1–0.3 mN/m. Use of the standard protocols and GLP procedures are recommended. Surface active impurities in substances may in some cases lead to false-positive surface tension measurements.

R.7.1.6.6 Endpoint specific information in the registration dossier / in IUCLID

Materials and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied;
- test material identity: apart from general issues, if surface tension of active impurities affects results, it should be noted.

Results and discussion

- surface tension value and unit (preferably mN/m or N/m but other units are also acceptable);
- concentration of the solution*¹⁶;
- age of solution*;
- type of water or solution used*;
- results from repeated measurements with varied equilibrium time (of the solution);
- several measurement results should be provided to assess the possible time-dependency of the measurement. Equilibration times may vary from minutes to hours. Measurements should be sufficient to prove that a constant surface tension was reached;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on surface tension can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.10	VII 7.6	Surface tension	E.4.11	3.9

¹⁶ *As indicated in test A.5. Surface tension is described in Council Regulation (EC) No 440/2008.

R.7.1.7 Water solubility

Advice to registrants with regard to nanomaterials characterisation of water solubility can be found in *Appendix R7-1 Recommendations for nanomaterials applicable to: Chapter R7a Endpoint specific guidance* of the [Guidance on IR&CSA](#), section 2.2.1 Water solubility.

R.7.1.7.1 Type of property

Water solubility is a significant parameter for a number of reasons:

- the mobility of a test substance is largely determined by its solubility in water. In general, highly soluble substances are more likely to be distributed by the hydrological cycle;
- water soluble substances gain access to humans and other living organisms;
- knowledge of the water solubility is a prerequisite for setting up test conditions for a range of fate (e.g. biodegradation, bioaccumulation) and effects studies;
- it is also used to derive other environmental parameters, such as K_{ow} , K_{oc} and Henry's Law Constant ([Appendix R.7.1-1](#)). It is also used as input for some QSAR models;
- water solubility is used as a regulatory trigger for waiving certain physicochemical and ecotoxicological endpoints.

R.7.1.7.2 Definition

'The solubility of a substance in water is specified by the saturation mass concentration of the substance in water at a given temperature. The solubility in water is specified in units of mass per volume of solution. The SI unit is kg/m^3 (grams per litre may also be used)' (see Regulation (EC) No 440/2008, A.6, section 1.2).

Mixtures of organic compounds, e.g. petroleum substances, behave differently from their single constituent compounds when brought into contact with water. Petroleum substances are typically hydrophobic and exhibit low solubility in water. However, reflecting the range of structures, constituent hydrocarbons will exhibit a wide range of water solubility. Therefore, water solubility measurements for these substances are loading rate dependent due to their complex composition. This water solubility behaviour impacts on both the conduct and interpretation of aquatic toxicity tests for these complex substances. The complex composition, and generally low water solubility, impact also on the choice and conduct of biodegradation studies.

Consequently, the above definition for solubility of a single substance in water is not applicable to substances which are multi-component, such as multi-constituent or UVCB substances, i.e. complex substances. The usually accepted meaning of 'solubility' in such cases is 'the composition of the aqueous solution formed at equilibrium under a defined set of conditions'. Temperature and the amount of substance added per unit volume of water (i.e. the 'loading') are the main factors to consider. It may not always be possible to establish that equilibrium of all components has been achieved; in these cases, time and type of agitation of the test vessels must also be described.

Similar testing issues also apply to inorganic compounds. Water solubility among compounds of the same metal may differ by several orders of magnitude. Differences in the solubility of metal compounds are related to the metal species and the characteristics of the aqueous medium. Highly soluble inorganic metal compounds can be assessed through the normal procedures. For sparingly soluble metal compounds, a solubility product can be calculated

thermodynamically (e.g. by using the Facility for Analysis of Chemical Thermodynamics ('F*A*C*T', FACT-Win version 3.05)). Although metals are generally insoluble, metals in the elemental state may react with water or a dilute aqueous electrolyte to form soluble or sparingly soluble cationic or anionic products. During this process the metal will oxidise, or transform, from the neutral or zero oxidation state to a higher oxidation state. The OECD Test Guidance on transformation/dissolution of metals and sparingly soluble metal compounds (OECD, 2001) can be used to determine the rate and extent to which metals and sparingly soluble metal compounds can produce soluble bioavailable ionic and other metal-bearing species in aqueous media under a set of standard laboratory conditions representative of those generally occurring in the environment. The outcomes of the transformation/dissolution tests are to be used for aquatic environmental hazard classification purposes.

R.7.1.7.3 Test method(s)

No single method is available to cover the whole range of solubility values in water, from relatively soluble to very low soluble substances. General test guidelines (OECD Method 105; EU Method A.6, Regulation (EC) No 440/2008) include two test methods which cover the whole range of solubility values but are not applicable to volatile substances. Water solubility determinations are normally run at 20 °C in distilled water according to standard test guidelines (OECD Method 105; EU Method A.6). Solubility data determined using these standard physico-chemical guidelines may differ if the test material is solubilised in either aqueous solutions containing salts or at different test temperatures (or both) (e.g. ecotoxicological test media).

The methods should be applied to essentially pure substances that are stable in water. Details of suitable methods are shown in [Table R.7.1-5](#).

A number of standardised methods are available for the determination of single substances and complex mixtures of liquids and solids. For metals and sparingly soluble inorganic metal compounds a specific water solubility approach was designed to measure transformation to the dissolved fraction under standard conditions. The test methods are not applicable to volatile substances. Care should be taken to ensure that the test substances examined are as pure as possible and their solubility levels are determined analytically using a specific analytical method wherever possible. Precautions should be taken to minimise degradation of the test substance, in particular if long periods of equilibration are required (e.g. 'slow stir' methods).

Measurement of water solubility does not usually impose excessive demands on chemical techniques. However, measurement of the solubility of sparingly soluble compounds requires extreme care to generate saturated solutions of the material without the introduction of dispersed material; invariably specific methods of analysis are able to determine the low levels (sub ppb-ppm) in solution. Reported water solubility data for such compounds can often contain appreciable errors.

Table R.7.1–5 Test methods for the determination of water solubility

Method details	Applications and requirements	Repeatability and sensitivity
Column elution method Based on elution of the test substance with water from a micro-column which is charged with an inert carrier material such as glass beads, silica gel or sand and an excess of test substance. The water solubility is determined when the mass concentration of the eluate is constant. The mass concentration of the test substance is determined analytically	Applicable to essentially pure substances only Used for low solubilities ($< 10^{-2}$ g/l) Organic substances, but not mobile oils or liquids	$< 30\%$; down to 1 µg/l
Flask method The test substance is dissolved in water at a temperature somewhat above the test temperature. When saturation is achieved the mixture is cooled and kept at the test temperature, stirring as long as necessary to reach equilibrium The mass concentration of the test substance is determined analytically	Applicable to essentially pure substances and also complex substances. Use of fast stirring techniques (300-400 rpm) appropriate for higher solubility ($> 10^{-2}$ g/l) test substances. Use of slow-stirring techniques (< 100 rpm) appropriate for low solubility ($< 10^{-2}$ g/l) test substances (Letinski et al, 2002) Requires equilibration study to determine the time taken to equilibrate the test substance and water	$< 15\%$; down to 1 µg/l
OECD series on Testing and Assessment Number 29 - Guidance Document on Transformation/Dissolution of Metals and Metal Compounds in Aqueous media.	Applicable to all metals and sparingly soluble inorganic metal compounds	/

R.7.1.7.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for water solubility:

'The study does not need to be conducted if:

- *the substance is hydrolytically unstable at pH 4, 7 and 9 (half-life less than 12 hours); or*
- *the substance is readily oxidisable in water.*

If the substance appears 'insoluble' in water, a limit test up to the detection limit of the analytical method shall be performed.'

For ionising substances, the pH-dependence of the water solubility should be known. At least the pH of the test water needs to be identified. In the context of marine risk assessment, when the pK_a is close to 8 it may be necessary to obtain realistic measurements using seawater.

For volatile compounds, it can be useful to have information on the vapour pressure.

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

Most physical properties, such as molecular weight, melting point, boiling point, density and water solubility can be obtained from commonly used environmental Handbooks, such as Verschueren's Handbook of Environmental Data on Organic Chemicals (1983), Howard's Handbook of Environmental Fate and Exposure Data, Vol. I and II (1990), Lide's CRC Handbook of Physics and Chemistry, Lange's Handbook of Chemistry, the Merck Index, the Aldrich Catalog, Kirk-Othmer Encyclopaedia of Chemical Technology and other handbook compilations such as Riddick *et al.* (1986).

Alternatively, searching on various environmental databases, such as HSDB (<http://www.toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?HSDB>), will provide summaries of chemical and physical properties of substances.

It is not unusual to find in the literature a wide range of solubilities for the same product. The oldest literature generally yields the highest solubility values: this is due to the fact that products were originally not as pure as they are nowadays and also non-specific methods were used which would not differentiate between the dissolved product and any impurities. Reported water solubility data for such compounds can often contain appreciable errors. Therefore, the reliability of data must be demonstrated by providing information on the identity and purity of the test substance, the methodology used to make the measurement, and whether or not this was performed to GLP standards.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Secondary data sources can be used in a WoE approach and they can collectively support the choice of a specific value for the water solubility. These secondary sources have to be based on a critical evaluation of peer-reviewed data and a consequent selection of a reliable and representative value for the water solubility. The use of Klimisch codes, can be extended to these secondary sources and a reliability code of (2) valid with restrictions should be assigned when using an authoritative secondary source.

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

For an organic solute to dissolve in water, firstly, the solute molecules must be separated from one another. Secondly, the solvent molecules must become partially separated from one another to create a cavity large enough to accommodate the solute. Once the solute occupies the cavity, there will be new attractive forces between solute and solvent. Finally, the water molecules in the solvation shell will form extra H-bonds to neighbouring water molecules. Thus, the water solubility depends not only on the affinity of a solute for water, but also on its affinity for its own structure. Molecules that are strongly bound to each other require considerable energy to separate them. This also means that such compounds have high melting points (for solids). Generally, solids with a high-melting temperature have poor solubility in any solvent.

Removal of a molecule from its crystal lattice means an increase in entropy, and this can be difficult to model accurately. For this reason, as well as the fact that the experimental error on solubility measurements can be quite high (generally reckoned to be about 0.5 log unit), the prediction of aqueous solubility is not as accurate as is the prediction of octanol/water partitioning. Nevertheless, many papers (Dearden 2006) and a book (Yalkowsky & Banerjee 1992) have been published on the prediction of aqueous solubility, as well as a number of reviews (Lyman 1990, ECETOC 1998, Reinhard & Drefahl 1999, Mackay 2000, Schwarzenbach *et al.* 2003, Dearden 2006). There are also a number of software programs available for that purpose (ECETOC 2003, Dearden 2006). Livingstone (2003) has discussed the reliability of aqueous solubility predictions from both QSPRs and commercial software.

It should be noted that there are various ways that water solubilities can be reported: in pure water, at a specified pH, at a specified ionic strength, as the undissociated species (intrinsic solubility), or in the presence of other solvents or solutes. Solubilities are also reported in different units, for example g/100 ml, mole/litre, mole fraction. The use of mole/litre is recommended, as this provides a good basis for comparison.

For solids, work has to be done to remove molecules from their crystal lattice, and the simplest way to account for this is to use what Yalkowsky and co-workers have termed the general solubility equation (GSE), which incorporates a melting point term to account for the behaviour of solids (Sanghvi *et al* 2003):

$$\log S_{\text{aq}} = 0.5 - \log K_{\text{ow}} - 0.01(\text{MP} - 25)$$

where MP is the melting point (°C). The melting point term is taken as zero for compounds melting at or below 25°C. Calculated log K_{ow} and MP values can be used in the GSE, although measured values are preferred. Aqueous solubilities of 1026 non-electrolytes, with a log S_{aq} range of -13 to +1 (S in mole L^{-1}), calculated with the GSE had a standard error of 0.38 log unit.

Good predictions for a large diverse data set have been obtained by the use of linear solvation energy descriptors (Abraham & Le 1999). These included two terms for polarity/polarisability, the sums of hydrogen bond donor and acceptor abilities of the solute molecule, and an expression of molecular volume

According to the Abraham and Le equation, the main factors controlling aqueous solubility seem to be hydrogen bond acceptor ability and molecular size, both of which are important elements in the molecular mechanisms of solubility.

Solubility can vary considerably with temperature, and it is important that solubility data are reported at a given temperature.

Grouping of substances and read-across approach

For the determination of the water solubility read-across is usually not possible. However interpolation may still be possible within homologous series.

Testing is technically not possible

For this endpoint, testing should almost always be possible and water solubility should usually be determined experimentally. Nonetheless, testing by the flask method might be precluded when the high viscosity of the saturated test solutions prevent from normal stirring. If it is technically not possible to conduct the study as a consequence of the properties of the substance (e.g. substances flammable in contact with water or substances readily oxidisable in water), testing may be omitted according to general rules for adaptation of the standard testing regime described in REACH Annex XI, Section 2.

Further adaptation possibilities

Not foreseen. However, for complex substances the information obtained from such testing is not relevant or of practical use, and therefore conducting the test may be waived where the data is irrelevant for subsequent assessments.

R.7.1.7.5 Impurities; uncertainties

The water solubility of the test substance can be considerably affected by the presence of impurities.

For a complex substance, the measured solubility is dependent on the amount of test substance added. In practical terms, solubility data are generated using at least two loading rates (e.g. 100 mg/l and 1000 mg/l). Accuracy in determining water solubility decreases as the water solubility of a test substance is reduced (e.g. as shown for reference substance data in OECD Method 105). When dealing with test substances with water solubilities of the order of < 10 µg/l, precautions need to be taken to avoid the introduction of dispersed material into the final extract.

Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

R.7.1.7.6 Endpoint specific information in the registration dossier / in IUCLID

Materials and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied;
- results from preliminary test (if any);
- chemical identity and impurities (preliminary purification step, if any);
- water temperature during saturation process;
- analytical method employed;
- any evidence of chemical instability;
- all information relevant for the interpretation of the results.

If Column Elution method:

- concentrations, flow rates and pH for each sample;
- mean and standard deviation of five samples at least;
- average for each of two successive runs at least;

- nature and loading of support material;
- solvent used.

If Flask method:

- pH of each sample;
- individual analytical determinations and the average;
- average of the values for different flasks.

Results and discussion & Applicant's summary and conclusion

- water solubility in (mg/l) at temperature (°C);
- pH value and concentration of test substance;
- description of solubility (if relevant);
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on water solubility can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Guide 3	Practical
4.8	VII 7.7	Water solubility	E.4.9	3.8	

R.7.1.7.7 References on water solubility

Abraham M.H. and Le J. (1999) The correlation and prediction of the solubility of compounds in water using an amended solvation energy relationship. *J. Pharm. Sci.* 88, 868-880.

Dearden J.C. (2006) *In silico* prediction of aqueous solubility. *Expert Opinion on Drug Discovery* 1, 31-52.

EC Method A6. "Water Solubility", Dir 92/69/EEC, Official Journal of the European Communities, O.J. L383 A)

EC Method A7. "Hydrolysis ", Dir 92/69/EEC, Official Journal of the European Communities, O.J. L383 A)

ECETOC Technical Report No. 74: *QSARs in the Assessment of the Environmental Fate and Effects of Chemicals*. ECETOC, Brussels, 1998.

ECETOC Technical Report No. 89: (Q)SARs: *Evaluation of the Commercially Available Software for Human Health and Environmental Endpoints with Respect to Chemical Management Applications*. ECETOC, Brussels, 2003.

Livingstone D.J. (2003) Theoretical property predictions. *Current Topics in Med. Chem.* 3, 1171-1192.

Letinski, D.J., Connolly, M.J., Peterson, D.R. and Parkerton, T.F. (2002) "Slow-stir water solubility measurements of selected alcohols and diesters", *Chemosphere*, 48, 257 – 265).

Mackay D. Solubility in water. In Boethling R.S. and Mackay D. (Eds.), *Handbook of Property Estimation Methods for Chemicals: Environmental and Health Sciences*. Lewis, Boca Raton, FL, 2000, pp. 125-139.

OECD Environmental Health and Safety Publications, 2000. Number 23. Guidance document on aquatic toxicity testing of difficult substances and mixtures.

OECD Environment, Health and Safety Publications, 2001. Series on Testing and Assessment, No. 29, Guidance Document on Transformation/Dissolution of Metals and Metal Compounds in Aqueous Media

OECD Guidelines for Testing of Chemicals – Method 105 "Water Solubility"

OECD Guidelines for Testing of Chemicals – Method 111 "Hydrolysis as a Function of pH"

Reinhard M. and Drefahl A. Handbook for Estimating Physico-chemical Properties of Organic Compounds. Wiley, New York, 1999.

Sanghvi T., Jain N., Yang G. and Yalkowsky S.H. (2003) Estimation of aqueous solubility by the general solubility equation (GSE) the easy way. *QSAR Comb. Sci.* 22, 258-262.

Schwarzenbach R.P., Gschwend P.M. and Imboden D.M. (2003) *Environmental Organic Chemistry*, 2nd edition, Wiley, Hoboken, NJ.

Yalkowsky S.H. and Banerjee S. (1992). Aqueous Solubility: Methods of Estimation for Organic Compounds. Marcel Dekker, New York.

R.7.1.8 Partition coefficient n-octanol/water

Advice to registrants with regard to nanomaterials characterisation of water solubility can be found in *Appendix R7-1 Recommendations for nanomaterials applicable to: Chapter R7a Endpoint specific guidance* of the [Guidance on IR&CSA](#), section 2.2.2 Partition coefficient n-octanol/water.

R.7.1.8.1 Type of property

The n-octanol/water partition coefficient (K_{ow}) is one of the key physicochemical parameters, and it is used in numerous estimation models and algorithms for environmental partitioning, sorption, bioavailability, bioconcentration, bioaccumulation and also human toxicity and ecotoxicity. As such K_{ow} is a critical parameter for chemical safety assessment, classification and labelling, and PBT assessment/screening (where required).

The generation of a K_{ow} value is required at all tonnage bands (i.e. > 1 t/y; information requirements according to REACH Annexes VII-X).

R.7.1.8.2 Definition

The n-octanol/water partition coefficient (K_{ow}) is defined as the ratio of the equilibrium concentrations of a dissolved substance in a two-phase system consisting of the largely immiscible solvents n-octanol and water. The property is moderately temperature-dependent and typically measured at 25°C. For further information on definition and units please see the Test Methods Regulation ((EC) No 440/2008), test method A.8, section 1.2.

R.7.1.8.3 Test method(s)

EU test method A.8 of the Test Methods Regulation ((EC) No 440/2008) describes two test procedures; a direct measurement *via* the Shake Flask method (OECD Test Guideline 107) and a correlation approach using the HPLC method (OECD Test Guideline 117). The Shake Flask method falls within the log K_{ow} range -2 to 4 and the HPLC method within the range 0 to 6. The applicability of the methods differ depending on the substance type and the amount of impurities in the test substance. Neither of the methods is applicable to surface active materials, for which an estimated value based on individual solubilities, or a calculated value along with calculation details should be provided. As with any endpoint and predictive method, the documentation and training set of the predictive method should be examined carefully to decide whether it is applicable to special categories of substances, such as zwitterionic or surface active substances.

Regardless of the method used, highly accurate measurements of log K_{ow} > ~5 are complicated by the fact that small amounts of octanol are entrained in the aqueous phase, leading to a potential underestimation of the measured log K_{ow} values. All of the direct methods for measuring log K_{ow} require quantifying the test material in either octanol or water and preferably in both matrices.

In addition, the OECD test guideline 123, Slow-stirring method, can be used to generate data for this endpoint.

Table R.7.1–6 Methods for determination of partition coefficient n-octanol/water

Method details	Repeatability	Applicability range
Shake Flask Method (EU A.8, OECD TG 107) The Shake Flask method is the default procedure. It is considered to give accurate results for low to medium hydrophobic substances. For substances with a high expected log K_{ow}, alternative methods are recommended. A suitable analytical method is needed to determine the concentration of the test material in the octanol and water phases. By applying mass balance considerations, it may be possible to measure the test material in only the less-soluble phase. However, this approach significantly decreases the reliability in the reported value. This technique is not suitable for surface active compounds (surfactants), or compounds that hydrolyse rapidly.	Three replicates should fall within $\pm 0.3 \log K_{ow}$	$-2 < \log K_{ow} < 4$
HPLC Method (EU A.8, OECD TG 117) This is a relatively quick way of estimating log K_{ow}. It is not measured directly, but from a correlation between log k (capacity factor) and log K_{ow} for a series of reference substances. It therefore depends on the quality of the log K_{ow} measurement of reference substances (often measured by the shake flask method). A series of reference compounds with similar chemical functionality to the test material should be used to generate the log k: log K_{ow} correlation. In general, the HPLC method is less sensitive to impurities than the shake flask method. The RP-HPLC is not recommended for strong acids and bases, metal complexes or surface active agents, or for measurements across very different classes of substances. The HPLC method is also very suitable for measuring the K_{ow} of mixtures of chemical homologues.	Three replicates should fall within $\pm 0.1 \log K_{ow}$	$0 < \log K_{ow} < 6$
Slow-Stirring Method (OECD TG 123) This is a more recent method developed as an alternative to the shake flask procedure (OECD TG 107, EU A.8). The advantage of slow stirring versus shaking is that emulsion formation will be reduced. The method requires a few days to reach equilibrium. The method may be difficult to adapt to a high throughput approach. As with the other direct methods, a suitable analytical method is needed to measure the concentration of the test material in the octanol and water phases. NB: Radiolabelled substances – which may be synthesised for use in other tests – can be very useful for accurate log K_{ow} determination.	Intralaboratory median standard deviation from $0.15 - 0.3 \log K_{ow}$ (Tolls <i>et al</i> , 2003).	Validation has shown that this method can also be used for very hydrophobic substances, up to Log K_{ow} 8.3 (OECD 2003, Tolls <i>et al</i> , 2003).
Estimation method based on individual solubilities in EU A.8 This method enables partition coefficients to be estimated based on the ratio of the solubility of the material in octanol and water. For some substances (e.g. some surfactants and pigments) it is technically not feasible (or good practice) to measure an octanol-water partition coefficient by OECD 107. For such substances it		

may be possible to obtain a ratio of the saturated water solubility (OECD 105) and saturated octanol solubility (no guideline currently available but based on the principles of OECD 105). This method however has the drawback that it does not include the interaction between the water and solvent phase (i.e. a substance with high K_{ow} is rather 'pushed out of the water' than 'pulled into octanol'). This explains the poor correlation typically observed between octanol solubility and K_{ow} (Dearden, 1990, Sijm <i>et al.</i>, 1999). The ratio was found to be somewhat more representative if one uses octanol/saturated water and water/saturated octanol. As such, a ratio estimation would be a less preferred yet acceptable alternative for the octanol/water partition coefficient (K_{ow}), but must be treated with caution as it would not have been derived in the same manner as other K_{ows} (OECD TG 107).		
---	--	--

R.7.1.8.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for n-octanol water partition coefficient:

'The study does not need to be conducted if the substance is inorganic. If the test cannot be performed (e.g. the substance decomposes, has a high surface activity, reacts violently during the performance of the test or does not dissolve in water or in octanol, or it is not possible to obtain a sufficiently pure substance), a calculated value for log P as well as details of the calculation method shall be provided.'

If experimental testing including estimation from the individual solubilities is not possible, log K_{ow} must normally be calculated by an appropriate numeric method based on the molecule's structure.

In case of rapid hydrolysis the registrant needs to provide evidence in the form of a hydrolysis endpoint study record (study summary) and should consider testing for the hydrolysis products instead, as information on the properties of (environmentally and toxicologically) relevant degradation products are needed for conducting the risk assessment of the substance to be registered.

Adaptation possibilities according to Annex XI to REACH

The reporting of the K_{ow} information cannot usually be waived (except for inorganic substances), because it is essential for CSA, classification and labelling and PBT assessments.

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

Log K_{ow} is a commonly documented property in substance databases. There are many public and commercial databases collecting log K_{ow} information and available on the internet. One example of publicly available database is the one in the freely downloadable software OECD QSAR Toolbox in the "Endpoint section" under "physico-chemical" properties (<http://www.qsartoolbox.org>).

Log K_{ow} information has to be submitted to ECHA as part of the dossier in IUCLID format (<http://iuclid.eu/index.php?fuseaction=home.iuclidHome>).

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Measured values are given precedence over calculated values. For organic substances experimentally derived high-quality K_{ow} values, or values which are evaluated in reviews and assigned *recommended values*, are preferred over other determinations of K_{ow} . Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

When no experimental data of high quality are available, or if experimental methods are known to be unreliable, valid (Q)SARs for log K_{ow} may be used e.g. in a weight-of-evidence approach. Due to the availability of large number of measured log K_{ow} values and robust QSAR models for this property, the QSARs can, in some cases, predict the partition coefficient of a molecule with higher accuracy compared to a single test. Such valid QSAR models may be used if they are restricted to substances for which their applicability is well characterised. In order to be used as a stand alone source of values to meet the data requirements of Annex VII, 7.8, the QSARs must meet the criteria set out in Annex XI, 1.3.

Grouping of substances and read-across approach

For the determination of the partition coefficient n-octanol/water read-across is usually not possible. However interpolation may still be possible within homologous series.

Testing is technically not possible

Testing should always be considered, if none of the waiving possibilities applies.

Further adaptation possibilities

Not foreseen.

R.7.1.8.5 Impurities; uncertainties

The effect of impurities in the test substance are discussed in the referenced test guidelines.

Difficult to test substances:

There are certain structural or physico-chemical properties that can make the accurate determination of K_{ow} or its measurement difficult. Difficult to test substances include poorly soluble, volatile, surface active, ionisable substances, mixtures of substances, as well as substances subject to rapid degradation due to such processes as phototransformation, hydrolysis, oxidation, or biotic degradation.

Guidance on regulatory compliant K_{ow} determination for ionisable substances and salts:

The K_{ow} is typically defined as the partition coefficient of the neutral, undissociated form of a substance. However, the relative extent to which an ionisable substance is likely to be dissociated in the environment (with pH usually in the range 5-9) can have a marked effect on

its physicochemical properties, especially the octanol-water partition coefficient and water solubility, which in turn affect fate and behaviour. As log K_{ow} is routinely used to predict bioconcentration/bioaccumulation potential, this aspect is especially important in a PBT context. For substances which dissociate within an environmentally relevant pH range (pK_a 5-9), values for K_{ow} must be derived for the neutral form, and preferably also for the dissociated form. In some cases a factor 4-5 has been recorded between the log K_{ow} of both species. The value for the dissociated molecule determined around a pH of 7 (sometimes referred to as D_{ow}) is considered more realistic for PBT and chemical safety assessment.

Based on practical experience the following guidance is provided:

Simple acids and bases in the normal pH range:

- The HPLC method is to be applied to acids and bases in their non-ionised forms, although the pH should be kept in the range 2 to 9 (however pH 5 to 9 is preferred).
- For the shake-flask method, the approach must be followed in which the study is conducted at a pH where the substance is not ionised, if possible, or at a pH where the extent of ionisation is minimised.
- Validated QSAR estimations may be useful for acids and bases.

Zwitterionic substances:

- For zwitterions, the shake-flask method should be used to develop a valid K_{ow} value. Even if the ionic charge pattern of the compound in octanol is not known, the value represents a practical and useful parameter. It is not justifiable to expect a full description of all the equilibria in both water and octanol. The pH of such a study should be 7 or the iso-electric point (pH value at which the molecule has no net electrical charge), as long as that point is in the range pH 5 to 9, so as to maximise the possibility of partition into octanol. There is no need to give both pH values.
- The HPLC method must not be used. The usual estimation methods should be valid, but particular care should be exercised.
- QSAR estimations may be useful provided that they are validated.

Salts of organic compounds:

- The shake-flask method should be used, usually at pH 7, or at any pH in the range 5 to 9 which maximises the potential for partition into octanol. For salts, the nature of the analytical method compared to the chemical composition will have to be considered. The ideal is to monitor cation and anion** individually in both phases. When only one half can be analysed, then the result must be understood as partial, even if it is the best that is achievable.
- Estimation by HPLC is not valid for the whole salt.
- QSAR methods will be valuable in assessing the properties of each half of the salt. Current estimation methods cannot estimate the K_{ow} of the ion pair.

Guidance on regulatory compliant K_{ow} determination for surfactants:

In many cases a calculated K_{ow} value based on the octanol and water solubilities will be the first choice for surfactants. It is also useful to compare a calculated with a measured value. For the calculation approaches, one needs to consider the pH of the system (which determines the

ionisation of the surfactant – see Section [R.7.1.17](#)). None of the experimental methods is very well suited for determining the K_{ow} of surface active substances. The shake flask method is the least suitable experimental method for surfactants. HPLC methodology may fail due to secondary interactions, and is sensitive to fluctuations of ionic strength. The slow stirring method in theory is the best, but still not demonstrated to be perfect. If using slow stirring, one needs to demonstrate a consistent result when starting with the surfactant in either phase, not just in the octanol. A working approach for surfactants might be the comparison of measured solubilities in octanol and water. However, it would then be prudent to take the critical micelle concentration in water (CMC) as a solubility limit, in order to avoid the artefact of unrealistically low K_{ow} values.

Guidance on regulatory compliant K_{ow} determination for mixtures:

It is possible that different components of mixtures have significantly different behaviour in the physico-chemical tests and therefore also *in vivo* and in the environment. It is therefore important to ensure that the results presented for the physico-chemical tests represent each component rather than the mixture being treated as a single component. For simple mixtures where the components are known and easily identifiable, this may mean presenting individual values for K_{ow} . For complex mixtures, the HPLC method is ideal for determination of K_{ow} , and a defined range of values should be presented, with an indication of the proportion of substance within a given range (e.g. > 90% of components have log K_{ow} in the range 4-5), to allow the significance of these results to be reflected in the risk assessment. The HPLC method is also recommended for petroleum products, which are typically mixtures.

R.7.1.8.6 Endpoint specific information in the registration dossier / in IUCLID

Materials and methods

Shake-flask method (EU A.8/OECD TG 107):

- equilibrium concentrations of the test substance in both phases;
- relative volumes of the two phases;
- analytical method(s).

Calculation method (EU A.8):

- identification of the method;
- working principle of the method;
- reference to the method;
- information on source chosen to justify K_{ow} values of fragments being manipulated;
- applicability of the method.

HPLC method (EU A.8/OECD TG 117):

- column(s) used;
- mobile phase (composition, buffer, pH);
- reference substances with respective K_{ow} values from the literature;
- concentrations measured.

Slow-stirring method (OECD TG 123):

- label purity of labelled substances and molar activity (where appropriate);
- sampling times;
- description of the test vessels and stirring conditions;
- number of replicates;

- temperature during the experiment;
- volumes of 1-octanol and water at the beginning, during and remaining after the test;
- determined concentrations of the test substance in 1-octanol and water as a function of time;
- description of the test vessels and stirring conditions (geometry of the stirring bar and of the test vessel, vortex height in mm, and when available: stirring rate) used;
- analytical methods used to determine the test substance (its repeatability and sensitivity) and the method limit of quantification;
- sampling times;
- pH of the aqueous phase and of the buffers used, when pH is adjusted for ionisable molecules;
- number of replicates;
- demonstration of mass balance;
- temperature and standard deviation or the range of temperature during the experiment;
- the regression of concentration ratio against time.

Results and discussion

- final value for log K_{ow};
- K_{ow} values and their mean;
- standard deviation of individual K_{ow} values;
- theoretical value when it has been calculated;
- temperature of the test solutions (°C);
- pH value(s) of the aqueous solution(s);
- composition and concentration of buffers;
- concentration of the stock solution;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used and reasons for it or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on partition coefficient can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.7	VII 7.8	Partition coefficient	E.4.8	3.7

R.7.1.9 Flash point

R.7.1.9.1 Type of property

The flash point is a property:

- for substance characterization;
- for the classification of flammable liquids;
- which is important for physical hazard assessment.

R.7.1.9.2 Definition

The flash point is the lowest temperature of the liquid (as measured in a prescribed manner) at a pressure corrected to 101.325 kPa, at which application of an ignition source causes the vapour of the liquid to ignite momentarily and the flame to propagate across the surface of the liquid under the specified conditions of test (see section 1.2, Test Method A.9).

R.7.1.9.3 Test method(s)

The EU test method A.9 – Flash point from the Regulation (EC) 440/2008 can be used. Suitable methods are listed in the CLP Regulation Annex I, 2.6.4.4, Table 2.6.3.

The method to be used has to be chosen taking into account the properties of the liquid (viscosity, halogenated compounds present) and the scope of the standard.

For substances with a high decomposition potential, a method using small amounts of liquid (e.g. EN ISO 3679: Determination of flash point - Rapid equilibrium closed cup method) is recommended to reduce the amount of substance under test.

For classification purposes it is recommended to use the mean of at least two test runs. If the experimentally determined flashpoint is found to be within $\pm 2^{\circ}\text{C}$ of the limiting criterion for classification or assigning a category when using a non-equilibrium method, it is recommended to repeat the determination with an equilibrium method.

R.7.1.9.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for flash point:

'The study does not need to be conducted if:

- *the substance is inorganic;*
- *the substance only contains volatile organic components with flash-points above 100°C for aqueous solutions; or*
- *the estimated flash-point is above 200°C; or*
- *the flash-point can be accurately predicted by interpolation from existing characterised materials.'*

The first point has to be further specified as:

- The substance is inorganic except where there are covalent bonds;

because some inorganic liquids with covalent bonds are flammable e.g. CS₂, N₂H₂, HCN.

The third point should only be applied when a well validated estimation model was used.

The fourth point should only be applied when there are enough reliable experimental data from existing characterised materials to be able to accurately interpolate to estimate the flash point.

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) which meet the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

For the determination of the flash point, weight of evidence is not possible.

(Q)SAR

For the determination of the flash point, QSAR approaches are discouraged for the purpose of classification / risk assessment, except where the mean absolute error of the QSAR is less than 2°C.

For non-halogenated liquids calculation based on the vapour pressure curve and lower explosion limit of the substance can be used as a screening test and a flashpoint need not be determined experimentally if the calculated value is at least 5°C higher than the relevant classification criterion.

Grouping of substances and read-across approach

For the determination of the flash-point read-across is usually not possible. However interpolation may still be possible within homologous series.

Testing is technically not possible

This applies if:

- the liquid is an explosive;
- the liquid is pyrophoric or self-reactive;
- decomposition occurs during the melting point study;
- some impurities have an impact on the ignition source in such a way as to distort/invalidate the results.

Testing should always be considered, if none of the waiving possibilities applies.

Further adaptation possibilities

The flash point does not need to be determined experimentally if conclusive and consistent literature data are available.

Data generated with the same tests and classification principles as specified in the CLP Regulation for flash point generated in conjunction with transport classification can satisfy the REACH requirements, but this needs to be checked on a case by case basis.

R.7.1.9.5 Impurities; uncertainties

Impurities can have a significant influence on the flash point. The influence depends on the amount and the vapour pressure of the impurity. Even if their concentration is below 0.5%, especially if their boiling point is substantially lower, they may have a strong effect on the flash point. Impurities with a higher boiling point will normally have no effect on the flashpoint. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

R.7.1.9.6 Endpoint specific information in the registration dossier / in IUCLID

Materials and methods

- reference to the standard or the test method applied;
- open cup or closed cup (for classification purposes only the closed cup methods are allowed);
- equilibrium or non-equilibrium method.

Results and discussion

- corrected flashpoint and unit;
- data on repeatability and reproducibility as given in the method;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on flash point can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.11	VII 7.9	Flash point	E.4.12	3.10

R.7.1.10 Flammability

Some of the information requirements according to REACH Annex VII were phrased in a way that they correspond to 'indications of danger' as given in Annex II of the DSD. For substances, classification and labelling according to CLP Regulation has been mandatory since 1 December 2010 (and will become mandatory for mixtures (preparations) from 1 June 2015, when the DPD will be repealed). Consequently properties associated with flammability are covered by classification of the substance according to the CLP Regulation. However, the physical hazards according to the CLP Regulation are structured completely differently from the physicochemical properties according to the DSD (and therefore also REACH, Annex VII). This means that for some of the CLP hazard classes an unambiguous assignment to one of the headlines (information requirements) in Annex VII to REACH is not possible. The assignment of hazard classes to the headline 'Flammability' as shown in the table below ([Table R.7.1-7](#)) must therefore only be understood as a means to structure this document in accordance with Annex VII to REACH. It has to be noted that self-reactive substances and organic peroxides are assigned to the headline 'Flammability' and only a cross reference is added under the headline 'Explosive properties' because these two hazard classes can have explosive and/or flammable properties.

Table R.7.1-7 Assignment of CLP hazard classes to the information requirement 'Flammability' according to REACH, Annex VII and correlation between the Test Method Regulation and the test method according to CLP and supporting link with the [Guidance on the Application of the CLP criteria](#).

Information requirement according to Art. 10 (a) (vi) of the REACH Regulation (EC) No. 1907/2006 (the no. in brackets is the respective no. in the table in Annexes VII to IX to REACH)	CLP Regulation (EC) No. 1272/2008 (the no. in brackets is the respective chapter no. in Annex I to CLP)	Chapter in revised R.7(a) guidance	Corresponding test method according to The Test Method Regulation (EC) No 440/2008	Corresponding test method according to CLP Regulation	Chapter in the Guidance on the application of the CLP Criteria (ex RIP 3.6)
Flammability (7.10)	Flammable gases ¹⁷ (2.2)*	R.7.1.10.1	A.11 Flammability (gases)	ISO 10156 EN 1839	2.2
	Flammable liquids (2.6)*	R.7.1.10.2	for liquids: see Flash point	see CLP, Annex I, Chapter 2.6.4.4, Table 2.6.3	2.6
	Flammable solids (2.7)*	R.7.1.10.3	A.10 Flammability (solids)	UN Test N.1	2.7

¹⁷ The 4th ATP to the CLP Regulation amends the criteria in the CLP Annex I, Section 2.2 Flammable gases by including subclassifications for chemically unstable gases.

	Self-reactive substances and mixtures (2.8)*	R.7.1.10.4	n.a.	UN Test series A to H	2.8
	Pyrophoric liquids (2.9)*	R.7.1.10.5	A.13 Pyrophoric properties of solids and liquids	UN Test N.3	2.9
	Pyrophoric solids (2.10)*	R.7.1.10.6		UN Test N.2	2.10
	Self-heating substances and mixtures (2.11)*	R.7.1.10.7	n.a.	UN Test N.4	2.11
	Substances and mixtures which in contact with water emit flammable gases (2.12)*	R.7.1.10.8	A.12 Flammability (Contact with water)	UN Test N.5	2.12
	Organic peroxides (2.15)*	R.7.1.10.9	n.a.	UN Test series A to H	2.15

* Note that regardless of whether the hazard class or category is listed in Article 14(4)(a) REACH the chemical safety assessment (where required) must be performed in accordance with Article 14(3) REACH. Furthermore, according to Article 10(a)(iv) of REACH the technical dossier of a registration of a substance under the REACH Regulation must include information on classification and labelling of the substance as specified in section 4 of Annex VI to the REACH Regulation.

In addition, it has to be noted that some substances have flammable properties which do not result in classification. Examples are the following:

- gases that do not have a flammable range at 20°C and standard pressure (and therefore are not classified as flammable gases) might have a flammable range at higher temperatures and/or pressure (e.g. ammonia);
- liquids that do not have a flash point (and therefore are not classified as flammable liquids) might have an explosion range (especially halogenated hydrocarbons).

Information about such properties should also be indicated in the dossier.

R.7.1.10.1 Flammable gases

Definition

'Flammable gas means a gas or gas mixture having a flammable range with air at 20°C and a standard pressure of 101.3 kPa' (Annex I to CLP, Section 2.2.1).

Classification criteria and relevant information

Flammable gases are classified into two categories depending on their flammability range (Annex I to CLP, Section 2.2.2. Table 2.2.1).

Detailed guidance on the classification criteria and the test method(s) can be found in the [Guidance on the Application of the CLP criteria](#), section 2.2¹⁸.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for flammability:

'The study does not need to be conducted:

- if the substance is a solid which possesses explosive or pyrophoric properties. These properties should always be considered before considering flammability; or*
- for gases, if the concentration of the flammable gas in a mixture with inert gases is so low that, when mixed with air, the concentration is all time below the lower limit; or*
- for substances which spontaneously ignite when in contact with air.'*

The relevant points can be paraphrased (first point is not relevant for this chapter), namely the study does not need to be conducted:

- if the concentration of the flammable gas in a mixture when mixed with air is below the lower limit;
- if the gas spontaneously ignites when in contact with air.

Gases that spontaneously ignite in contact with air are pyrophoric and are therefore flammable gases.

Adaptation possibilities according to Annex XI to REACH

- **Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH**

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Many gases are classified in Annex VI to CLP either as Flam. Gas 1 or Flam. Gas 2, and additional flammable gases are listed in the UN-RTDG whose classifications correspond to Flam. Gas 1 according to CLP.

¹⁸ The 4th ATP to the CLP Regulation amends the criteria in the CLP Annex I, Section 2.2 Flammable gases by including subclassifications for chemically unstable gases. Consequently the [Guidance on the Application of the CLP criteria](#), Part 2: Physical hazards has been restructured to take account of the 4th ATP, which applies to substances from 1 December 2014 and to mixtures from 1 June 2015. When the 4th ATP is applied a Guidance corrigendum will be made to delete the outdated sub-chapter 2.2.1 Flammable gases in the [Guidance on the Application of the CLP criteria](#).

- Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

For gases that are not classified in Annex VI to the CLP Regulation nor in the UN-RTDG, there is ample scientific literature giving the flammability range for most gases (e.g. IEC 60079-20-1 *Data for flammable gases and vapours, relating to the use of electrical apparatus* – (under revision)).

- (Q)SAR

At present (Q)SAR is generally not applicable for determination of explosion (/flammability) limits of gases.

- Grouping of substances and read-across approach

For the determination of the flammable gases read-across is usually not possible. However interpolation may still be possible within homologous series.

- Testing is technically not possible

Testing should always be considered, if none of the waiving possibilities applies.

Further adaptation possibilities

Further adaptation is possible for gases that are known to be non-flammable. Examples are nitrogen, the noble gases (helium, neon, argon, krypton, xenon), carbon dioxide and sulphur hexafluoride. As stated in Annex IX of REACH, when for certain endpoints, it is proposed to not provide information for other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used.

Impurities; uncertainties

Tests should be performed with the lowest concentration of impurities in the gas encountered in the normal manufacturing process and the moisture content should be less than or equal to 0.01 mol %. Utmost care should be taken in the selection of the key study(ies) and/or use of weight-of-evidence approaches that the data selected is representative of the substance being registered by the respective companies.

How to conclude on the DSD classification

All gases with a flammability range in air are classified 'Extremely flammable F+ ; R12' according to DSD, unless classified differently according to Annex VI, Table 3.2 of the CLP Regulation. This means that all gases classified as flammable gases according to CLP (either Category 1 or 2) are classified as 'Extremely flammable F+; R12'.

Endpoint specific information in the registration dossier/in IUCLID

Material and methods:

- description of the apparatus and dimensions or reference to the standard or the test method applied;
- test temperature;
- tested concentrations.

Results and discussion & Applicant's Summary and conclusion (interpretation of results)

- indicate lower and upper explosion limits in % volume;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on flammability can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.13	VII 7.10	Flammability	E.4.14	3.12

Further information / references

For the testing of flammable gases according to CLP classification requirements, refer also to the [Guidance on the Application of the CLP criteria](#), section 2.2, and in Directive 2008/47/CE.

R.7.1.10.2 Flammable liquids

Definition

Flammable liquid means a liquid which may form flammable/explosive vapour-air mixtures. Within the CLP Regulation 'Flammable liquid' means a liquid having a flashpoint of not more than 60°C (see CLP Annex I, section 2.6.1).

Classification criteria and relevant information

Flammable liquids are classified in three categories according to the criteria of the CLP Regulation (see CLP Annex I, section 2.6, table 2.6.1) based on their boiling point and their flash point. Derogation is possible (see CLP Annex I, section 2.6.4.5) for Flam. Liquid Cat. 3 having a flashpoint above 35°C based on the information on sustained combustibility. Furthermore, gas oils, diesel and light heating oils having a flash point between $\geq 55^{\circ}\text{C}$ and $\leq 75^{\circ}\text{C}$ may be regarded as Category 3 flammable liquids according to the CLP Regulation (CLP Annex I, section 2.6, footnote to table 2.6.1).

In addition EUH018 - '*In use may form flammable/explosive vapour-air mixture*' has to be assigned to substances classified under the CLP Regulation which may form flammable/explosive vapour-air mixtures although they do not have a flash point e. g. CH_2Cl_2 , $\text{C}_2\text{H}_3\text{Cl}_3$. In such cases it is possible to make the decision on whether flammable/explosive vapour-air mixture may be formed based on either the determination of explosion limits according to EN 1839 or the determination of explosion points according to EN 15794. It is sufficient to determine either the lower explosion limit or the lower explosion point.

Detailed guidance on the classification criteria and the test method(s) can be found in the [Guidance on the Application of the CLP criteria](#), section 2.6.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

The entries 'flammability' (7.10), 'boiling point' (7.3) and 'flashpoint' (7.9) are the relevant ones. For the latter two entries, see their respective relevant sections in this document.

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for flammability:

'The study does not need to be conducted:

- if the substance is a solid which possesses explosive or pyrophoric properties. These properties should always be considered before considering flammability; or*
- for gases, if the concentration of the flammable gas in a mixture with inert gases is so low that, when mixed with air, the concentration is all time [i.e. 'always'] below the lower limit; or*
- for substances which spontaneously ignite when in contact with air.'*

The relevant points can be paraphrased (first two points are not relevant for this chapter), namely the 3rd point specifies that for flammability, Annex VII to REACH does not require testing for substances which spontaneously ignite when in contact with air.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR

To be used as a stand alone value to meet the data requirements of Annex VII, 7.8, QSAR models must meet the criteria set out in Annex XI, 1.3. The entries 'boiling point' (7.3) and 'flashpoint' (7.9) are also the relevant ones, therefore please check under each respective QSAR sub-section for more information.

Sustained Combustibility:

No (Q)SAR exists currently.

For further reference see also the [Guidance on the Application of the CLP criteria](#), section 2.6.

- [Grouping of substances and read-across approach](#)

The entries 'boiling point' (7.3) and 'flashpoint' (7.9) are again the relevant ones. For both these entries, see their respective sections in this document.

Sustained Combustibility:

For the determination of the sustained combustibility read-across is usually not possible. However interpolation may still be possible within homologous series.

- [Testing is technically not possible](#)

Testing is not possible if:

- the liquid is an explosive;
- the liquid is pyrophoric or self-reactive.

Testing should always be considered if none of the waiving possibilities applies.

Further adaptation possibilities

Use of data on boiling point, flashpoint when determined with a closed cup method, explosion limits or lower explosion point from validated literature (see below chapter Further information/ references) is possible. Data on boiling point generated in relation to transport classification may also satisfy the Annex XI requirements. Data on flashpoint generated in relation to with transport classification may satisfy the Annex XI requirements if closed cup methods have been used. However care has to be taken in cases where there is no transport classification as 'flammable liquid', because certain substances can form flammable/explosive vapour-air mixtures although they do not have a flash point.

As stated in Annex IX of REACH, when for certain endpoints, it is proposed to not provide information for other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used.

Impurities; uncertainties

Boiling point:

Impurities will influence the boiling point of the main component. The influence depends on the amount and boiling point of the impurity. The higher the amount and the higher the difference between the boiling points of the main component and the impurity, the higher the influence.

Flashpoint:

Special care has to be taken when a sample contains impurities with a lower boiling point than the main component. Even if their concentration is below 0.5%, especially if their boiling point is substantially lower, they may have a strong effect on the test result. Impurities with a higher boiling point will normally have no effect on the flashpoint.

Sustained combustibility:

Impurities with lower boiling point may influence the ability to sustain combustion. However it is not yet possible to quantify the influence of impurities.

How to conclude on the DSD classification

Based on the data on boiling point and flashpoint the DSD classification according to the respective DSD criteria is possible. Simplified direct translation between CLP classification and DSD classification is not possible, see figure below ([Figure R.7.1-1](#)).

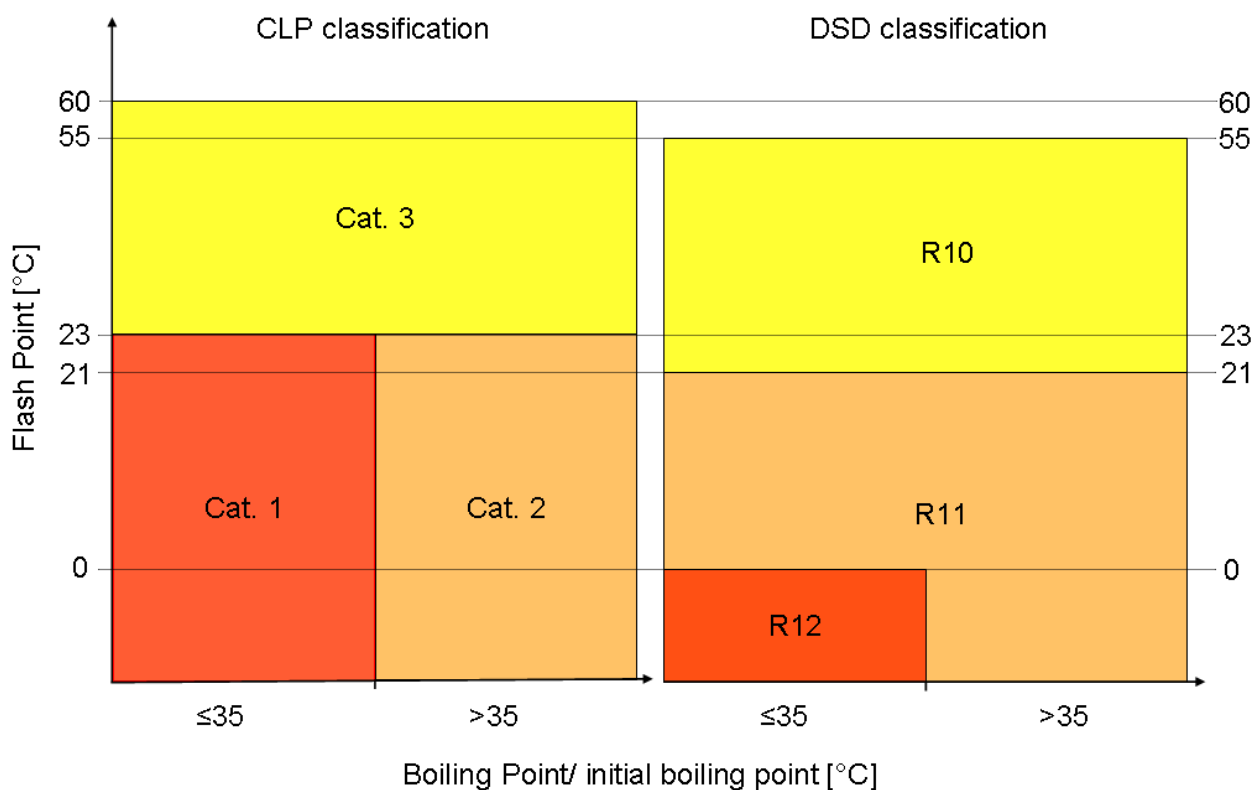


Figure R.7.1-1 Comparison of the DSD and the CLP classification

Substances exempted from classification in Cat. 3 because of their flashpoint and behaviour when tested for sustained combustibility can be exempted from being classified under DSD as R10, if they don't show additional dangerous properties relevant for classification.

Endpoint specific information in the registration dossier/in IUCLID

The physicochemical studies reporting data relevant for classification as a flammable liquid (flashpoint and boiling point) are to be reported in the relevant IUCLID endpoint records.

Material and methods

See chapter [R.7.1.9](#) Flash point and [R.7.1.3](#) Boiling point.

Results and discussion

- corrected flashpoint and unit;
- data on repeatability and reproducibility as given in the method;
- boiling point value (°C) as measured;
- pressure value and unit;
- rate of temperature increase;
- decomposition (if applicable);
- measurement uncertainty if available;

- boiling point value in °C (corrected to standard pressure, except where the boiling point was determined at reduced pressures) (as above, but in a separate block of fields);
- if available explosion limits;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on flammability can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.13	VII 7.10	Flammability	E.4.14	3.12

Further information / references

See also R.7.1.3 Boiling point and 0 Flash point. For testing of flammable liquids according to CLP classification requirements refer also to the [Guidance on the Application of the CLP criteria](#), section 2.6.

R.7.1.10.3 Flammable solids

Definition

'A flammable solid means a solid which is readily combustible, or may cause or contribute to fire through friction. Readily combustible solids are powdered, granular, or pasty substances or mixtures which are dangerous if they can be easily ignited by brief contact with an ignition source, such as a burning match, and if the flame spreads rapidly' (see CLP Regulation, Annex I, section 2.7.1).

Classification criteria and relevant information

Solid substances and mixtures are classified as flammable in two categories according to their burning behaviour (see the CLP Regulation, Annex I, section 2.7) using UN Test N.1 as described in section 33.2.1 of the UN-MTC.

Chapter 2.7 of the [Guidance on the Application of the CLP criteria](#) gives detailed information on the CLP classification of flammable solids, the UN Test N.1 and the relation to the DSD and the UN-RTDG regulations.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for flammable solids:

'The study does not need to be conducted:

- *if the substance is a solid which possesses explosive or pyrophoric properties. These properties should always be considered before considering flammability; or*

- *for gases, if the concentration of the flammable gas in a mixture with inert gases is so low that, when mixed with air, the concentration is all time [i.e. always] below the lower limit; or*
- *for substances which spontaneously ignite when in contact with air.'*

Concerning the first indent, testing for flammability of a solid is a part of classification in CLP Regulation. Refer also to the [Guidance on the Application of the CLP criteria](#), section 2.7 on classification requirements. For substances having explosive properties, testing for a classification as a flammable solid may be waived. This applies to substances and mixtures classified as explosives, organic peroxides and self-reactive substances and mixtures.

Second indent is not applicable for this endpoint.

With regards to the third indent, substances which spontaneously ignite when in contact with air are pyrophoric substances as defined by the CLP Regulation (see the [Guidance on the Application of the CLP criteria](#), section 2.10). Such substances are not classified as flammable solids but as pyrophoric solids under the CLP Regulation.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

Literature data – even if available – should not be used since flammability strongly depends on particle size, surface treatment and other parameters.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

If available data from an A.10 test method indicate that a classification as a flammable solid does not apply (result: not highly flammable), no more testing is necessary. However, if the A.10 test method has come to the conclusion 'highly flammable', it will be necessary to also determine the influence of the wetted zone as described in the UN Test N.1.

- Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR

At present (Q)SAR is generally not applicable for flammable solids. Application of (Q)SAR is not possible.

- Grouping of substances and read-across approach

At present, grouping and read across are not applicable.

- Testing is technically not possible

Testing should always be considered if none of the waiving possibilities applies.

Further adaptation possibilities

As stated in Annex IX of REACH, when for certain endpoints, it is proposed to not provide information for other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used.

If a suitable screening test clearly shows that the substance is not flammable, further testing is not necessary (see also the [Guidance on the Application of the CLP criteria](#), section 2.7.4.2). An example for a suitable screening test is the burning index as described in VDI guideline (VDI Guideline, 1990) if a burning index of 3 or less is found, the substance should not be classified as a flammable solid and no further testing is required.

Substances and mixtures classified according to the CLP Regulation as explosives, organic peroxides, self-reactive substances and mixtures as well as pyrophoric or oxidising solids should not be considered for classification as flammable solids (see the [Guidance on the Application of the CLP criteria](#), section 2.7.3).

However, if a substance gives a positive result in UN Test Series 1 or 2 as described in the UN-MTC, but is exempted from classification as an explosive on the basis of UN Test Series 6, a test for classification as a flammable solid should be performed.

Impurities; uncertainties

Impurities do not tend to have a large effect on the flammability of a solid. However, if a solid which is not flammable in the pure state contains flammable organic liquids or organometallic impurities it may burn more rapidly and thus become flammable. Therefore utmost care should be taken in the selection of the key study(ies) and during use of weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

How to conclude on the DSD classification

Any substance found to be a flammable solid according to CLP Regulation has to be classified as 'F; R11' according to the DSD.

Endpoint specific information in the registration dossier/in IUCLID

Material and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied.

Solid flammability:

- indicate if preliminary and/or main test performed;
- moisture content;
- particle size and distribution (if available) (see [R.7.1.14](#) Granulometry).

Results and discussion

- indicate burning time;
- pass/non pass of the wetted zone (in the case of the UN Test N.1);
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on flammability can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.13	VII 7.10	Flammability	E.4.14	3.12

Further information / references

VDI guideline 2263, part 1, (1990): 'Test methods for the Determination of the Safety Characteristics of Dusts'.

For testing of flammable solids according to CLP classification requirements, refer also to the [Guidance on the Application of the CLP criteria](#), section 2.7.

R.7.1.10.4 Self-reactive substances and mixtures

In the CLP Regulation self-reactive substances are a distinct hazard class. Self-reactive substances are classified into one of the seven categories of 'Types A to G' according to the classification criteria given in section 2.8.2.3 of Annex I of CLP. In the Dangerous Substances Directive (67/548/EEC) no hazard class for 'self-reactive substances' is defined. Nevertheless, self-reactive substances were also classified as dangerous according to the DSD, e.g. as flammable or as substances with explosive properties.

As mentioned below under the sub-section 'Definition', self-reactive substances are excluded from testing as explosives according to Test Series 1 to 8 in Part I of the UN-MTC (see R.7.1.11.1 Explosives). In Test Series A to H however, no tests on sensitivity to impact (solids and liquids) and friction (solids only) are included. For the risk assessment and the safe use and handling, data according to the EU test method A.14 as described in Regulation (EC) No 440/2008, if available, or UN Test 3 (a) (ii) BAM Fallhammer and Test 3 (b) (i) BAM friction apparatus (see R.7.1.11) should be part of the hazard communication in the registration dossier (REACH Annex VII, 7.11) and the safety data sheet.

Definition

The definition of a self-reactive substance is given in section 2.8.1 of Annex I to CLP Regulation:

'Self-reactive substances or mixtures are thermally unstable liquid or solid substances or mixtures liable to undergo a strongly exothermic decomposition even without participation of oxygen (air). This definition excludes substances and mixtures classified according to this Part as explosives, organic peroxides or as oxidising. A self-reactive substance or mixture is regarded as possessing explosive properties when in laboratory testing the formulation is liable to detonate, to deflagrate rapidly or to show a violent effect when heated under confinement.'

Background information and guidance on the definition is given in the [Guidance on the Application of the CLP criteria](#), sections 2.8.1 and 2.8.2.

Classification criteria and relevant information

Classification principles are given in CLP Regulation Annex I, sections 2.8.2 and 2.8.4. Background information and guidance on relevant aspects regarding the classification is given in the [Guidance on the Application of the CLP criteria](#), sections 2.8.4, 2.8.5 and 2.8.6.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Only self-reactive substances, as defined in the section definition, have to be tested according to the UN-MTC, Part II test series A - H.

CLP Annex I, section 2.8.2.1 provides the following specific rules for adaptation of the standard information requirement for self-reactive substances and mixtures.

'Any self-reactive substance or mixture shall be considered for classification in this class as a self-reactive substance or mixture unless:

- a. *they are explosives, according to the criteria given in 2.1;*
- b. *they are oxidising liquids or solids, according to the criteria given in 2.13 or 2.14, except that mixtures of oxidising substances, which contain 5 % or more of combustible organic substances shall be classified as self-reactive substances according to the procedure defined in 2.8.2.2;*
- c. *they are organic peroxides, according to the criteria given in 2.15;*
- d. *their heat of decomposition is less than 300 J/g; or*
- e. *their self-accelerating decomposition temperature (SADT) is greater than 75 °C for a 50 kg package¹⁹.*

Adaptation possibilities according to Annex XI to REACH

- *Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH*

A number of already tested and classified substances and mixtures are listed in UN-RTDG, section 2.4.2.3.2.3. Available information may originate from the classification for transport. More details are given in the [Guidance on the Application of the CLP criteria](#), sections 1.7.2.1 and 2.8.6.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

¹⁹ See UN RTDG, sub-sections 28.1, 28.2, 28.3 and Table 28.3.

- [Weight of evidence](#)

For the determination of the self-reactive substances and mixtures, weight of evidence is not possible.

- [\(Q\)SAR](#)

At present (Q)SAR is generally not applicable for determination of self-reactive substances. Application of (Q)SAR is not possible.

- [Grouping of substances and read-across approach](#)

At present grouping and read-across are not applicable.

- [Testing is technically not possible](#)

A few of substances can, for safety reasons, only be handled and tested in diluted form, see the substances and mixtures listed in UN-RTDG, section 2.4.2.3.2.3.

Further adaptation possibilities

Not foreseen.

Impurities; uncertainties

Minor impurities can have an influence on thermal stability. Background information and guidance on these aspects is given in the [Guidance on the Application of the CLP criteria](#), section 2.8.4.3.

How to conclude on the DSD classification

In the DSD self-reactive substances are not covered. They may be classified in other DSD classes (e.g. explosive substance, flammable solid or liquid). See also the [Guidance on the Application of the CLP criteria](#), section 2.8.6.1.

What information is required in the registration dossier in IUCLID

Material and methods

- see UN-MTC, Part II, classification procedures and test series A-H.

Results and discussion

The following data on self-reactive substances should be submitted:

- type of self-reactive substance;
- decomposition energy (value and method of determination);
- SADT (Self accelerating decomposition temperature) together with the volume the SADT relates to;
- detonation properties (Yes/Partial/No);
- deflagration properties (Yes rapidly/Yes slowly/No);
- effect of heating under confinement (Violent/Medium/Low/No);
- explosive power if applicable (Not low/Low/None).

For assigning the type of self-reactive substance, the list of currently assigned self-reactive substances according to the 2.4.2.3.2.3 of the UN-RTDG can be used, in cases where the assignment was based on test(s) according to the UN-MTC. The relevant underlying test data may be collected from the respective UN documents from the UN Committee of experts on the

transport of dangerous goods, from test reports produced by competent authorities or industry, or from other reliable sources.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

The following example ([Figure R.7.1–2](#)) shows how the data mentioned above could be documented in the chemical safety report (CSR):

Figure R.7.1–2 Example 2,2'-Azodi (isobutyronitrile)

UN Test Series A to H	Test method	Results + Evaluation	Remarks
Propagation of detonation	A.5	"yes"	Apparent density (kg/m ³): 366 Fragmented length (cm): 40
Propagation of deflagration #1	C.1	"yes, slowly"	68 ms
Propagation of deflagration #2	C.2	"no"	
Effect of heating under defined confinement #1	Koenen E.1	"violent"	Limiting diameter 3.0 mm Type of fragmentation: F
Effect of heating under defined confinement #2	DPVT E.2	"medium"	Limiting diameter 5.5 mm
Explosive power	F.4	"not Low"	Average net expansion (cm ³): 18
SADT	H.4	50°C	500 ml Dewar vessel
Competent Authority approval number	<i>Example from UN Recommendations on the Transport of Dangerous Goods, Manual of Tests and Criteria</i>		

Reference to other ECHA Guidance Documents

A template data set does not currently exist in IUCLID for the hazard class 'self-reactive substances'. As long as there is no specific section available in IUCLID the test results in IUCLID section 4.23 'Additional physico-chemical information' under the endpoint title 'Self-reactive substances' should be inserted. In the CSR the information should be included under flammability.

Further information / references

Background information and guidance on classification testing, additional testing and available information is given in the [Guidance on the Application of the CLP criteria](#), section 2.8.

R.7.1.10.5 Pyrophoric liquids

Definition

The definition of a pyrophoric liquid is given in the section 2.9.1 of Annex I to CLP Regulation:

'Pyrophoric liquid means a liquid substance or mixture which, even in small quantities, is liable to ignite within five minutes after coming into contact with air.'

Background information and guidance on the definition is given in the [Guidance on the Application of the CLP criteria](#), sections 2.9.1 and 2.9.2.

Classification criteria and relevant information

Classification principles are given in CLP Regulation Annex I, section 2.9.2.

The criterion for a pyrophoric liquid is as follows: *'The liquid ignites within 5 min when added to an inert carrier and exposed to air, or it ignites or chars a filter paper on contact with air within 5 min.'*

Background information and guidance on relevant aspects regarding the classification is given in the [Guidance on the Application of the CLP criteria](#), sections 2.9.1, 2.9.2, 2.9.3 and 2.9.4.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Other flammability tests do not have to be performed as well as the determination of the self-ignition temperature, if the substance is a pyrophoric substance. However, flammability in contact with water may be relevant.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

The UN Test N.3 of the UN-MTC is also used for classification according to the regulations on the transport of dangerous goods (ADR and RID). If the liquid in question has been classified as belonging to Class 4.2, packing group I of the ADR/RID on the basis of UN Test N.3 results, it is a pyrophoric liquid according to CLP criteria. Packing group I of the ADR/RID directly corresponds to Category 1 of the CLP.

According to the DSD, the A.13 method of Regulation (EC) 440/2008 is used for the assessment of pyrophoric properties for liquids and liquids. This method is identical to the UN Test N.3.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR

Application of (Q)SAR is not possible, however assessment of the chemical structure may be used to exclude pyrophoric properties of a substance. Such an assessment of chemical structure, in conjunction with experience in manufacture and handling, could also formally form part of a weight-of-evidence argument.

- Grouping of substances and read-across approach

Assessment of the chemical structure may be used to anticipate pyrophoric properties of a substance.

- Testing is technically not possible

Testing should always be considered if none of the waiving possibilities applies. Due to pyrophoric properties a number of other tests on physicochemical, toxicological and ecotoxicological endpoints cannot be conducted.

Further adaptation possibilities

Not foreseen.

Impurities; uncertainties

More background information and guidance on this and other aspects is given in the [Guidance on the Application of the CLP criteria](#), section 2.9.

How to conclude on the DSD classification

Because the test methods of DSD and CLP are identical for this endpoint there is no difference in classification, see also the [Guidance on the Application of the CLP criteria](#), section 2.9.6.

Endpoint specific information in the registration dossier IUCLID

Material and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied.

Note that in this case the experience in handling may be sufficient.

Results and discussion

- whether ignition occurs when poured or whether the filter paper is charred;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on flammability can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.13	VII 7.10	Flammability	E.4.14	3.12

Further information / references

Background information and guidance on classification testing, additional testing and available information is given in the [Guidance on the Application of the CLP criteria](#), section 2.9.

R.7.1.10.6 Pyrophoric solids

Definition

The definition of a pyrophoric solid is given in CLP Regulation Annex I, section 2.10.1.

'Pyrophoric solid means a solid substance or mixture which, even in small quantities, is liable to ignite within five minutes after coming into contact with air.'

Background information and guidance on the definition is given in the [Guidance on the Application of the CLP criteria](#), sections 2.10.1 and 2.10.2.

Classification criteria and relevant information

Classification principles are given in CLP Regulation Annex I, section 2.10.2.

The criterion for a pyrophoric solid is as follows: *'The solid ignites within 5 minutes of coming into contact with air.'*

Background information and guidance on relevant aspects regarding the classification is given in the [Guidance on the Application of the CLP criteria](#), sections 2.10.1, 2.10.2, 2.10.3 and 2.10.4.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Other flammability tests do not have to be performed in addition to the determination of the self-ignition temperature, if the substance is a pyrophoric substance. However, flammability in contact with water may be relevant.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

The UN Test N.2 of the UN-MTC is also used for classification according to the regulations on the transport of dangerous goods (ADR and RID). If the solid in question has been classified as belonging to Class 4.2, packing group I of the ADR/RID on the basis of UN Test N.2 results, it is a pyrophoric solid according to CLP Regulation criteria. Packing group I of the ADR/RID directly corresponds to Category 1 of CLP.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

According to the DSD, the A.13 method of Regulation (EC) 440/2008 is used for the assessment of pyrophoric properties for solids and liquids. This method is identical to the N.2 test method.

- **Weight of evidence**

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- **(Q)SAR**

Application of (Q)SAR is not possible, however assessment of the chemical structure may be used to exclude pyrophoric properties of a substance. Such an assessment of chemical structure, in conjunction with experience in manufacture and handling, could also formally form part of a weight-of-evidence argument.

- **Grouping of substances and read-across approach**

Assessment of the chemical structure may be used to anticipate pyrophoric properties of a substance.

- **Testing is technically not possible**

Testing should always be considered if none of the waiving possibilities applies. Due to pyrophoric properties a number of other tests on physicochemical, toxicological and ecotoxicological endpoints cannot be conducted.

Further adaptation possibilities

Not foreseen.

Impurities; uncertainties

Particle size may play an important role. More background information and guidance on this and other aspects is given in the [Guidance on the Application of the CLP criteria](#), section 2.10.

How to conclude on the DSD classification

Because the test methods of DSD and CLP Regulation are identical for this endpoint there is no difference in classification, see also the [Guidance on the Application of the CLP criteria](#), section 2.10.6.

Endpoint specific information in the registration dossier/in IUCLID

Material and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied;
- particle size and distribution (if practicable);

Note that in this case experience in handling may be sufficient.

Results and discussion

- whether ignition occurs when poured;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on flammability can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.13	VII 7.10	Flammability	E.4.14	3.12

Further information / references

Background information and guidance on classification testing, additional testing and available information is given in the [Guidance on the Application of the CLP criteria](#), section 2.10.

R.7.1.10.7 Self-heating substances and mixtures

Definition

For solids and liquids adsorbed onto a large surface, self-heating may occur by reaction with air with subsequent ignition. According to the section 2.11.1.1 of Annex I to CLP Regulation:

'A self-heating substance or mixture is a liquid or solid substance or mixture, other than a pyrophoric liquid or solid, which, by reaction with air and without energy supply, is liable to self-heat; this substance or mixture differs from a pyrophoric liquid or solid in that it will ignite only when in large amounts (kilograms) and after long periods of time (hours or days).'

Classification criteria and relevant information

Self-heating substances and mixtures are classified in two categories according to the criteria of the CLP Regulation (see section 2.11, table 2.11.1). In general, self-heating occurs only for solids in contact with air. The [Guidance on the Application of the CLP criteria](#), section 2.11 gives detailed background information about this phenomenon.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of the REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for self-ignition temperature.

'The study does not need to be conducted:

- if the substance is explosive or ignites spontaneously with air at room temperature; or*
- for liquids non flammable in air, e.g. no flash point up to 200°C, or*
- for gases having no flammable range, or*
- for solids, if the substance has a melting point < 160°C, or if preliminary results exclude self-heating of the substance up to 400°C.'*

The first indent specifies that no data is required for substances which is explosive or ignites spontaneously with air at room temperature.

Second and third indent are not applicable for this endpoint.

With regards to fourth indent, for the purposes of REACH, no data are required for solids classified as:

- pyrophoric; or
- explosive, unstable or division 1.1 to 1.6; or
- organic peroxide; or
- self-reactive substance.

Further, no data are required for substances with a melting point below 160°C. This means also that liquids do not have to be tested for this endpoint for the purposes of this regulation. Annex VII of REACH also allows waiving '*if preliminary results exclude self-heating of the substance up to 400°C*'. This refers to Test Method Regulation 440/2008, method A.16. However, the criteria are not very clear, and therefore it is recommended to instead refer to the CLP Regulation classification criteria, if applicable, and to waive otherwise.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

Literature data – even if available – should not be used since self-heating strongly depends on particle size, surface treatment and other parameters.

The use of existing data is possible provided that the test has been carried out by a qualified institution. If available data from a test according to method A.16 indicate that a classification as a self-heating substance does not apply, no more testing is necessary. However, the interpretation of the A.16 test method data in terms of the CLP criteria requires appropriate expert knowledge.

- Weight of evidence

For the determination of the self-heating substances and mixtures, weight of evidence is not possible.

- (Q)SAR

At present (Q)SAR is generally not applicable for self-heating substances and mixtures. Application of QSAR is not possible.

- Grouping of substances and read-across approach

At present grouping and read-across are not applicable.

- Testing is technically not possible

In some cases, exothermic decomposition may occur when performing the test, and special care will be necessary with respect to performing the tests and interpreting the results; see the

[Guidance on the Application of the CLP criteria](#), section 2.11.4.4.3. In such cases, it may not be possible to determine these properties.

Further adaptation possibilities

According to the UN-MTC, the classification procedure for self-heating substances or mixtures need not be applied if the results of a screening test can be adequately correlated with the classification test and an appropriate safety margin is applied. Examples of screening tests are:

- a. the Greuer Oven test (VDI guideline, 1990) with an onset temperature 80 K above the reference temperature for a volume of 1 litre;
- b. the Bulk Powder Screening Test (Gibson *et al.*, 1985) with an onset temperature 60 K above the reference temperature for a volume of 1 litre.

As stated in Annex IX of REACH, when for certain endpoints, it is proposed to not provide information for other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used. The [Guidance on the Application of the CLP criteria](#), section 2.11.4.2 should be consulted for details about waiving and screening criteria.

Impurities; uncertainties

Particle size may play an important role. More background information and guidance on this and other aspects is given in the [Guidance on the Application of the CLP criteria](#), section 2.11.

How to conclude on the DSD classification

This hazard class is not defined in DSD, therefore translation is not possible.

Endpoint specific information in the registration dossier/in IUCLID

Material and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied;
- indicate if preliminary and/or main test performed;
- moisture content;
- particle size and distribution (if available).

Results and discussion

- indicate temperature rise obtained for the individual tests and classification result.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on flammability can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.13	VII 7.10	Flammability	E.4.14	3.12

Further information / references

ECHA guidance document *the [Guidance on the Application of the CLP criteria](#)* gives in section 2.11 detailed information on the self-heating property, the CLP-classification, the relevant test method and the relation to the DSD and the UN-RTDG.

VDI guideline 2263, part 1 (1990): 'Test methods for the Determination of the Safety Characteristics of Dusts'.

Gibson, N. Harper, D.J. Rogers (1985): 'Evaluation of the fire and explosion risks in drying powders', *Plant Operations Progress*, 4 (3), 181-189.

R.7.1.10.8 Substances which in contact with water emit flammable gases

Definition

The CLP Regulation, Annex I, section 2.12.1 provides the following definition:

'Substances or mixtures which, in contact with water, emit flammable gases means solid or liquid substances or mixtures which, by interaction with water, are liable to become spontaneously flammable or to give off flammable gases in dangerous quantities.'

Classification criteria and relevant information

Classification according to the CLP Regulation is required if the gas produced upon contact with water ignites spontaneously and/or if the reaction rate with which the flammable gas is produced is ≥ 1 l/kg h.

If the gas produced ignites spontaneously, this does not necessarily imply that the gas produced is pyrophoric but this generally is the case if the heat of reaction is sufficient to result in ignition of the gas.

The test method for classification of substances and mixtures which in contact with water emit flammable gases is described in the UN-MTC (UN Test N.5, see Section 33.4). This method is referred to in Annex I, Part 2 of the CLP Regulation and it is strongly recommended to use this method and not to apply test method A.12 of the Test Methods Regulation if new testing is carried out. UN Test N.5 foresees dividing into three categories depending on the violence and rate of the reaction whereas test method A.12 does not allow any further dividing of the substances. Furthermore, the results of both methods might differ slightly due to some differences in the testing procedure (for these differences see the [Guidance on the Application of the CLP criteria](#), Section 2.12.6). Therefore unnecessary testing can be avoided by applying only UN Test N.5 because it leads to more detailed information (and has in any case to be applied for other purposes such as classification and transport).

Data which is based on the classification according to DSD may be available. There are, however, differences between the methods UN Test N.5 and A.12 which should be considered. They are described in detail in the [Guidance on the Application of the CLP criteria](#), section 2.12.6.

Detailed guidance on the test method itself can be found in the [Guidance on the Application of the CLP criteria](#), section 2.12.4.4.1.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for flammability.

'The study does not need to be conducted:

- if the substance is a solid which possesses explosive or pyrophoric properties. These properties should always be considered before considering flammability; or*
- for gases, if the concentration of the flammable gas in a mixture with inert gases is so low that, when mixed with air, the concentration is all time below the lower limit; or*
- for substances which spontaneously ignite when in contact with air.'*

The first point is valid with regard to explosive substances because they are not classified as substances which in contact with water emit flammable gases. In that case testing can be waived.

The other waiving possibilities are not applicable with regard to substances which in contact with water emit flammable gases.

The first point is not correct with regard to pyrophoric substances because pyrophoric substances can be classified as substances which in contact with water emit flammable gases based on UN Test N.5 which is referred to by CLP. UN Test N.5 explicitly requires testing of pyrophoric substances under nitrogen (see UN-MTC, section 33.4.1.3.1).

The second point is not applicable because gases do not fall under the hazard class of substances which in contact with water emit flammable gases.

For the same reasons, the last point (waiving would be possible for substances which spontaneously ignite when in contact with air) is also not valid in this case.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH*

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence*

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR*

There are currently no QSPR models for predicting whether a substance in contact with water emits flammable gases and if so what the gas evolution rate is.

- Grouping of substances and read-across approach*

At present grouping and read-across are not applicable.

- Testing is technically not possible*

Testing should always be possible if none of the waiving possibilities applies. If the substance is known to be soluble in water to form a stable solution, or if it is clearly known that it does

not react with water, e.g. because it is manufactured or washed with water, testing is not necessary.

Further adaptation possibilities

Classification in certain hazard classes do not foresee the assignment of further physical hazard classes or at least normally do not match with classification in this hazard class:

Substances that are classified as explosives, self-reactives or organic peroxides are not classified in this hazard class (or any other physical hazard class). For explosives this is considered through the first point of the adaptation possibilities according to REACH Annex VII, column 2 (see above).

Oxidizing substances are generally not considered for flammability and therefore are also not classified in this hazard class (there may be some exceptions, however).

As stated in Annex IX of REACH, when for certain endpoints, it is proposed to not provide information for other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used.

Impurities; uncertainties

The descriptions of the methods UN Test N.5 and A.12 are not very detailed and therefore allow for technical variations such as with regard to the apparatus used or the procedure. In particular, the testing protocol does not prescribe a specific method for measuring the gas evolution rate. An interlaboratory comparison for this test method has shown that laboratories - based on the freedom the description of the test methods gives - apply different approaches when performing this test. Furthermore, the interlaboratory comparison showed that the test results vary in a rather wide range. It therefore has to be kept in mind that this test method has a non-negligible uncertainty with regard to trueness and precision. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

Sea water may be a particular case of interest (in case of maritime transport).

How to conclude on the DSD classification

Substances which in contact with water emit flammable gases would be classified as 'F; R15' under DSD (the sum of categories 1 to 3 corresponds to 'F; R15').

Endpoint specific information in the registration dossier/in IUCLID

Material and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied;
- particle size and distribution.

Results and discussion

- indicate whether full test was performed or whether it was terminated at a particular step/stage;
- substance identity of evolved gas;
- indicate whether the gas evolved ignites spontaneously;

- rate of gas evolution (unless the test has been terminated);
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on flammability is found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.13	VII 7.10	Flammability	E.4.14	3.12

Further information / references

The ECHA document [Guidance on the Application of the CLP criteria](#) gives in its section 2.12 detailed information on substances and mixtures which, in contact with water, emit flammable gases, their CLP-classification, the relevant test method and the relation to the DSD and the transport of dangerous goods regulations.

Janès *et al.*, 'Towards the improvement of UN N.5 test method intended to the characterization of substances which in contact with water emit Flammable Gases', submitted in revised form to the Journal of Loss Prevention in the Process Industries.

Interlaboratory test on the method UN Test N.5 / EC A.12 'Substances which, in contact with water, emit flammable gases' 2007, Kunath, K., Lüth, P., Uhlig, S., ISBN 978-3-9814634-1-5, http://www.bam.de/de/service/publikationen/publikationen_medien/short_report_rv_un_n_5.pdf.

R.7.1.10.9 Organic peroxides

In the Dangerous Substances Directive (DSD) (67/548/EEC) organic peroxides were classified on the basis of their chemical structure either as explosive or as oxidising. In general, organic peroxides have only weak oxidising properties or do not show oxidizing properties at all. In the CLP Regulation organic peroxides are a distinct hazard class. Organic peroxides are classified in one of the seven categories of 'Types A to G' according to the classification criteria given in Section 2.15.2 of Annex I, of CLP.

As mentioned below under sub-section Definition, organic peroxides are excluded from testing as explosives according to Test Series 1 to 8 in Part I of the UN-MTC (see [R.7.1.11.1 Explosives](#)). In Test Series A to H however, no tests on sensitivity to impact (solids and liquids) and friction (solids only) are included. For the risk assessment and the safe use and handling, data according to the EU test method A.14 as described in Regulation (EC) No 440/2008, if available, or UN Test 3 (a) (ii) BAM Fallhammer and Test 3 (b) (i) BAM friction apparatus (see [R.7.1.11](#)) should be part of the hazard communication in the registration dossier (REACH Annex VII, 7.11) and in the safety data sheet.

Definition

The definition of an organic peroxide is given in CLP Annex I, section 2.15.1:

'Organic peroxides means liquid or solid organic substances which contain the bivalent -O-O- structure and may be considered derivatives of hydrogen peroxide, where one or both of the hydrogen atoms have been replaced by organic radicals. The term organic peroxide includes organic peroxide mixtures (formulations) containing at least one organic peroxide. Organic peroxides are thermally unstable substances or mixtures, which can undergo exothermic self-accelerating decomposition. In addition, they can have one or more of the following properties:

- (i) be liable to explosive decomposition;*
- (ii) burn rapidly;*
- (iii) be sensitive to impact or friction;*
- (iv) react dangerously with other substances.*

An organic peroxide is regarded as possessing explosive properties when in laboratory testing the mixture (formulation) is liable to detonate, to deflagrate rapidly or to show a violent effect when heated under confinement.'

Background information and guidance on the definition is given in [Guidance on the Application of the CLP criteria](#), sections 2.15.1 and 2.15.2.

Classification criteria and relevant information

The Classification principles are given in CLP Annex I, sections 2.15.2 and 2.15.4. Background information and guidance on relevant aspects regarding the classification is given in [Guidance on the Application of the CLP criteria](#), sections 2.15.3, 2.15.4, 2.15.5, 2.15.6 and 2.15.7.

Adaptation of the standard testing regime

[Adaptation possibilities according to column 2 of Annex VII to REACH](#)

Only organic peroxides, as defined in CLP, Annex I, section 2.15.1 definition, have to be tested according to the UN-MTC, Part II test series A - H.

CLP Annex I, section 2.15.2.1. provides the following specific rules for adaptation of the standard information requirement for organic peroxides.

'Any organic peroxide shall be considered for classification in this class, unless it contains:

- (a) not more than 1.0% available oxygen from the organic peroxides when containing not more than 1.0% hydrogen peroxide; or*
- (b) not more than 0.5% available oxygen from the organic peroxides when containing more than 1.0% but not more than 7.0% hydrogen peroxide.*

 **NOTE:** The available oxygen content (%) of an organic peroxide mixture is given by the formula:

$$16 \times \sum_i^n \left(\frac{n_i \times c_i}{m_i} \right)$$

where:

n_i = number of peroxygen groups per molecule of organic peroxide i ;

c_i = concentration (mass %) of organic peroxide i ;

m_i = molecular mass of organic peroxide i .

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

A number of already tested and classified substances and mixtures are listed in the UN-RTDG, 2.5.3.2.4.

Available information may especially originate from the classification for transport. In the DSD organic peroxides were classified as oxidizing substances, by definition. More details are described in the [Guidance on the Application of the CLP criteria](#), sections 1.7.2.1 and 2.15.6. If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence

For the determination of the organic peroxides, weight of evidence is not possible. Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR

At present QSAR is generally not applicable for organic peroxides. Application of (Q)SAR is not possible.

- Grouping of substances and read-across approach

At present grouping and read across are not applicable.

- Testing is technically not possible

A number of substances can, for safety reasons, only be handled and tested in diluted form, see the substances and mixtures listed in UN TDG, 2.5.3.2.4. Testing should always be considered if none of the waiving possibilities applies.

Further adaptation possibilities

Not foreseen.

Impurities; uncertainties

Minor impurities can have an influence on thermal stability. Background information and guidance on these aspects is given in [Guidance on the Application of the CLP criteria](#), section 2.15.4.

How to conclude on the DSD classification

In the DSD organic peroxides are classified as oxidizing substances and a few of them as having explosive properties.

Endpoint specific information in the registration dossier/in IUCLID

Material and methods

- See UN MTC, Part II, classification procedures and test series A-H.

Results and discussion

The following data on organic peroxides should be submitted:

- if testing is waived, the reasons for waiving must be documented in the dossier;
- type of organic peroxide;
- SADT (Self accelerating decomposition temperature) together with the volume the SADT related to;
- detonation properties (Yes/Partial/No);
- deflagration properties (Yes rapidly/Yes slowly/No);
- effect of heating under confinement (Violent/Medium/Low/No);
- explosive power, if applicable (Not low/Low/None).

The following example ([Figure R.7.1–3](#)) shows how data mentioned above could be documented in the CSR:

Figure R.7.1–3 Example: Di-tert-butyl peroxide

UN Test Series A to H	Test method	Results + Evaluation	Remarks
Propagation of detonation	A.1	"No"	Fragmented length (cm): 16
Propagation of deflagration #1	C.1	"Yes, slowly "	Maximum pressure (kPa): > 2070 Time for a pressure rise from 690 to 2070 kPa (ms): 100
Propagation of deflagration #2	C.2	"No"	deflagration rate (mm/s): 0.27
Effect of heating under defined confinement #1	Koenen E.1	"No"	Limiting diameter (mm): < 1.0 Type of fragmentation (and pieces): O
Effect of heating under defined confinement #2	DPVT E.2	"Medium"	Limiting diameter (mm): 3.5
Explosive power	F.3	"Not Low"	Expansion (cm ³ /10 g test sample): 28
Explosive power	F.4	"Not Low"	Average net expansion (cm ³): 12
SADT	H.4	80°C	500 ml Dewar vessel
Competent Authority approval number	<i>Example from UN Recommendations on the Transport of Dangerous Goods, Manual of Tests and Criteria</i>		

For assigning the Type of organic peroxide, the list of currently assigned organic peroxides according to section 2.5.3.2.4 of the UN RTDG can be used, in case the assignment was based on a test according to the UN MTC. The relevant underlying test data may be collected from the respective UN documents from the UN Committee of experts on the

transport of dangerous goods, from test reports produced by either competent authorities or industry, or from other reliable sources (such as e.g. the dedicated database 'DATATOP').

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

A Template data set in IUCLID does not exist for the hazard class 'organic peroxides'. As long as there is no specific section in IUCLID the test results in section 4.23 'Additional physico-chemical information' should be inserted under the endpoint title 'organic peroxides'. The information on organic peroxides should not be included in IUCLID section 4.15 'Oxidising properties'. In the registration dossier the information should be included under flammability.

Further information / references

Background information and guidance on classification testing, additional testing and available information is given in [Guidance on the Application of the CLP criteria](#), section 2.15.

Data from the 'DATATOP' database can be obtained from the gatekeeper of this database TNO, Department Energetic Materials, Lange Kleiweg 137, 2288GJ, Rijswijk The Netherlands.

Various national guidelines which provide guidance and outline safe standards for handling and storage of organic peroxides for the assignment of organic peroxides to storage groups are available e.g. Netherlands Directive: Publication Series on Dangerous Substances 8 (PGS 8) Storage of Organic Peroxides, UK HSE: The storage and handling of organic peroxides - Guidance Note CS21 or German guideline: BGV B4.

R.7.1.11 Explosive properties

Some of the information requirements according to the REACH Regulation, Annex VII were phrased such that they correspond to 'indications of danger' as given in Annex II of DSD. For substances, classification and labelling according to the CLP Regulation has been mandatory since 1 December 2010 (and will become mandatory for mixtures (preparations) from 1 June 2015, when the DSD and DPD will be repealed). Consequently, explosive properties are covered by classification of the substance according to the CLP Regulation. However, the physical hazards according to CLP are structured completely differently from the physico-chemical properties according to the DSD (and therefore also REACH, Annex VII). This means that for some of the CLP hazard classes an unambiguous assignment to one of the headlines (information requirements) in Annex VII to REACH is not possible. The assignment of hazard classes to the headline 'Explosive properties' as shown in Table below ([Table R.7.1-8](#)) must therefore only be understood as a means to structure this document in accordance with Annex VII to REACH. It has to be noted that self-reactive substances and organic peroxides are primarily assigned to the headline 'Flammability' and only a cross reference to corresponding sub-chapter under heading 'Flammability' is included in the sub-chapters on 'Explosive properties' below because these two hazard classes can have explosive and/or flammable properties.

Table R.7.1-8 Assignment of CLP hazard classes to the information requirement 'Explosive properties' according to REACH, Annex VII and correlation between the Test method Regulation and the test method according to CLP and supporting link with the [Guidance on the Application of the CLP criteria](#).

Information requirement according to Art. 10 (a) (vi) of the REACH Regulation (EC) No. 1907/2006 (the no. in brackets is the respective no. in the table in Annexes VII to IX to REACH)	CLP Regulation (EC) No. 1272/2008 (the no. in brackets is the respective chapter no. in Annex I to CLP)	Chapter in revised R.7(a) guidance	Corresponding test method according to the Test Method Regulation (EC) No. 440/2008	Corresponding test method according to the CLP Regulation	Chapter in the Guidance on the Application of the CLP Criteria (ex RIP 3.6)
Explosive properties (7.11)	Explosives (2.1)*	R.7.1.11.1	A.14 Explosive properties	UN Test series 1 to 3 (further test series 4 to 6 are necessary for classification)	2.1
	Self-reactive substances and mixtures (2.8)*	R.7.1.11.2 See R.7.1.10.4	n.a.	A.14 (existing data only)	2.8
	Organic peroxides (2.15)*	R.7.1.11.3 See R.7.1.10.9	n.a.	A.14 (existing data only)	2.15

* Note that regardless of whether the hazard class or category is listed in Article 14(4)(a) REACH the chemical safety assessment must be performed in accordance with Article 14 (3) of REACH. Furthermore, according to Article 10(a)(iv) of REACH the technical dossier of a registration of a substance under the REACH Regulation must include information on classification and labelling of the substance as specified in section 4 of Annex VI to the REACH Regulation.

In addition, it has to be noted that some substances have explosive properties which do not result in classification. Examples are the following:

- substances with a positive result in UN Test Series 1 or 2 but which are exempted from the classification as explosives based on their packaging in UN Test Series 6;
- substances which are mechanically sensitive only. These are substances with a sensitiveness to impact (determined by UN Test Series 3 (a) (ii)) of 40 J or less and/or a sensitiveness to friction (determined by Test Series 3 (b) (i)) of 360 N or less for substances and mixtures which may have explosive properties based on the screening procedure according to Appendix 6, Part 3 of the UN-MTC and which are not classified as explosives, self-reactive or organic peroxide.

Such substances may be classified in other hazard classes (e.g. as flammable solids, oxidizing solids, corrosive to metals) or even not at all. Information about such explosive properties should be indicated in the dossier as well.

R.7.1.11.1 Explosives

Please note that explosive atmospheres as, for example, created by flammable liquids and by powders are not the subject of this chapter.

Definition

The following definitions are provided in CLP Annex I, section 2.1.1:

'An explosive substance or mixture is a solid or liquid substance or mixture of substances which is in itself capable by chemical reaction of producing gas at such a temperature and pressure and at such a speed as to cause damage to the surroundings. Pyrotechnic substances are included even when they do not evolve gases.'

A pyrotechnic substance or mixture is a substance or mixture of substances designed to produce an effect by heat, light, sound, gas or smoke or a combination of these as the result of non-detonative self-sustaining exothermic chemical reactions.

An unstable explosive is an explosive substance or mixture which is thermally unstable and/or too sensitive for normal handling, transport and use.

An explosive article is an article containing one or more explosive substances or mixtures.

A pyrotechnic article is an article containing one or more pyrotechnic substances or mixtures.

An intentional explosive is a substance, mixture or article which is manufactured with a view to producing a practical, explosive or pyrotechnic effect.'

Organic Peroxides and Self Reactive Substances may also have explosive properties and should be screened. See chapter [R.7.1.11.3](#) for Organic peroxides and chapter [R.7.1.11.2](#) for Self Reactive Substances and Mixtures.

Intentional explosive

Council Directive 93/15/EEC of 5 April 1993 lays down rules for the harmonisation of the provisions relating to the placing on the market and supervision of explosives for civil uses.

Directive 2007/23/ EC on the placing on the market of pyrotechnic articles establishes rules designed to achieve the free movement of pyrotechnic articles in the internal market while, at the same time, ensuring a high level of protection of human health and public security and the protection and safety of consumers and taking into account the relevant aspects related to

environmental protection. Pyrotechnic articles (CLP, Annex I, Section 2.1.1.2) are classified as explosives for CLP and as class 1 for transport (see UN-RTDG). According to Article 9 and Annex II of Directive 2007/23/EC the conformity assessment procedures are carried out by notified bodies, which have to issue an EC type-examination certificate to the applicant. All data included in the EC type-examination certificate are sufficient for the information requirements under the REACH Regulation.

Classification criteria and relevant information

Substances, mixtures and articles of this class are classified as an unstable explosive on the basis of the flowchart in Annex I to CLP Regulation, Figure 2.1.2. The test methods are described in Part I of the UN-MTC.

Explosives, which are not classified as an unstable explosive, must be classified in one of the six Divisions referred to in paragraph 2.1.2.2 of Annex 2.1 to the CLP Regulation, based on the results of the tests laid down in Table 2.1.1 on Test Series 2 to 8 in Part I of the UN-MTC. If explosives are unpackaged or repacked in packaging other than the original or similar packaging, they must be retested. If a substance gives a positive result in any of the test series 1 or 2 this should be mentioned in the REACH registration dossier for the substance, even if it would not be classified as an 'Explosive' in Test Series 6.

The test methods used for deciding on provisional acceptance into the class of explosives are grouped into four series, numbered 1 to 4 (see CLP Annex I, Figure 2.1.2).

It may be important for the safety of testers that certain tests, using small amounts of material, be conducted first before proceeding to test with larger quantities. Therefore it is highly recommended to start the testing procedure with Test Series 3, because these tests involve relatively small sample sizes, which reduces the risk to personnel.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for explosive properties.

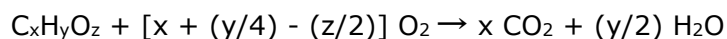
'The study does not need to be conducted if:

- *there are no chemical groups associated with explosive properties present in the molecule, or*
- *the substance contains chemical groups associated with explosive properties which include oxygen and the calculated oxygen balance is less than -200, or*
- *the organic substance or a homogenous mixture of organic substances contains chemical groups associated with explosive properties, but the exothermic decomposition energy is less than 500 J/g and the onset of exothermic decomposition is below 500°C, or*
- *for mixtures of inorganic oxidising substances (UN Division 5.1) with organic materials, the concentration of the inorganic oxidising substance is:*
 - *less than 15%, by mass, if assigned to UN Packaging Group I (high hazard) or II (medium hazard),*
 - *less than 30%, by mass, if assigned to UN Packaging Group III (low hazard).*

 *Note: Neither a test for propagation of detonation nor a test for sensitivity to detonative shock is required if the exothermic decomposition energy of organic materials is less than 800 J/g.'*

 **Note on the use of the Oxygen Balance:**

The oxygen balance is calculated for the chemical reaction:



Using the formula:

$$\text{Oxygen balance} = -1600 [2x + (y/2) - z] / \text{molecular weight};$$

The oxygen balance was developed for compounds containing only nitrate groups and it applies only to organic substances. Extending its use to molecules with other oxygen containing groups should be done with care. As an example the presence of hydroxyl-groups will strongly affect the oxygen balance towards higher values, whereas this group does not contribute to explosive properties. In addition the presence of for instance halogens tends to decrease the flammability and explosivity but this is not accounted for.

Please also check Appendix 6, Section 3 of the UN-MTC.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met. Application of weight of evidence is possible with substances where explosive properties can clearly be excluded. Weight of evidence should be accompanied with extensive and reliable literature references.

- (Q)SAR

There is currently no QSPR/(Q)SAR software known with sufficient accuracy and reliability to assist in assessing (potential) explosive properties. DSC testing is cheap and fast and is strongly recommended to identify potential hazards connected with the substance.

- Grouping of substances and read-across approach

An assessment of chemical structure would formally form part of a column 2 waiver. For further information please refer to the [Guidance on the Application of the CLP criteria](#), Part 2: Physical Hazards, Section 2.1 Explosives.

- Testing is technically not possible

Testing should always be considered if none of the waiving possibilities applies. Testing for explosives may be omitted if it is technically not possible to conduct the study as a consequence of the properties of the substance.

Further adaptation possibilities

Testing may be waived if there are no chemical groups associated with explosive properties present in the molecule. The potential generation of explosive atmospheres by flammable gases/liquids or combustible solids is not considered an explosive property and should therefore not be reported under this heading.

As stated in Annex IX of REACH, when for certain endpoints, it is proposed to not provide information for other reasons than those mentioned in column 2 of that Annex or in Annex XI of REACH, this fact and the reasons must also be clearly stated. Such an approach may then be used.

Impurities; uncertainties

Small amounts of other compounds may enhance or suppress the chemical reaction that gives the explosive property to a substance. Therefore impurities may considerably influence the explosive properties of a substance. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

How to conclude on the DSD classification

For DSD explosives are substances and preparations which may explode under the effect of flame or which are more sensitive to shocks or friction than dinitrobenzene.

Reclassification of substances classified as explosive according to DSD:

Under the regime of the old DSD, testing of explosive properties was achieved by performing test method A.14. For classification purposes under the CLP Regulation this test is not adequate in the case of a negative result for thermal sensitivity. The test method A.14 stops with a limiting diameter of 2 mm, while UN Test E.1 proceeds to down to a 1 mm orifice. Testing according to the CLP Regulation is the same as that described in Part I of the UN-MTC. This is why the translation table of Annex VII of the CLP Regulation states that there is no direct translation possible for classification from (E, R2) and (E, R3) to CLP criteria.

Therefore, if the screening procedure of section 2.1.4.2 of the CLP Regulation identifies a substance or mixture to be a potential explosive, appropriate data are required for classification.

Moreover, if data from performing test method A.14 or the UN Test series 3 tests 3a or 3b indicate that a substance is sensitive to impact or friction such information should be provided in the REACH registration dossier.

Endpoint specific information in the registration dossier/in IUCLID

Material and methods

- reference to the standard and the test method applied;
- description of the substance that was tested.

Results and discussion

- if testing is waived, the reasons for waiving must be documented in the dossier;
- if testing is not waived then the tests done according to the UN Test Manual and the outcome (explosive or not explosive) must be documented in the dossier. The mechanical sensitivity test according to UN Test Series 3a and 3b must be done and documented if UN Test Series 1 or 2 give a positive result. If data according to test method A.14 are available, then the results can be used instead of UN Test series 3a and 3b.

An example is given below ([Figure R.7.1–4](#)) of how summarised results from the application of the class 1 procedure for the hypothetical substance 'New explosive substance' could be presented.

Figure R.7.1–4 Results from application of the class 1 acceptance procedure

1. Name of substance	New explosive substance
2. General data	2.1 Composition : technically pure 2.2 Physical form : Fine crystalline powder 2.3 Colour : Yellow
3. Box 2	Is the substance manufactured with the view to producing a practical explosive or pyrotechnic effect? 3.1 Answer : No
4. Box 3	4.1 Propagation of Detonation : UN-Test A.1 Result : "-", no propagation of detonation 4.2 Effect of heating under confinement: 4.2.1 Koenen test (test 1(b)) Result : "+", 4.2.2 Time/pressure test (test 1(c)(i)) Result : "-", no effect on ignition under confinement 4.5 Exit : Go to Box 4
5. Box 4	Is it an explosive substance? 5.1 Answer from Test Series 1 : Yes 5.2 Exit : Go to box 5
6. Box 5	6.1 Sensitivity to shock : based on the test result of UN-Test A.1 Result "-" 6.2 Effect of heating under confinement: Koenen test (test 2(b)): limiting diameter 2,5 mm Result: "+" 6.3 Exit : Go to Box 6
7. Box 6 :	Is the substance too insensitive for acceptance into Class 1? 7.1 Answer from Test Series 2 : No 7.2 Conclusion : Substance to be considered for Class 1 (box 8) 7.3 Exit : Go to Box 9

8. Box 9	Test Series 3 8.1 Thermal Stability: based on the DSC measurement data Result: thermally stable 8.2 Impact sensitivity : BAM fallhammer test (test 3(a)(ii)) Result : "-", not too dangerous to transport in form tested 8.3 Friction sensitivity : BAM friction test (test 3(b)(i)) Result : "-", not too dangerous to transport in form tested 8.4 Exit : Go to box 10
9. Box 10	Is the substance thermally stable? 9.1 Answer from DSC data : Yes 9.2 Exit : Go to box 11
10. Box 11	Is the substance too dangerous for transport in the form in which it was tested? 10.1 Answer from Test Series 3 (a)(ii) and 3 (b)(i): No 10.2 Exit : Go to box 18
11. Conclusion	PROVISIONALLY ACCEPT INTO CLASS 1 11.1 Exit : Apply the Class 1 assignment procedure

Figure R.7.1–5 Results from the application of the class 1 assignment procedure

1. Box 19	Is the substance a candidate for Division 1.5? 1.1 Answer : No 1.2 Exit : Go to box 25
2. Box 25	2.1 UN-Tests 6(a) and 6(c) were not conducted because the substance showed no propagation of detonation in the UN-Test A.1 and also no propagation of deflagration in the UN-test 1(c)(ii). 2.2 UN-Test 6 (c) Sample conditions: 1 × 30 kg fibre drum Observations: Only slow burning with black smoke and soot occurred. 2.3 Exit : Go to box 26
3. Box 26	Is the result a mass explosion? 3.1 Answer from Test Series 6 : No 3.2 Exit : Go to box 28
4. Box 28	Is the major hazard that from dangerous projections? 4.1 Answer from Test Series 6 : No 4.2 Exit : Go to box 30
5. Box 30	Is the major hazard radiant heat and/or violent burning but with no dangerous blast or projection hazard? 5.1 Answer from Test Series 6 : No 5.2 Exit : Go to box 32

6. Box 32	Is there nevertheless a small hazard in the event of ignition or initiation? 6.1 Answer from Test Series 6 : No 6.2 Exit : Go to box 35
7. Box 35	Is the substance or article manufactured with the view to producing a practical explosive or pyrotechnic effect? 7.1 Answer : No 7.2 Exit : Go to box 38
8. Conclusion	NOT CLASS 1 8.1 Exit : Consider for another class/division

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on explosiveness can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.14	VII 7.11	Explosiveness	E.4.15	3.13

Further information / references

Further information about classification and testing for explosives can be found in the [Guidance on the Application of the CLP criteria](#), section 2.1.

Gharagheizi F. Quantitative structure-property relationship for prediction of the lower flammability limit of pure compounds. Energy & Fuels 22 (2008) 3037-3039.

Gharagheizi F. A new group contribution-based model for estimation of lower flammability limit of pure compounds. J. Haz. Mat. 170 (2009a) 595-604.

R.7.1.11.2 Self-reactive substances and mixtures

Self-reactive substances are primarily assigned to the headline 'Flammability' therefore please also refer to chapter [R.7.1.10.4](#).

The sensitivity of self-reactive substances to impact (solids and liquids) and friction (solids only) may be of importance for the safe handling of the substances, in the event that these substances have pronounced explosive properties. If data according to EU test method A.14 as described in Regulation (EC) No 440/ 2008 are available, then this information should be part of the hazard communication in the registration dossier (REACH Annex VII, 7.11).

R.7.1.11.3 Organic peroxides

Organic peroxides are primarily assigned to the headline 'Flammability' therefore please also refer to chapter [R.7.1.10.9](#).

The sensitivity of organic peroxides to impact (solids and liquids) and friction (solids only) may be of importance for the safe handling of the substances, in the event that these substances have pronounced explosive properties. If data according to EU test method A.14 as described in Regulation (EC) No 440/ 2008 are available, then this information should be part of the hazard communication in the registration dossier (REACH Annex VII, 7.11).

R.7.1.12 Self-ignition temperature

The terminology used in Annex VII of REACH is not very precise. Therefore, some guidance in interpretation appears necessary:

For liquids and gases, the term '**auto-ignition**' instead of 'self-ignition' is generally used. Auto-ignitability is of high importance for the assignment of temperature classes in explosion protection (i. e. ATEX in Europe) of plants and equipment.

For solids and liquids adsorbed on a large surface, **self-heating** may occur by reaction with air with subsequent ignition. According to the CLP Regulation, Annex I, section 2.11, a self-heating substance or mixture is a liquid or solid substance or mixture, other than a pyrophoric liquid or solid, which, by reaction with air and without energy supply, is liable to self-heat; this substance or mixture differs from a pyrophoric liquid or solid in that it will ignite only when in large amounts (kilograms) and after long periods of time (hours or days). Therefore solids are considered under self heating substances in the chapter below.

Table R.7.1–9 Assignment of CLP hazard classes to the information requirement 'Self ignition temperature' according to REACH, Annex VII and the Test Method Regulation.

Information requirement according to Art. 10 (a) (vi) of the REACH Regulation (EC) No. 1907/2006 (the no. in brackets is the respective no. in the table in Annexes VII to IX to REACH)	CLP Regulation (EC) No. 1272/2008 (the no. in brackets is the respective chapter no. in Annex I to CLP)	Chapter in revised R.7(a) guidance	Corresponding test method according to The Test Method Regulation (EC) No. 440/2008	Corresponding test method according to CLP Regulation	Chapter in the <i>Guidance on the Application of the CLP Criteria</i> (ex RIP 3.6)
Self ignition temperature (7.12)	For gases and liquids*	R.7.1.12.1	A.15 Auto-ignition temperature (liquids and gases)	n.a.	n.a.
	For solids * Note: the UN Test N.4 is preferable to generate the information for this endpoint. Refer to R.7.1.10.7	R.7.1.12.2 , R.7.1.10.7	A.16 Relative self-ignition temperature for solids	n.a.	Section 2.11

* Note that regardless of whether the hazard class or category is listed in Article 14 (4) (a) of REACH, the chemical safety assessment (when required) must be performed in accordance with Article 14 (3) of REACH. Furthermore, according to Article 10 (a) (iv) of REACH the technical dossier of a registration for a substance under the REACH Regulation must include information on classification and labelling of the substance as specified in section 4 of Annex VI to the REACH Regulation.

R.7.1.12.1 Auto-ignition

Type of property

For liquids and gases, the term '**auto-ignition**' instead of 'self-ignition' is generally used. Auto-ignitability is of high importance for the assignment of temperature classes in explosion protection (i. e. ATEX in Europe) of plants and equipment. In this chapter, only the **auto-ignition** phenomena will be discussed.

Definition

The degree of auto-ignitability is expressed in terms of the auto-ignition temperature. The auto-ignition temperature is the lowest temperature at which the test substance will ignite when mixed with air under the conditions defined in the test method.

Test method(s)

For testing Auto-ignition temperature, method A.15 of Regulation (EC) 440/2008 should be used, which references several national and international standards (e.g. EN 14522, etc.). The test procedure is applicable to gases, liquids and vapours which, in the presence of air, can be ignited by a hot surface.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for self-ignition temperature.

'The study does not need to be conducted:

- if the substance is explosive or ignites spontaneously with air at room temperature; or*
- for liquids non flammable in air, e.g. no flash point up to 200°C; or*
- for gases having no flammable range, or*
- for solids, if the substance has a melting point $\leq 160^{\circ}\text{C}$, or if preliminary results exclude self-heating of the substance up to 400°C.'*

This means:

For gases:

Only gases classified as flammable according to the CLP Regulation have to be considered.

For liquids:

The auto-ignition temperature should be determined according to Directive EC 440/2008, method A.15. No data are required for liquids classified as:

- pyrophoric; or
- explosive, unstable or division 1.1 to 1.6; or
- organic peroxide; or
- self-reactive substance.

Further, the auto-ignition temperature does not have to be determined for liquids having no flash point up to 200°C. In practice, liquids with a boiling point above 350°C will not have a flash point below 200°C. Therefore, determination of the auto-ignition temperature is not necessary in such cases if the flash point is not known.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence

For the determination of the auto-ignition temperature, the weight of evidence approach is not possible. Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR

For the determination of the auto-ignition temperature, (Q)SAR approaches are strongly discouraged for the purpose of classification/ risk assessment.

- Grouping of substances and read-across approach

For the determination of the auto-ignition temperature read-across is usually not possible. However interpolation may still be possible within homologous series.

However, it is not possible to read across from methyl compounds to ethyl and propyl compounds and vice versa.

- Testing is technically not possible

Testing should always be considered if none of the waiving possibilities applies. Substances which decompose below room temperature or which react vigorously with moisture may be difficult to test. In such cases, the test may be waived due to technical reasons.

Further adaptation possibilities

Not foreseen.

Impurities; uncertainties

The auto-ignition temperature can be considerably reduced by the presence of catalytic impurities. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

Endpoint specific information in the registration dossier / in IUCLID

Material and methods

- description of the apparatus or reference to the standard or the test method applied;
- quantity of sample used.

Results and discussion

- the value or the range of the auto-ignition temperature;
- if testing is waived, the reasons for waiving must be documented in the dossier.

For liquids/gases: observations (e.g decomposition with air, reactions with moisture, etc.)

For solids see the below chapter [R.7.1.12.2](#).

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on auto flammability (self-ignition temperature) can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.12	VII 7.12	Auto flammability	E.4.13	3.11

R.7.1.12.2 Self-heating substances

For solids and liquids adsorbed on a large surface, **self-heating** may occur by reaction with air with subsequent ignition. According to the CLP Regulation Annex I, section 2.11 the following definition is provided:

'A self-heating substance or mixture is a liquid or solid substance or mixture, other than a pyrophoric liquid or solid, which, by reaction with air and without energy supply, is liable to self-heat; this substance or mixture differs from a pyrophoric liquid or solid in that it will ignite only when in large amounts (kilograms) and after long periods of time (hours or days).'

The ECHA [Guidance on the Application of the CLP criteria](#) gives in Chapter 2.11 detailed information on the self-heating property, the CLP-classification, the relevant test method and the relation to the DSD and the transport of dangerous goods regulations.

See Section [R.7.1.10.7](#) of this guidance document for further details and information.

R.7.1.13 Oxidising properties

Some of the information requirements according to REACH Annex VII were phrased such that they correspond to 'indications of danger' as given in Annex II of DSD. For substances, classification and labelling according to the CLP Regulation has been mandatory since 1 December 2010 (and will become mandatory for mixtures (preparations) from 1 June 2015, when the DSD and DPD will be repealed). Consequently, information requirements on oxidising properties are inherently covered by classification of the substance according to the CLP Regulation. However, the physical hazards according to CLP Regulation are structured completely differently from the physicochemical properties according to DSD (and therefore also REACH, Annex VII). This means that for some of the CLP hazard classes an unambiguous assignment to one of the headlines (information requirements) in Annex VII to REACH is not possible. The assignment of hazard classes to the headline 'oxidising properties' as shown in the table below ([Table R.7.1-10](#)) must therefore only be understood as a means to structure this document in accordance with Annex VII to REACH.

Table R.7.1-10 Assignment of CLP hazard classes to the information requirement 'Oxidising properties' according to REACH, Annex VII and correlation between the Test method Regulation and the test method according to CLP and supporting link with the [Guidance on the Application of the CLP criteria](#).

Information requirement according to Art. 10 (a) (vi) of the REACH Regulation (EC) No. 1907/2006 (the no. in brackets is the respective no. in the table in Annexes VII to IX to REACH)	CLP Regulation (EC) No. 1272/2008 (the no. in brackets is the respective chapter no. in Annex I to CLP)	Chapter in revised R.7(a) guidance	Corresponding test method according to The Test Method Regulation, Regulation (EC) No. 440/2008	Corresponding test method according to CLP Regulation	Chapter in the Guidance on the Application of the CLP Criteria (ex RIP 3.6)
Oxidising properties (7.13)	Oxidising gases (2.4) *	R.7.1.13.1	n.a.	ISO 10156	2.4
	Oxidising liquids (2.13) *	R.7.1.13.2	A.21 Oxidising properties (liquids)	UN Test O.2	2.13
	Oxidising solids (2.14) *	R.7.1.13.3	A.17 Oxidising properties (solids)	UN Test O.1	2.14

* Note that regardless of whether the hazard class or category is listed in Article 14 (4)(a) of REACH the chemical safety assessment (when required) must be performed in accordance with Article 14 (3) REACH. Furthermore, according to Article 10(a)(iv) of REACH the technical dossier of a registration of a substance under the REACH Regulation must include information on classification and labelling of the substance as specified in section 4 of Annex VI to the REACH Regulation.

R.7.1.13.1 Oxidising gases

Definition

The following definition of oxidising gases is provided in CLP Annex I, section 2.4.1.:

'Oxidising gas means any gas or gas mixture which may, generally by providing oxygen, cause or contribute to the combustion of other material more than air does.'

The criteria *'more than air does'* is further defined in a Note under Table 2.4.1 in Section 2.4.1 as *'having an oxidising power greater than 23.5 % as determined by a method specified in ISO 10156 as amended'*.

Classification criteria and relevant information

All oxidising gases are classified as oxidising gas, Category 1 (Ox. Gas 1, H270). Detailed guidance on the classification criteria and the test method(s) can be found in the [Guidance on the Application of the CLP criteria](#), section 2.4.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for oxidising properties.

'The study does not need to be conducted if:

- the substance is explosive, or*
- the substance is highly flammable, or*
- the substance is an organic peroxide, or*
- the substance is incapable of reacting exothermically with combustible materials, for example on the basis of the chemical structure (e.g. organic substances not containing oxygen or halogen atoms and these elements are not chemically bonded to nitrogen or oxygen, or inorganic substances not containing oxygen or halogen atoms).*

The full test does not need to be conducted for solids if the preliminary test clearly indicates that the test substance has oxidising properties.

Note that as there is no test method to determine the oxidising properties of gaseous mixtures, the evaluation of these properties must be realised by an estimation method based on the comparison of the oxidising potential of gases in a mixture with that of the oxidising potential of oxygen in air.'

According to above indents, the study therefore does not need to be conducted if the gas:

- is classified as highly flammable; or*
- does not contain oxygen, fluorine and/or chlorine which are chemically bonded to elements other than carbon or hydrogen.*

The other above cited indents are not relevant for this endpoint.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

There are not many gases that are oxidising. Most oxidising gases are identified as such in the UN-RTDG and in ISO 10156: 2010 *Gas cylinders - Gases and gas mixtures: - Determination of fire potential and oxidizing ability for the selection of cylinder valve outlets*.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence

There is no known scientific literature that refers to test results for gases that are not classified in ISO 10156 nor in the UN-RTDG.

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR

At present (Q)SAR is generally not applicable for the determination of oxidising limits of gases. Application of (Q)SAR is not possible. However, assessment of the chemical structure may be used to exclude oxidising behaviour of a substance. Possibly, this relation could be exploited in the development of future QSPR methods.

- Grouping of substances and read-across approach

For the determination of the oxidising gases read-across is usually not possible. However interpolation may still be possible within homologous series.

- Testing is technically not possible

Testing should always be considered if none of the waiving possibilities applies.

Further adaptation possibilities

Not foreseen.

Impurities; uncertainties

The normal level of impurities in the technical grade of oxidising gases does not impact the result of the test. Tests should be performed with the lowest concentration of impurities in the gas encountered in the normal manufacturing process and the moisture content should be less than or equal to 0.01 mol%. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

How to conclude on the DSD classification

All gases with a positive test result according to the test method described in ISO 10156 are classified 'Oxidising O, R8'.

Endpoint specific information in the registration dossier/in IUCLID

Material and methods

- reference to the standard applied.

Results and discussion

- if the test is positive indicate that the gas is 'oxidising';
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on oxidising properties can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.15	VII 7.13	Oxidising properties	E.4.16	3.14

Further information / references

Further information about classification and testing for oxidising gases can be found in the ECHA *Guidance on the application of CLP criteria*, section 2.4.

The test method is described in ISO 10156. The test is qualitative. If reaction is observed during the test, the gas to be evaluated is oxidizing.

For several gases, a 'coefficient of oxygen equivalency' (Ci) has been deduced from the explosion ranges observed during the tests. The Ci factors are listed in ISO 10156 along with the list of oxidising gases.

R.7.1.13.2 Oxidising liquids

Definition

The following definition of oxidising liquids is provided in CLP Annex I, section 2.13.1.:

'Oxidising liquid means a liquid substance or mixture which, while in itself not necessarily combustible, may, generally by yielding oxygen, cause, or contribute to, the combustion of other material.'

Classification criteria and relevant information

According to the CLP Regulation, a liquid is classified as an oxidising liquid if, in testing according to the UN Test O.2 of the UN-MTC (Part III, Section 34) it is at least as oxidising as a 65 % aqueous solution of nitric acid. The CLP Regulation has three categories for Oxidising Liquids. The category is also determined through the UN Test O.2, by comparison to various reference oxidisers.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for oxidising properties.

'The study does not need to be conducted if:

- the substance is explosive, or*
- the substance is highly flammable, or*
- the substance is an organic peroxide, or*
- the substance is incapable of reacting exothermically with combustible materials, for example on the basis of the chemical structure (e.g. organic substances not containing oxygen or halogen atoms and these elements are not chemically bonded to nitrogen or oxygen, or inorganic substances not containing oxygen or halogen atoms).*

The full test does not need to be conducted for solids if the preliminary test clearly indicates that the test substance has oxidising properties.

Note that as there is no test method to determine the oxidising properties of gaseous mixtures, the evaluation of these properties must be realised by an estimation method based on the comparison of the oxidising potential of gases in a mixture with that of the oxidising potential of oxygen in air.'

The first indent states that explosive substances should not be tested for oxidising properties. For instance, organic substances with oxidising functional groups may be explosive and should first undergo the screening procedures for explosive properties in Annex 6 of the UN-MTC to rule out possible explosive behaviour. Such substances may also be thermally unstable and show self-reactive behaviour. Substances that have been classified as Explosives according to the CLP Regulation or have been assigned risk phrases R2 or R3 according to the DSD, should normally not be tested for oxidising properties, since they are known to be explosive.

The second indent states that highly flammable substances do not have to be tested for oxidising properties. While it is not very clear what 'highly flammable' means in this case (whether it is or is not intended to mean 'extremely flammable' and 'flammable'), liquids that have a low flash point, or which are pyrophoric, are rarely oxidising. This implies that liquids classified as Flammable Liquids category 1 or 2, or as Pyrophoric Liquids, according to the CLP Regulation, normally do not need to be tested for oxidising properties. This corresponds to classification with risk phrases R12, R11 or R17 according to the DSD. If they contain oxidising functional groups, such substances may instead show self-reactive or explosive behaviour.

The third indent states that organic peroxides should not be tested for oxidising properties. Organic peroxides are distinguished by their chemical structure, and should be treated according to the procedures for the hazard class Organic Peroxides of the CLP Regulation, see Section [R.7.1.10.9](#) of this document.

Waiving according to the fourth indent relies on the absence of particular molecular structural features. The wording is more precise in section 2.13.4 of Annex I to the CLP Regulation, which is in principle the same as the wording as in section 6 of Appendix 6 to the UN-MTC.

The last two paragraphs above quoted from Column 2 Specific rules for adaptation from Column 1 are not applicable for this endpoint.

According to Section 2.13.4.1 of Annex I to the CLP Regulation, an organic liquid does not have to be assessed for oxidising properties if:

- a. *'the substance does not contain oxygen, fluorine or chlorine; or*
- b. *the substance contains oxygen, fluorine or chlorine and these elements are chemically bonded only to carbon or hydrogen.'*

For inorganic liquids, assessment of oxidising properties does not have to be done if the substance does not contain any oxygen or halogen atoms, according to section 2.13.4.2 of Annex I to the CLP Regulation.

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

The UN Test O.2 of the UN-MCT is also used for classification according to the UN-RTDG, and consequently also in the various regulations on transport of dangerous goods e.g. ADR and RID. A liquid that has been classified as belonging to Division 5.1 according to the regulations on transport of dangerous goods on the basis of results from the UN Test O.2, is an Oxidising Liquid according to the criteria of the CLP Regulation.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence

For the determination of whether a liquid is an oxidising liquid, weight of evidence is not possible. Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR

At the time of writing, no reliable (Q)SAR-methods exist for sufficiently accurate predictions of oxidising properties. As explained above, however, assessment of the chemical structure may be used to exclude oxidising behaviour of a substance. Possibly, this relation could be exploited in the development of future QSPR-methods. Such an assessment of chemical structure would formally form part of a Column 2 adaptation justification.

- Grouping of substances and read-across approach

For the determination of whether a liquid is an oxidising liquid, read-across is usually not possible. However interpolation may still be possible within homologous series.

- Testing is technically not possible

Testing should always be considered, if none of the waiving possibilities applies. Some oxidising substances may decompose when heated. Substances may occasionally react with

cellulose in other ways than through oxidation of the cellulose (e.g. through breaking chemical bonds within the cellulose). See also section 2.13.4.4 of Annex I to the CLP Regulation.

Further adaptation possibilities

Not foreseen.

Impurities; uncertainties

Minor impurities will usually not influence the test, unless they are very strong oxidisers. Expert judgement should be used to determine whether impurities may have an effect. Therefore utmost care should be taken in the selection of the key study(ies), or weight-of-evidence approaches, that the data selected is representative of the substance being registered by the respective companies.

A few substances may show other reactions than pure oxidation of the cellulose, or may decompose. If this is suspected, expert judgement should be sought. See also section 2.13.4.4 of Annex I to the CLP Regulation.

How to conclude on the DSD classification

Any substance classified as an oxidising liquid according to the CLP-criteria should normally be classified with risk phrase R8 or R9 according to the DSD. The DSD-criteria for classification with risk phrase R9 are not very precise, but if the CLP classification is Category 1, the substance should be classified with risk phrase R9 if the reaction with cellulose is violent, e.g. if spontaneous ignition occurs in the test.

In the DSD, the A.21 test method of Regulation (EC) 440/2008 is used for the assessment of oxidising properties of liquids. This method is in principle identical to the UN Test O.2 of the UN-MTC used in the CLP Regulation. However, the DSD does not make any division corresponding to the categories of the CLP, and therefore only one reference substance is used in the A.21 test method. Since the CLP Regulation method is used for classification of substances, it is strongly advisable to use the UN Test O.2 instead of the A.21 test method. This is because the O.2 test method will also give more detailed information on the oxidising behaviour of a substance (or mixture), since more reference mixtures are used.

Endpoint specific information in the registration dossier/in IUCLID

Material and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied.

Results and discussion

- indicate the results of the spontaneous ignition test;
- indicate the mean pressure rise time for the test substance;
- indicate the mean pressure rise time for the reference substance(s);
- interpretation of results;
- if testing is waived, the reasons for waiving must be documented in the dossier.

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on oxidising properties can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.15	VII 7.13	Oxidising properties	E.4.16	3.14

Further information / references

The ECHA guidance document [Guidance on the Application of the CLP criteria](#) gives in Chapter 2.13 detailed information on the oxidising property, the CLP-classification, the UN Test O.2 and the relation to the DSD and the transport of dangerous goods regulations.

R.7.1.13.3 Oxidising solids

Definition

The following definition of oxidising solids is provided in CLP Annex I, section 2.14.1:

'Oxidising solid means a solid substance or mixture which, while in itself is not necessarily combustible, may, generally by yielding oxygen, cause, or contribute to, the combustion of other material.'

Classification criteria and relevant information

According to the CLP Regulation, a solid is classified as an oxidising solid if in testing according to the UN Test O.1 of the UN-MTC (Part III, Section 34), it is at least as oxidising as potassium bromate in a 3:7 mixture with cellulose. The test is based on the burning behaviour of a mixture of cellulose and the tested solid. The CLP Regulation has three categories for oxidising solids. The category is also determined through the UN Test O.1 in the UN-MTC by comparison to reference mixtures of cellulose and potassium bromate²⁰.

Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for oxidising properties.

'The study does not need to be conducted if:

- the substance is explosive, or*
- the substance is highly flammable, or*
- the substance is an organic peroxide, or*

²⁰ At the time of writing, work is in progress at the UN-level to modify Test O.1: Test for oxidising solids. This includes changing the reference substance and introducing a gravimetric method for the measurement. For further information, see document UN/SCEGHS/23/INF.17 available at the following link: <http://www.unece.org/fileadmin/DAM/trans/doc/2012/dgac10c4/UN-SCEGHS-23-INF17.doc-UN-SCETDG-41-INF.43e.pdf> .

- *the substance is incapable of reacting exothermically with combustible materials, for example on the basis of the chemical structure (e.g. organic substances not containing oxygen or halogen atoms and these elements are not chemically bonded to nitrogen or oxygen, or inorganic substances not containing oxygen or halogen atoms).*

The full test does not need to be conducted for solids if the preliminary test clearly indicates that the test substance has oxidising properties.

Note that as there is no test method to determine the oxidising properties of gaseous mixtures, the evaluation of these properties must be realised by an estimation method based on the comparison of the oxidising potential of gases in a mixture with that of the oxidising potential of oxygen in air.'

The first indent states that explosive substances should not be tested for oxidising properties. For instance, organic substances with oxidising functional groups may be explosive and should first undergo the screening procedures for explosive properties in Annex 6 of the UN-MTC to rule out possible explosive behaviour. Such substances may also be thermally unstable and show self-reactive behaviour. Substances that have been classified as Explosives according to the CLP-regulation or have been assigned risk phrases R2 or R3 according the DSD, should normally not be tested for oxidising properties, since they are known to be explosive.

The second indent states that highly flammable substances do not have to be tested for oxidising properties. While it is not very clear what 'highly flammable' means in this case (whether it is or is not intended to mean 'extremely flammable' and 'flammable'), solids classified as Flammable Solids or as Pyrophoric Solids according to the CLP-regulation are rarely oxidising. This corresponds to classification with risk phrases R11 or R17 according to the DSD. If they contain oxidising functional groups, such substances may instead show self-reactive or explosive behaviour.

The third indent states that organic peroxides should not be tested for oxidising properties. Organic peroxides are distinguished by their chemical structure, and should be treated according to the procedures for the hazard class Organic Peroxides of the CLP-regulation, see Section [R.7.1.10.9](#) of this document.

Waiving according to the fourth indent relies on the absence of particular molecular structural features. The wording is more precise in section 2.14.4 of Annex I to the CLP-regulation, which is in principle the same as the wording as in Section 6 of Appendix 6 to the UN-MTC.

The first note under last indent from REACH Annex VII, which allows waiving of further testing, namely '*[...] if the preliminary test clearly indicates that the test substance has oxidising properties*' is relevant only when using the A.17 test method of Regulation (EC) 440/2008, which is **not** the preferred test method since it belongs to the DSD classification system. The UN Test O.1 used for classification according to the CLP Regulation does not include any preliminary test.

The last note taken from Column 2 'Specific rules for adaptation from Column 1' is not applicable for this endpoint. For inorganic solids, assessment of oxidising properties does not have to be done if the substance does not contain any oxygen or halogen atoms, according to section 2.14.4.2 of Annex I to the CLP Regulation.

According to section 2.14.4.1 of Annex I to the CLP-regulation, an organic solid does not have to be assessed for oxidising properties if:

- 'the substance does not contain oxygen, fluorine or chlorine; or*
- the substance contains oxygen, fluorine or chlorine and these elements are chemically bonded only to carbon or hydrogen.'*

Adaptation possibilities according to Annex XI to REACH

- Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

The UN Test O.1 of the UN-MTC is also used for classification according to the UN-RTDG, and consequently also in the various regulations on transport of dangerous goods e.g. ADR and RID. A solid that has been classified as belonging to Division 5.1 according to the regulations on transport of dangerous goods on the basis of results from the UN Test O.1, is an oxidising solid according to the criteria of the CLP Regulation.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

- Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

- (Q)SAR

At the time of writing, no reliable (Q)SAR-methods exist for sufficiently accurate predictions of oxidising properties. As explained above, however, assessment of the chemical structure may be used to exclude oxidising behaviour of a substance. Possibly, this relation could be exploited in the development of future (Q)SPR-methods. Such an assessment of chemical structure would formally form part of a Column 2 adaptation argument.

- Grouping of substances and read-across approach

For the determination of the oxidising solids read-across is usually not possible. However interpolation may still be possible within homologous series.

- Testing is technically not possible

Testing should always be considered if none of the waiving possibilities applies. Some substances may decompose upon heating. Substances may occasionally react with cellulose in other ways than through oxidation of the cellulose.

Further adaptation possibilities

Not foreseen.

Impurities; uncertainties

The UN Test O.1 is (currently) performed using the unaided eye as measuring instrument. Only by expert judgement and thorough experience can the result of the test be correctly judged, and even then uncertainties may arise.

Minor impurities will usually not influence the test, unless they are very strong oxidisers. Expert judgement should be used to determine whether impurities may have an effect.

A few substances may show other reactions than pure oxidation of the cellulose, or may decompose. If this is suspected, expert judgement should be sought. Particle size and size distribution can have an influence on the test results.

How to conclude on the DSD classification

Any substance classified as an oxidising solid according to the CLP Regulation criteria should normally be classified with risk phrase R8 or R9 according to the DSD. The DSD-criteria for classification with risk phrase R9 are not very precise, but if the CLP Regulation classification is Category 1, the substance should be classified with risk phrase R9 if the reaction with cellulose is violent.

In the DSD, the A.17 test method of Regulation (EC) 440/2008 is used for the assessment of oxidising properties of solids. Although the principle of this method is to a large extent the same as that of the UN Test O.1 of the UN-MTC, the experimental set-up, reference substance (barium nitrate) and measured quantity differ. Furthermore, the DSD does not make any division corresponding to the categories of the CLP. Since the CLP Regulation is used for classification of substances, it is not advisable to use the A.17 method (which belongs to the DSD classification system). Instead, the UN Test O.1 should be used, which will also give more detailed information on the oxidising behaviour of a substance (or mixture), since more reference mixtures are used.

Endpoints specific information in the registration dossier/in IUCLID

Material and methods

- description of the apparatus and dimensions or reference to the standard or the test method applied;
- particle size and distribution.

Results and discussion

- if testing is waived, the reasons for waiving must be documented in the dossier.

If the UN test O.1 was used:

- indicate if a vigorous reaction was observed;
- indicate the maximum burning time for the test mixture;
- indicate the maximum burning time for the reference mixtures;
- interpretation of results, including any relevant special observations;
- estimated accuracy of the result (including bias and precision).

If A.17 test method was used:

- indicate if in the preliminary test, a vigorous reaction was observed;
- indicate the maximum burning rate for the test mixture;
- indicate the maximum burning rate for the reference mixture;
- interpretation of results, including any relevant special observations;
- estimated accuracy of the result (including bias and precision).

Any deviation from the guideline method used (and reasons for it) or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on oxidising properties can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.15	VII 7.13	Oxidising properties	E.4.16	3.14

Further information / references

The ECHA [Guidance on the Application of the CLP criteria](#) gives in Chapter 2.14 detailed information on the oxidising property, the CLP-classification, the UN Test O.1 and the relation to the DSD and the transport of dangerous goods regulations.

R.7.1.14 Granulometry

Advice to registrants with regard to nanomaterials characterisation of granulometry can be found in *Appendix R7-1 Recommendations for nanomaterials applicable to: Chapter R7a Endpoint specific guidance* of the [Guidance on IR&CSA](#), section 2.2.3 Granulometry.

R.7.1.14.1 Type of property

Granulometry is not a specific physico-chemical property of a substance. The original particle size distribution is highly dependent on the industrial processing methods used and can also be affected by subsequent environmental or human transformations. Particle size is usually measured in micrometers ($= 10^{-6}$ m; μm ; 'microns').

Granulometry is of considerable importance for the toxic properties of a substance as it influences aspects such as:

- the route of exposure of humans and toxicity by inhalation;
- the choice of route of administration for animal testing;
- the efficiency of uptake in an organism;
- the distribution in the environment.

Granulometry is of importance for combustible dusts as it influences aspects such as the likelihood to form combustible/explosive dust - air mixtures.

In general all powder materials have a range of particle sizes (particle size distribution), a presentation of the particle size distribution (e.g. using a histogram of the particle size vs. mass, particle size vs. number of particles, etc.) is therefore necessary to interpret the data.

For inhalation exposure it is well known that the human toxicity will be related with the place of deposition into the respiratory tract. The location of deposition mainly depends on the properties of the particle (size, shape, density etc) that are commonly taken into account considering the aerodynamic diameter of the particle (see definition below). Thus, the general approach has been to use mass fractions (e.g. health related fractions as defined by EN 481 or the EPA PM Fractions). For instance, in Europe, from the publication of the EN 481 the OELs for powder materials have been defined for one or several fractions (inhalable, thoracic or respirable).

Photocentrifuge method - the method of determining the particle size distribution, which is described in ISO 13318-2:2007, is applicable to powders that can be dispersed in liquids, powders that are present in slurry form and some emulsions. Typical particle size range for analysis is from about 0.1 μm to 5 μm . The method is applicable to powders in which all particles have the same density and comparable shapes and do not undergo chemical or physical change in the suspension liquid. It is usually necessary that the particles have a density higher than that of the liquid.

Light extinction liquid-borne particle counter - in ISO 21501-3:2007 a calibration and verification method for a light extinction liquid-borne particle counter (LSLPC) is described, which is used to measure the size and particle number concentration of particles suspended in liquid. The light extinction method is based on single particle measurements and the typical size range of particles measured by this method is between 1 μm and 100 μm .

Light scattering liquid-borne particle counter - in ISO 21501-2:2007 a calibration and verification method for a light scattering liquid-borne particle counter (LSLPC) is described, which is used to measure the size and particle number concentration of particles suspended in liquid. The light scattering method is based on single particle measurements and the typical size range of particles measured by this method is between 0.1 μm and 10 μm .

Centrifugal X-ray method - the method of determining the particles size distribution described in ISO 13318-3:2004 is applicable to powders which can be dispersed in liquids or powders which are present in slurry form. The typical particle size range for analysis is from 0.1 μm to 5 μm . The method is applicable to powders in which all particles have the same effective density, chemical composition and comparable shapes.

The CEN document, EN 481 'Workplace Atmospheres – size fraction definitions for measurement of airborne particles' (CEN 1993) provides definitions of the inhalable, thoracic and respirable size fractions, and target specifications (conventions) for sampling instruments to measure these fractions. The current standard defines sampling conventions for particle size fractions which are to be used in assessing the possible health effects resulting from inhalation of airborne particles in the workplace. The different particle sizes defined in EN 481 are:

- inhalable fraction (the mass fraction of particles that can be inhaled by nose and mouth. Particles >100 μm are not included in the inhalable convention;
- thoracic fraction (the mass fraction of the inhaled particles that passes the larynx). The convention for thoracic fraction sets that 50% of the particles in air with an aerodynamic diameter of 10 μm belong to the thoracic fraction;
- respirable fraction (the mass fraction of the inhaled particles that reaches the alveoli) The convention for respirable fraction sets that 50% of particles with an aerodynamic diameter of 4 μm belong to the respirable fraction.

R.7.1.14.2 Definitions

Aerodynamic diameter: the diameter of a sphere of density 1 g.cm^{-3} with the same terminal velocity (falling speed) due to gravitational force in calm air as the particle under the prevailing conditions of temperature, pressure and relative humidity (CEN, 1993). The aerodynamic diameter is used to compare particles of different sizes, shapes and densities and it is a useful parameter to predict where in the respiratory tract such particles may be deposited. It is used in contrast to 'optical', 'measured' or 'geometric' diameters which are representations of actual diameters which in themselves cannot be related with the deposition within the respiratory tract.

Particle diffusion diameter: for particles of aerodynamic diameter less than 0.5 μm , the particle diffusion diameter should be used instead of the particle aerodynamic diameter. For diffusion, the appropriate *equivalent diameter* is the diffusion (mobility) diameter. This is defined as the diameter of a sphere with the same diffusion coefficient as the particle under the prevailing conditions of temperature, pressure and relative humidity.

The parameter of interest is the effective hydrodynamic radius, or effective Stoke's radius R_s . Particle size distribution (effective hydrodynamic radius) requires information on water insolubility. Fibre length and diameter distributions require information on the fibrous nature of the product and on stability of the fibrous shape under electron microscope conditions.

A fibre: is a water insoluble particle with an aspect ratio (length/diameter > 3) and diameter < 100 μm . Fibres of length < 5 μm need not be considered.

Particle: Minute piece of matter with defined physical boundaries. (ISO/TS 27687:2008)

Agglomerate: A collection of weakly bound particles of aggregates or mixtures of the two where the resulting external surface area is similar to the sum of the surface areas of the individual components (ISO/TS 27687:2008).

Aggregate: Particle comprising strongly bonded or fused particles where the resulting external surface area may be significantly smaller than the sum of calculated surface areas of the individual components (ISO/TS 27687:2008).

R.7.1.14.3 Test methods

Many methods are available for particle size measurements, but none of them is applicable to the entire size range (see [Table R.7.1-11](#)). Sieving, microscopic sedimentation and elutriation techniques are most commonly employed. Methods for determining particle size distribution are designed to provide information on the transportation and sedimentation of insoluble particles in water and air. An integrated testing strategy (ITS) detailing the appropriate methods for determination of particle size distribution of respirable and inhalable particles is shown in [Figure R.7.1-6](#).

Details of methods for determining particle size distribution and for fibre length and diameter distributions are outlined in OECD TG 110 and in the 'Guidance Document on the Determination of Particle Size Distribution, Fibre Length and Diameter Distribution of Chemical Substances' (JRC, 2002).

The particle size distribution is carried out on the material under investigation and not as airborne dust.

The measurement principle of the method used will determine what kind of diameter of the particle can be determined: for instance, optical diameter when using light scattering or aerodynamic diameter when using impactors. Methods which determine the mass median aerodynamic diameter (MMAD) need the generation of representative test atmospheres using suitable generation equipment and correct sampling techniques. They can be used in case of airborne particles (dusts, smokes, fumes), nebulised particles (wet aerosol) or dispersed particles (dry aerosol).

Figure R.7.1–6 Integrated testing strategy for granulometry

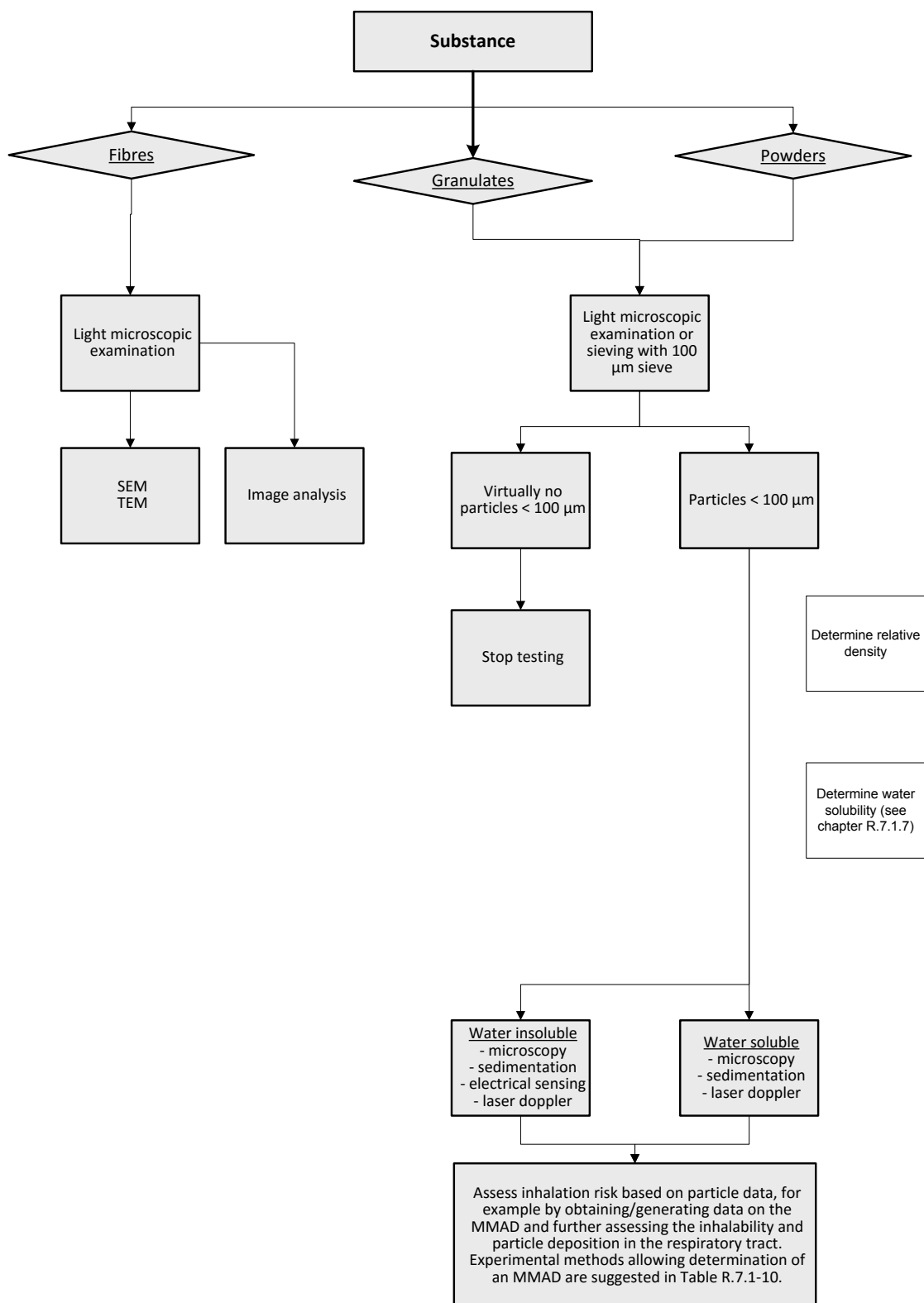


Table R.7.1–11 Methods to determine particle size distribution of a material

Method and details	Material and size range	MMAD
Microscopic examination It is preferable to prepare samples directly in order not to influence shape and size of the particles. This method determines size distribution of particles.	Particles of all kinds Size range: 0.5–5000 microns (light microscope) and <0.1–10 microns (SEM/TEM)	MMAD cannot be determined
Sieving Sieving using wire-mesh sieves and perforated sheet metal sieves is not suitable to determine the distribution of particles of respirable and inhalable size since their range is only 100–10,000 microns. Micro mesh sieves (range 5–100 micron) may give better results. However, since these sieves are generally operated in combination with mechanical or ultrasonic vibration, modification of median size and form may result. Sieving not suitable to determine distribution of particles of respirable size, but might be suitable to determine bigger particles.	Dry powders/granulates Size range: 100–10,000 microns (wire mesh/metal sieves) and 5–100 (micromesh)	MMAD cannot be determined
Sedimentation (gravitational settling) Method is based on gravitational settling of particles in liquid and the effective hydrodynamic radius is determined. Effective hydrodynamic radius distribution should be measured 3x with no two values differing by >20%. Requires sufficient numbers of radius intervals be used to resolve the radius distribution curve. Binary or ternary mixtures of latex spheres (2–100 microns) are recommended as calibration material. Method might be suitable to determine the distribution of particles of respirable and inhalable size.	Dry powders/granulates Size range: 2–200 microns	MMAD cannot be determined
Electrical Sensing Zone (e.g. Coulter) method Samples are suspended in an electrolytic solution. As the particle is drawn through an aperture, the change in conductance gives a measure of particle size. The important parameter is the settling velocity in the liquid phase, which depends on both density and diameter. Particles having a density of several g/cm³ can be determined. Applicable to particles that are complete electrical isolators in the fluid. Difference in density between particles and fluid must not be too large. Method might be suitable to determine the distribution of particles of respirable and inhalable size.	Dry powders/granulates (non-conducting) Size range: 1–1000 microns	MMAD cannot be determined

Phase Doppler Anemometry Expensive technique. Particle size distribution can be measured either in air or in liquid. The method presupposes that the particles are spherical with known refractive index. Method might be suitable to determine the distribution of particles of respirable and inhalable size.	Dry powders/granulates Size range: 0.5-80 microns (in air); 0.5-1000 microns (in liquid)	MMAD cannot be determined
Determination of fibre length and diameter distributions Light microscopy used to examine likelihood of fibres present by comparing similarities to known fibrous or fibre releasing substances or other data. Extreme care required during sample preparation to avoid fibre breaking and clumping. Care should also be taken to avoid contamination by airborne fibres. Samples might be prepared by (a) producing suspensions in water by gentle hand agitation or vortex mixing or (b) transfer of dry material onto copper tape either directly or by spraying of the dry fibres by use of atomiser or pipette. Length and diameter distributions should be measured independently at least twice and at least 70 fibres counted. No two values in a given histogram interval should differ by > 50% or 3 fibres, whichever is larger. The presence of long thin fibres would indicate a need for further, more precise measurements.	Fibrous products Size range: diameters as small as 0.1 micron and as large as 100 micron and lengths as small as 5 micron and as large as 300 micron	

It is advantageous to have accurate information about the propensity of materials to produce airborne dust (the *dustiness* of the material). No single method of dustiness testing is likely to represent and reproduce the various types of processing and handling used in industry. The measurement of dustiness depends on the test apparatus used, the properties of the dust and various environmental variables (i.e the dustiness is not a measurement of the 'dust as it is') . There are a number of methods for measuring the dustiness of bulk materials, based on the health related aerosol fractions defined in EN 481. Two methods (the rotating drum method and the continuous drop method) are detailed in EN 15051 'Workplace atmospheres – Measurement of the dustiness of bulk materials – Requirements and reference test methods' (CEN, 2006).

Dustiness is a relative term (derived from the amount of dust emitted during a standard test procedure). This is dependent on the method chosen, the condition and properties of the tested bulk material, and various environmental variables in which the tests are carried out. Thus, the two methods in EN 15051 may provide different results (the methods are intended to simulate handling processes) The standard is currently under revision (draft of European standard available) and the final publication is expected for 2013. The standard has been divided in 3 parts (a general part and one part for each of the methods). The methods ([Table R.7.1-12](#)) as described in the standard are used to determine dustiness in terms of the health related fractions defined by EN 481. Further analyse (e.g. analysing the contents on the dust collection stages) can be used to obtain the particle sizedistribution. These methods require the generation of representative test atmospheres using suitable generation equipment and correct sampling techniques.

Table R.7.1–12 Methods to generate/sample airborne dispersed or nebulised particles

Method and details	Material and size range	MMAD
Cascade impaction Cascade impactors can be used to obtain the size distribution of an aerosol (i.e in this context a dust cloud). Air samples are drawn through a device which consists of several stages on which particles are deposited on an impactation substrate. Particles will impact on a certain stage depending on their aerodynamic diameter . The cut- off size can be calculated from the jet velocities at each stage by weighing each stage before and after sampling and the MMAD derived from these calculations. This is a well established technique to measure the size distribution of particles (allowing calculating any mass fraction). Some models are specifically designed to give the 3 health related fractions defined by the EN 481. Please also check ISO/TR 27628:2007, which contains specific information on methods for bulk aerosol characterization and single particle analysis while using cascade impaction method.	Particles in an aerosol Size range: 0.1-20 and 0.5-80 microns	MMAD can be determined <i>via</i> an appropriate coupled analytical technique.
Laser scattering/diffraction In general, the scattering of the incident light gives distinct pattern which are measured by a detector. This technique is particle property dependent – i.e. material has unique scattering and diffraction properties which are also particle size dependent. It is important to calibrate the instrument with similar material (of the same size range as the material to be measured). Laser scattering techniques are suitable for geometric particles, viz spheres, cubes and monocrystals. Particle size will be established optically. The MMAD can be calculated by means of a calculation correction. Further information about corrections and limitations of the methods can be found in CEN/TR 16013-1 and CEN/TR 16013-2. Please also check ISO 13320:2009 Particle size analysis – Laser diffraction methods taking into account the possible limitations of the method, a the technique assumes a spherical particle shape in its optical model. The resulting particle size distribution is different from that obtained by methods based on other physical principles (e.g. sedimentation, sieving).	Particles of all kind Size range: 0.1 um to 3 mm (with special instrumentation and conditions, the size range can be extended above 3 mm and below 0.1 mm)	MMAD can be determined.

Rotating drum method (prEN 15051-2) This method is based on size selective sampling of an airborne dust cloud produced by the repeated lifting and dropping of a material in a rotating drum. Air drawn through the drum passes through a specially designed outlet and a 3-stage fractionating system consisting of two porous polyurethane foams and a membrane filter. The mass of dust collected on each collection stage is determined gravimetrically to give a direct measure of the biologically relevant size fractions. This method simulates a wide range of material handling processes in industry and determines the biologically relevant size functions of a material in the airborne state. This method is suitable to determine the respirable thoracic or inhalable fractions.	Dry powders/granulates/friable products Size range: 0.5-10,000 microns	MMAD cannot be determined.
Continuous drop method (prEN 15051-3) This method is based on the size selective sampling of an airborne dust cloud produced by the continuous single dropping of material in a slow vertical air current. The dust released by dropping material is conducted by the airflow to a sampling section where it is separated into the inhalable and respirable fractions. This method is suitable to determine the respirable and inhalable fractions.	Dry powders/granulates/friable products Size range: 0.5-10,000 microns	MMAD can be determined.

Table R.7.1–13 Methods that measure inhalable fractions only or that give no detailed distributions

Method and details	Material and size range	MMAD
Elutriation Particles are drawn out on a column at varying velocity. The velocity is used to calculate particle size and the weight of the remaining sample at a particular velocity is used to calculate the distribution. The method is limited to particles >15 microns. The method is not suitable to determine the distribution of particles of respirable size, but might be suitable to determine the distribution of particles of inhalable size	Dry powders/granulates Size range: 15-115 microns	MMAD cannot be determined.
Air jet sieve Air is aspirated through a weighted sample on a fine sieve and the weight loss measured. The method is capable of estimating the non-floatable fraction of the material under investigation. Aggregation of the particles will result in unreliable values. In addition, since the lower detection limit is only 10 micron, this method is not suitable to determine the distribution of particles of respirable size. The method is not suitable to determine the distribution of particles of the respirable fraction, but might be suitable to determine the distribution of particles between 10 and 10,000 microns..	Particles of all kind Size range: 10-10,000 microns	MMAD cannot be determined.
Cyclons The use of a cyclone is a simple approach to determining whether respirable and/or inhalable particles are present in the test atmospheres by constructing the cyclone cut off points at 4.25 and 100 microns. By measuring the weight of particles which pass through the cyclone it can be decided whether more sophisticated methods have to be applied to determine the size distribution of the particles smaller than 10 micron. This method is suitable to determine the respirable, thoracic or inhalable fraction.	Particles of all kind Size range: 0.1-200 microns	MMAD cannot be determined.

R.7.1.14.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VII to REACH

Column 2 of REACH Annex VII provides the following specific rules for adaptation of the standard information requirement for granulometry.

'The study does not need to be conducted if the substance is marketed or used in a non solid or granular form.'

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

As the granulometry of a substance is highly dependent on the industrial processing methods and possibly also on handling of the material, any published data on granulometry will be pertinent only to the particular sample or process.

There are a number of web sites and electronic databases that include compilations of and evaluations of data on particle properties. However, there appear to be a limited number of reference books that provide particle size data.

The equivalence of the various national and international standard methods for particle size distribution has not been tested and is not known.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

In some situations where data is available from multiple sources (e.g. information on particle size distribution of different batches, or information from different methods), a weight of evidence approach may be used. Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

There are no QSPR/(Q)SAR tools available for predicting particle size and the data will therefore need to be experimentally determined. Application of (Q)SAR is not possible.

Grouping of substances and read-across approach

At present grouping and read across are not applicable.

Testing is technically not possible

Testing should always be considered, if none of the waiving possibilities applies. Testing should always be possible for solids or granular substances.

Further adaptation possibilities

Not foreseen.

R.7.1.14.5 Impurities; uncertainties

There is a particular problem in relation to sedimentation and Coulter counter measurements. The effect of impurities on particle shape should be considered when measuring fibre length and diameter distributions.

The small quantities used as samples must be representative of product batches comprising many kilograms; therefore sampling and sample handling require great care.

Great care should also be taken due to the fact that non-conducting particles in a non-conducting liquid may be electrically charged resulting in non-representative settling of particles of a certain size. In addition, in the process of particle size distribution determination, it is very important to take the electrostatic charge of the particles into account. Electrostatically charged particles behave differently and may influence sampling.

It is useful to distinguish between aggregates and agglomerates. While an aggregate is held together by strong forces and may be considered to be permanent, agglomerates are held together with weak forces and may break up under certain circumstances. As small particles often form agglomerates, sample pre-treatment (e.g. the addition of dispersing agents, agitation or low-level ultrasonic treatment) may be required before the primary particle size can be determined. However, great care must be taken to avoid changing the particle size distribution.

R.7.1.14.6 Endpoint specific information in the registration dossier / in IUCLID

Material and methods

- sample preparation, such as any sonication, grinding, or addition of dispersion agents (if any);
- if a suspending medium is used (e.g. sedimentation test): indicate type of medium, temperature, pH, concentration and solubility of the substance in the suspending medium;
- the type of method used.

Results and discussion

- in the particle size field: mean and standard deviation;
- in the particle size distribution at different passages field: size and distribution;
- approximate information on particle shape (e.g. spherical, platelike, needle shaped) if available;
- for fibres: indicate both length and diameter of fibres.

Any deviation from the guideline method used or any other special consideration should be reported. In cases where there is more than one source of data, the endpoint summary under results and discussion should provide a justification for the selection of the key study chapter.

Reference to other ECHA Guidance Documents

Further detailed guidance on particle size distribution (Granulometry) can be found in the following chapters:

IUCLID Section	REACH Annex	Endpoint title	IUCLID 5 End User Manual Chapter	ECHA Practical Guide 3
4.5	VII 7.14	Particle size distribution (Granulometry)	E.4.6	3.5

R.7.1.14.7 Further information / references

CEN 1993	EN 481: Workplace atmospheres. Size fraction definitions for measurement of airborne particles
CEN 2006	EN 15051: Workplace atmospheres. Measurement of the dustiness of bulk materials – Requirements and reference test methods
JRC (2002)	"Guidance Document on the Determination of Particle Size Distribution, Fibre Length and Diameter Distribution of Chemical Substances", ISBN 92-894-3704-9, EUR 20268 EN, http://publications.jrc.ec.europa.eu/repository/handle/111111111/5555
OECD TG 110	Test No. 110: Particle size distribution/fibre length and diameter distributions
prEN 15051-1 rev	Workplace exposure - Measurement of dustiness of bulk materials - Part 1: Requirements and choice of test methods
prEN 15051-2	Workplace exposure - Measurement of the dustiness of bulk materials - Part 2: Rotating drum method
prEN 15051-3	Workplace exposure - Measurement of the dustiness of bulk materials - Part 3: Continuous drop method
(ISO/TS 27687:2008)	Nanotechnologies-Terminology and definitions for nano-objects- Nanoparticle, nanofibre, and nanoplate
CEN/TR 16013-1:2010	Workplace exposure. Guide for the use of direct-reading instruments for aerosol monitoring. Choice of monitor for specific applications
CEN/TR 16013-2:2010	Workplace exposure. Guide for the use of direct-reading instruments for aerosol monitoring. Evaluation of airborne particle concentrations using optical particle counters

R.7.1.15 Adsorption/Desorption

Advice to registrants with regard to nanomaterials characterisation of adsorption/desorption can be found in *Appendix R7-1 Recommendations for nanomaterials applicable to: Chapter R7a Endpoint specific guidance* of the [Guidance on IR&CSA](#), section 2.2.4 Adsorption/desorption.

R.7.1.15.1 Type of property

Adsorption/desorption is not a specific physicochemical property of a substance. This property indicates the binding capacity (or 'stickiness') of a substance to solid surfaces, and so is essential for understanding environmental partitioning behaviour.

Information on adsorption/desorption is an essential input to environmental exposure models, because:

- adsorption to suspended matter can be an important physical elimination process from water in sewage treatment plants (STPs). This in turn may mean that sewage sludge, if spread to land, is a major source of the substance in soil;
- adsorption to suspended matter in receiving waters affects both the concentration in surface water and the concentration in sediment;
- desorption of a substance from soil directly influences its mobility and potential to reach surface or groundwaters.

Consequently, information on adsorption/desorption is also an important factor in test strategies for assessing toxicity to sediment- or soil-dwelling organisms.

Substances that adsorb strongly to biological surfaces (e.g. gills, skin, etc.) may lead to toxic effects in higher organisms after biomagnification.

The information is also relevant for assessing environmental persistence. For example: degradation rates in sediment and soil are also assumed to be reduced by default if a substance is highly sorptive (since it is less bioavailable to microorganisms). This may lead to consideration of soil/sediment simulation testing in some cases.

Finally, there may be practical implications for test performance: Substances that adsorb strongly to surfaces can be difficult to test in aquatic systems.

R.7.1.15.2 Definition

Adsorption is caused by temporary (reversible) or permanent bonding between the substance and a surface (e.g. due to van der Waals interactions, hydrogen bonding to hydroxyl groups, ionic interactions, covalent bonding, etc.). The OECD guidances offer further information (OECD 2000a, OECD 2000b, OECD 2001, OECD 2002).

The organic carbon normalized adsorption coefficient (K_{oc}) is the ratio of a substance concentration sorbed in the organic matter component of soil or sediment to that in the aqueous phase at equilibrium. In other words, $K_{oc} = K_d/f_{oc}$, where K_d is the distribution coefficient for adsorption, and f_{oc} the organic carbon content – the fraction organic carbon present in the soil or sediment. In turn, K_d is the experimental ratio of a substance's concentration in the soil (C_s) to that in the aqueous phase (C_{aq}) at equilibrium; namely $K_d = C_s/C_{aq}$. The organic matter normalized distribution coefficient (K_{om}) is similarly defined, but refers to the organic matter content of soil rather than the organic carbon content (OECD, 2000a).

R.7.1.15.3 Test method(s)

The adsorption of a substance to sewage sludge, sediment and/or soil can be measured or estimated using a variety of methods, which are tabulated in

[Table R.7.1-14](#) in order of increasing complexity. The dissociation constant (if appropriate) should be known before testing. Information on vapour pressure, solubility in water and organic solvents, octanol-water partition coefficient and stability/degradability is also useful.

Table R.7.1-14 Methods for the measurement of adsorption

Method and Description	Applicability/Notes
Adsorption control within an inherent biodegradability test (OECD TG 302B) Estimate of the extent of adsorption to STP sludge made from the elimination level in a Zahn-Wellens inherent biodegradation test. (e.g. OECD TG 302B). 3-hour value recommended. Values beyond 24 hours not normally used. Where data are not available for adsorption up to 24 hours, data from time scales beyond this can only be used if adsorption is the only removal mechanism, with an upper limit of 7 days.	Highly adsorptive substances that are water soluble
HPLC method: OECD TG 121; EU C.19: Estimation of the Adsorption Coefficient (K_{oc}) on Soil and on Sewage Sludge using High Performance Liquid Chromatography (HPLC) (Original Guideline, adopted 22 January 2001) Calibration with reference substances (preferably structurally related to the test substance) of known K_{oc} allows the K_{oc} of the test substance to be estimated. Test substance K_{oc} value should lie within the calibration range of the reference substances.	Measurement of log K_{oc} in the range 1.5 to 5.0. Validated for several chemical types, see test guideline for details. Poorly soluble and volatile substances as well as mixtures. Ionisable substances: test both ionised and unionised forms in appropriate buffer solutions where at least 10 % of the test compound will be dissociated within pH range 5.5 to 7.5. May not be suitable for: substances that react with the column, solvent or other test system components; surface active substances; substances that interact in a specific way with inorganic soil components such as clay minerals; inorganic compounds; moderate to strong acids and bases.
Batch test of adsorption of substances on activated sludge (ISO 18749) Screening method to determine the degree of adsorption of substances on activated or primary sludge in sewage treatment plants (ISO, 2004). The method does not differentiate between adsorption and other elimination methods (such as complex formation, flocculation, precipitation, sedimentation or biodegradation).	Suitable for substances that: - are water soluble, or allow for stable suspensions/dispersions/emulsions, - are not significantly removed by abiotic processes (e.g. stripping/foaming), - do not de-flocculate activated sludge, - are not readily biodegradable, and - have a sufficiently sensitive analytical method.

Sediment and soil adsorption/desorption isotherm (OPPTS 835.1220) Screening method according to US-EPA guideline (OPPTS, 1996) using three soil types.	
Batch equilibrium method (OECD TG 106; EU C.18: Absorption – Desorption Using a Batch Equilibrium Method (Updated Guideline, adopted 21 January 2000) Test uses a range of actual soils and so represents a more realistic scenario than the HPLC (OECD 121) method.	Used for substances with K_{oc} values that cannot be reliably determined using other techniques (e.g. surfactants). Requires a quantitative analytical method for the substance, reliable over the range of test concentrations. For ionisable substances, soil types should cover a wide range of pH. Adjustments for poorly soluble substances given in the test guideline.
OECD TG 312: Leaching in Soil Columns (Original Guideline, adopted 13 April 2004) K_d values can be derived from column leaching studies.	Appropriate study design to estimate K_d values particularly for unstable test substances that degrade significantly during the equilibrium time of 'shake flask' sorption studies
Simulation tests and direct field measurement: including OECD guidance document no. 22 (OECD, 2000b). Monolith lysimeters can be used to study the fate and behaviour of substances in an undisturbed soil profile under outdoor conditions. They allow for monitoring of the volume of leaching/drainage water as well as the concentrations of a substance and its transformation products. They are mainly used in pesticide studies. Field leaching studies can also be carried out where hydrodynamically isolated soil layers are analysed <i>in situ</i>. Although such studies are the most realistic, their reproducibility and representativity may be limited (e.g. due to the effects of large-scale soil structure, weather events, the soil conditions at the time of application, etc.). Since data from these methods are unlikely to be encountered for the vast majority of industrial substances, they are not considered further here. Further information can be found in guidance for pesticide registration.	

R.7.1.15.4 Adaptation of the standard testing regime

Adaptation possibilities according to column 2 of Annex VIII and IX to REACH

Screening information on adsorption (and desorption) is required for substances manufactured or imported in quantities of 10 t/y or more. Depending on the results, further information (for example, a test) may be required for substances manufactured or imported in quantities of 100 t/y or more.

Column 2 of REACH Annexes VIII and IX provides two exemptions.

'The study does not need to be conducted if:

- based on the physicochemical properties the substance can be expected to have a low potential for adsorption (e.g. the substance has a low octanol water partition coefficient), or*
- the substance and its relevant degradation products decompose rapidly.'*

Or in other words, the substance and its relevant degradation products decompose rapidly. Therefore, if a substance hydrolyses, it might be more appropriate to also determine the degree of adsorption of the hydrolysis products.

In practice, a cutoff value of $\log K_{ow} = 3$ can be applied for adsorption potential. However, caution should be exercised in using this criterion, as substances that are water soluble and have a low octanol-water partition coefficient do not necessarily always have a low adsorption potential. A *measured* adsorption coefficient is usually needed for ionising substances, since it is important to have information on pH-dependence (cationic substances in particular generally adsorb strongly). Similarly, measured values will normally be needed for surface active substances (e.g. surfactants), because K_{ow} values (predicted or measured) are likely to be poor predictors of adsorption for these types of substance. For ionisable substances, partition coefficients should also be corrected according to the pH of the environment being assessed (see Annex 2). For complex mixtures (e.g. UVCBs), a single value of K_{oc} will not be definitive. In such cases a range of values or a representative value can be given, depending on the substance.

Adaptation possibilities according to Annex XI to REACH

Use of existing data: Data on physical-chemical properties from experiments not carried out according to GLP or the test methods referred to in Article 13 (3) of REACH

For all organic substances manufactured or supplied in quantities of 10 tonnes per year or more, the K_{oc} should be estimated using read-across or QSPR methods as a first step. If the property is likely to be a significant determinant in the calculation of risk (e.g. following a sensitivity analysis), then a test should be conducted to provide a more reliable value for substances manufactured or supplied in quantities of 100 t/y or more. In general, confirmatory testing would not be expected for non-ionising substances with a $\log K_{ow}$ value below 3, or for substances that degrade rapidly (in which case the degradation products may be more relevant). The HPLC method may be used as a first step in testing, with the batch equilibrium method being considered only if more definitive data become necessary for the Chemical Safety Assessment. Column leaching studies might be an option under some circumstances (e.g. for unstable test substances that degrade significantly during the equilibrium time of shake flask sorption studies).

If estimation methods are not appropriate (e.g. because the substance is a surfactant or ionisable at environmentally-relevant pH), then a batch equilibrium test may need to be considered at the 10 tonnes per year band, and would be essential at the 100 tonnes per year band.

If experimental data are available (study reports or literature data) meeting the criteria in section 1.1.1 of Annex XI to REACH, these could be used to meet the endpoint data requirements. If an estimation method is used as a source of information according to Column 2 of Annex VII, the QSAR model must meet the criteria set out in section 1.3 of Annex XI to REACH.

Weight of evidence

Where no single source of existing data (study reports, QSAR, literature data) is considered sufficiently reliable, thus not fully meeting the criteria in section 1.1.1 of Annex XI to REACH, or where several sources of similar reliability with deviating results exist, a weight of evidence approach may be used. The criteria in section 1.2 of Annex XI to REACH must then be met.

(Q)SAR

Soil sorption (K_{oc}) of organic non-ionic substances can often be estimated from their octanol-water partition coefficient (K_{ow}), as well as from other properties such as aqueous solubility.